
Muon Dynamics as a Spectral Wasserstein Flow

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Abstract

Gradient normalization can stabilize deep-learning optimization, and spectral normalizations are especially natural for matrix-shaped parameter blocks; Muon is the motivating example. We study an idealized deterministic, continuous-time, vanishing-momentum version of this idea in the mean-field regime, where wide models are represented by probability measures on parameter space. Starting from normalized matrix flows, we introduce Spectral Wasserstein distances indexed by norms γ on positive semidefinite matrices: the trace norm recovers the classical W_2 distance, the operator norm gives the Muon geometry, and Schatten norms interpolate between them. We develop a static Kantorovich formulation, a max–min robust-cost representation, and Gaussian reductions extending the Bures formula. For monotone norms, we prove equivalence with a Benamou–Brenier formulation and obtain a gradient-flow interpretation of normalized mean-field training dynamics. Numerical experiments illustrate the resulting flows for MMD objectives, Gaussian reductions, two-layer ReLU models, and shallow attention.

1 Introduction

Spectrally normalized optimizers such as Muon replace the Euclidean geometry of gradient descent by a matrix-aware geometry adapted to block-shaped parameters. This paper develops a mathematical framework for such optimizers in very wide, possibly infinite-width, models. The central object is a Spectral Wasserstein geometry in which their idealized dynamics become mean-field gradient flows.

Normalized gradient descent and mean field dynamics. On the optimization side, a growing literature studies normalized first-order methods. The recent framework of Pethick et al. [2025] is particularly relevant because it treats norm-constrained linear minimization oracles as a general language for normalized gradient methods and includes spectral normalizations as special cases. Earlier works such as Cutkosky and Mehta [2020] and Murray et al. [2019] show how normalized gradient methods change the optimization dynamics even in nonconvex settings. For deep architectures, matrix-aware normalizations are especially natural: Shampoo is an early influential example of tensor/matrix-aware preconditioning [Gupta et al., 2018], and Muon has become a leading example of spectral normalization in large-scale training [Jordan et al., 2024, Liu et al., 2025]. This connects naturally with the mean-field description of wide neural networks through probability measures on parameter space, which underlies the landscape analysis of two-layer networks by Mei et al. [2018], the optimal-transport convergence analysis of over-parameterized models by Chizat and Bach [2018], and the metric-gradient-flow viewpoint developed in Ambrosio et al. [2008]. Our work keeps this mean-field perspective but changes the underlying metric from Euclidean Wasserstein geometry to matrix-aware Spectral Wasserstein geometries.

Generalized optimal distances. On the transport side, our work is closest in spirit to generalized coupling costs and weak transport, as developed for instance by Gozlan et al. [2017], Backhoff-Veraguas et al. [2019], and Backhoff-Veraguas and Pammer [2022]. It is also related to covariance-dependent transport costs [Burger et al., 2025], to matrix-valued optimal transport [Ning et al., 2015,

Chen et al., 2018], and to the Bures–Wasserstein geometry of covariance matrices [Bhatia et al., 2019]. Another useful perspective is to robustify the Wasserstein distance by optimizing over the cost itself. The closest antecedent is the subspace robust Wasserstein distance of Paty and Cuturi [2019], which corresponds to the special case of our spectral Wasserstein distance W_γ obtained when γ is a Ky Fan norm [Fan, 1951]. Maximizing over costs is also connected to metric learning in Wasserstein discriminant analysis [Flamary et al., 2018], ground metric learning [Cuturi and Avis, 2014], and congestion models for transportation networks [Carlier et al., 2008]. The opposite direction, minimizing over costs, appears, for instance, in Sebbouh et al. [2024]; this leads to a concave minimization problem useful for Gromov–Wasserstein-type structure. By contrast, our robustification is a concave maximization over admissible quadratic costs, leading to a convex static problem.

Contributions. The construction has three steps: a matrix normalization defines a descent direction; its force-covariance formula extends that direction to arbitrary measures; and a transport distance supplies the corresponding geometry. We first derive spectrally normalized matrix flows and their mean-field continuity equations in Section 2. We then introduce the static Spectral Wasserstein cost in Section 3, compare it with W_2 in Proposition 3.2, establish its max–min robust-cost representation in Theorem 3.4, and derive its Gaussian reduction in Proposition 3.6. For monotone norms, Section 4 proves the equivalence between the static and Benamou–Brenier formulations in Theorem 4.2, yielding the metric property in Corollary 4.3 and explicit geodesics in Corollary 4.4. Section 5 identifies the normalized PDE as the corresponding metric gradient flow and discusses MLP, shallow-attention, MMD, and linear-network examples. The code used to reproduce the numerical experiments is available at <https://github.com/gpeyre/muon-dynamics>.

Scope. This paper does not attempt to explain why Muon-type algorithms outperform vanilla gradient descent for particular architectures, especially attention layers. Its goal is to provide the mathematical formalism needed to investigate such questions in the regime of very wide networks. The resulting transport geometries also raise theoretical questions that extend beyond neural-network training. The finite-particle identity is exact, while the gradient-flow characterization is conditional on stated regularity assumptions. We do not prove convergence of particle systems to the limiting PDE, global PDE well-posedness, or convergence of practical stochastic, momentum-based Muon implementations.

2 Spectral Gradient Flow and Mean Field Dynamics

Spectral gradient flows on matrices. This section introduces the normalized dynamics studied in the paper, whose geometry is governed by a matrix gauge γ . The “Muon-type” descent corresponds to the case where γ is the operator norm. We impose monotonicity throughout the main text. It is used in the static–dynamic equivalence of Theorem 4.2 and is crucial for the Spectral Wasserstein gradient-flow interpretation in Section 5. Appendix C explains what can fail without monotonicity, while Appendix E.4 gives a different extension beyond convex gauges.

Definition 2.1 (Monotone gauge γ and norm $N(V)$). *Unless explicitly stated otherwise, γ denotes the restriction to $\mathbb{S}_+^d$ of a norm on $\mathbb{S}^d$. We assume that this restriction is monotone: whenever $0 \preceq S \preceq S'$, one has $\gamma(S) \leq \gamma(S')$. The matrix norm induced by γ is $N(V) := \gamma(V^\top V)^{1/2}$; for $\gamma = \gamma_p$, one has $N(V) = \|V\|_{S_{2p}}$. As shown in Proposition A.3, $N(V)$ is indeed a norm when γ is monotone.*

The canonical examples used throughout this article are the Schatten gauges: if $\lambda(S)$ denotes the vector of eigenvalues of $S \succeq 0$, then $\gamma_p(S) := \|S\|_{S_p} = \|\lambda(S)\|_{\ell_p}$, $p \in [1, \infty]$. Other examples include the Ky Fan norms [Fan, 1951], $\gamma(S) := \sum_{i=1}^k \lambda_i(S)$, obtained by summing the k largest eigenvalues. They interpolate between the operator norm ($k = 1$) and the trace norm ($k = d$) and recover the subspace-robust Wasserstein setting of Paty and Cuturi [2019]. Although the main text assumes that γ is convex, many metric and Benamou–Brenier results extend to the larger class of possibly nonconvex root-norm gauges introduced in Appendix E.4.

Definition 2.2 (Linear Minimization Oracle (LMO)). *For $G \in \mathbb{R}^{n \times d}$, the matrix LMO associated with γ is any minimizer $J_\gamma(G) \in \arg\min_V \{ \langle G, V \rangle_F + \frac{1}{2} N(V)^2 \}$.*

The name is a slight abuse of the usual terminology: a classical LMO minimizes a linear functional on a fixed ball. Here it is followed by an optimal scalar step. Indeed, if N^* is the dual norm and

$G \neq 0$, then

$$U_G \in \operatorname{argmin}_{N(U) \leq 1} \langle G, U \rangle_F \implies J_\gamma(G) = N^*(G)U_G. \quad (1)$$

This follows by minimizing $-rN^*(G) + r^2/2$ over $r \geq 0$; $J_\gamma(0) = 0$. We use the same convention for the Measure-LMO below.

Let $F : \mathbb{R}^{n \times d} \rightarrow \mathbb{R}$ be a smooth objective. Following Pethick et al. [2025], it yields both the discrete step and the ODE flow

$$X_{k+1} = X_k + \tau J_\gamma(\nabla F(X_k)) \quad \text{and} \quad \dot{X}_t = J_\gamma(\nabla F(X_t)). \quad (2)$$

For $p = 1$, $J_{\gamma_1}(G) = -G$, so one recovers the usual gradient flow; for $p = \infty$, J_{γ_∞} is the trace-scaled partial polar factor of G , giving a simplified ‘‘Muon’’ ODE. Explicit Schatten formulas are given in Appendix A. The scalar factor matters: the unscaled polar update and this squared-norm flow follow the same deterministic paths, away from stationary points, but with different time parametrizations. Our metric fixes the latter clock.

Mean-field lift and normalized continuity equation. Given particles $x_1, \dots, x_n \in \mathbb{R}^d$, write $X := [x_1^\top; \dots; x_n^\top] \in \mathbb{R}^{n \times d}$ and $\mu_X := n^{-1} \sum_i \delta_{x_i}$. We consider permutation-invariant objectives lifted from measures, $F(X) := n\mathcal{F}(\mu_X)$. This scaling applies, in particular, to the two-layer MLP and attention models discussed in Section 5.

Definition 2.3 (Measure-LMO). *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$. For $v \in L^2(\mu; \mathbb{R}^d)$, define $\mathcal{N}_\mu(v)^2 := \gamma(\int v(x)v(x)^\top d\mu(x))$. For $g \in L^2(\mu; \mathbb{R}^d)$, the measure-LMO is any member of $\mathcal{J}_\gamma^\mu(g) \in \operatorname{argmin}_{v \in L^2(\mu; \mathbb{R}^d)} \{ \int g(x) \cdot v(x) d\mu(x) + \frac{1}{2} \mathcal{N}_\mu(v)^2 \}$.*

For an empirical measure $\mu_X = n^{-1} \sum_i \delta_{x_i}$ transported by particle velocities stacked into a matrix V , one has $\mathcal{N}_{\mu_X}(v)^2 = n^{-1} N(V)^2$. If G denotes the matrix with rows $g(x_i)^\top$, then the rows of $\mathcal{J}_\gamma^{\mu_X}(g)$ coincide with those of $J_\gamma(G)$. For a more general measure μ , however, it is not a priori clear how to compute $\mathcal{J}_\gamma^\mu(g)$. The following proposition reduces this infinite-dimensional LMO to a finite-dimensional matrix problem.

Proposition 2.4 (Finite-dimensional Measure-LMO). *Let $S_g := \mathcal{S}_\mu(g) = \int g(x)g(x)^\top d\mu(x)$ and let P_g be the orthogonal projector onto $\operatorname{range}(S_g)$. If $S_g = 0$, then $\mathcal{J}_\gamma^\mu(g) = 0$. Otherwise, let*

$$A_g \in \operatorname{argmin}_{A = P_g A} \left\{ \operatorname{tr}(S_g^{1/2} A) + \frac{1}{2} \gamma(A^\top A) \right\}.$$

Then

$$\mathcal{J}_\gamma^\mu(g)(x) = A_g^\top S_g^{\dagger/2} g(x)$$

is a valid Measure-LMO.

Proof. Associate to each field $h \in L^2(\mu; \mathbb{R}^d)$ the operator $T_h : \mathbb{R}^d \rightarrow L^2(\mu)$, $(T_h a)(x) = a \cdot h(x)$. Then $T_h^* T_h = \int h h^\top d\mu$ and $\int g \cdot h d\mu = \langle T_g, T_h \rangle_{\text{HS}}$. Writing the polar decomposition $T_g = U_g S_g^{1/2}$, set $A = U_g^* T_h$ and $B = (I - U_g U_g^*) T_h$. Then $P_g A = A$, $T_h^* T_h = A^\top A + B^* B$, and $\langle T_g, T_h \rangle_{\text{HS}} = \operatorname{tr}(S_g^{1/2} A)$. Monotonicity of γ shows that deleting B cannot increase the objective. Conversely, every $A = P_g A$ is realized by $T_h = U_g A$, or $h(x) = A^\top S_g^{\dagger/2} g(x)$. The finite-dimensional objective is continuous and coercive, so a minimizer exists. If $S_g = 0$, then $g = 0$ μ -almost everywhere and the unique optimal field is zero. $\square$

For Schatten gauges, this finite-dimensional LMO gives the explicit formula in Corollary A.6; in particular, it covers the Muon endpoint.

Following Ambrosio et al. [2008], define the Wasserstein force by

$$g_\mu(x) := \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}(x),$$

where the first variation satisfies

$$\left. \frac{d}{d\varepsilon} \mathcal{F}(\mu + \varepsilon \sigma) \right|_{\varepsilon=0} = \int \frac{\delta \mathcal{F}}{\delta \mu} d\sigma$$

for signed perturbations σ of zero total mass. The force g_μ is a cotangent quantity, not the spectral velocity: the latter is $\mathcal{J}_\gamma^\mu(g_\mu)$. The normalized mean-field dynamics is then the PDE

$$\partial_t \mu_t + \operatorname{div}(\mu_t \mathcal{J}_\gamma^{\mu_t}(g_{\mu_t})) = 0. \quad (3)$$

When $\gamma(S) = \operatorname{tr}(S)$, $\mathcal{J}_\gamma^\mu(g) = -g$, and this reduces to the classical W_2 gradient-flow PDE. For discrete measures, one recovers ODE (2).

Proposition 2.5 (Finite-width interpretation). *Assume that the normalized continuity equation starting from μ_{X_0} preserves empirical measures, and write $\mu_t = n^{-1} \sum_i \delta_{x_i(t)}$ and $X_t := [x_1(t)^\top; \dots; x_n(t)^\top]$. As long as the atoms remain pairwise distinct, X_t solves ODE (2) if and only if μ_{X_t} solves (3) weakly. The same statement holds through collisions when coincident atoms are assigned the same selected velocity. The proof is in Appendix A.*

Extending the global-in-time PDE well-posedness theory of Chizat and Bach [2018], which covers the trace case $p = 1$ under smoothness and growth assumptions on the feature map, to general Schatten geometries remains a difficult open problem. The Measure-LMO nevertheless defines the same evolution rule for empirical and continuous distributions, without a prescribed number of particles. For suitable continuous families it can be substantially simpler: affine velocities preserve Gaussian distributions and close the dynamics on their moments (Appendix F). The major question left open by the construction above is whether (3) is the gradient flow of an actual distance on probability measures. The next two sections answer this question: first by defining a static Spectral Wasserstein cost, and then by proving that, for monotone norms, this cost admits a Benamou–Brenier dynamic formulation whose infinitesimal norm is precisely $\mathcal{N}_\mu$.

Remark 2.6 (Connection with Muon). *Appendix B relates the normalized-flow viewpoint to practical ingredients of Muon. Finitely many Newton–Schulz iterations replace the exact LMO by a polynomial spectral map generated by an explicit potential. Momentum can be modeled by an exponentially averaged force variable, leading in the mean-field limit to a phase-space Liouville equation; this is no longer a gradient flow on the position marginal alone, and thus lies partly outside the Spectral Wasserstein geometry developed below. Blockwise normalization, in contrast, is exactly covered by a block gauge. The same appendix distinguishes these deterministic extensions from noisy minibatch updates, for which normalization and expectation do not commute.*

3 Static Spectral Wasserstein Geometry

Kantorovich cost. This section introduces the Spectral Wasserstein transport cost. The key point is that the correct object is coupling-based and depends on a norm γ acting on the global displacement covariance.

Definition 3.1 (Static Spectral Wasserstein). *For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, let $\Pi(\mu, \nu)$ be the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν , and define*

$$W_\gamma(\mu, \nu)^2 := \inf_{\pi \in \Pi(\mu, \nu)} \gamma \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)(y - x)^\top d\pi(x, y) \right).$$

The Monge restriction, the possible strictness of the coupling relaxation, and the obstruction to the triangle inequality for non-monotone norms are discussed in Appendix C. In the monotone case, the full metric property is proved later through the dynamic formulation in Corollary 4.3.

Comparison with W_2 and cost-robust formulation. We now show that W_γ has the same topology as W_2 . The proof, explicit Schatten constants, and sharpness examples are given in Appendix A.2.

Proposition 3.2 (Norm comparison). *Let $0 < c_\gamma \leq C_\gamma < \infty$ satisfy $c_\gamma \operatorname{tr}(S) \leq \gamma(S) \leq C_\gamma \operatorname{tr}(S)$ for every $S \succeq 0$; such constants always exist. Then $\sqrt{c_\gamma} W_2(\mu, \nu) \leq W_\gamma(\mu, \nu) \leq \sqrt{C_\gamma} W_2(\mu, \nu)$.*

The comparison bounds of Proposition 3.2 already imply separation and topological equivalence with W_2 . What is not immediate from the static formulation alone is the triangle inequality. To address this, and to clarify the geometry of W_γ , Theorem 3.4 below recasts W_γ as a cost-robust Wasserstein distance. This reformulation has two immediate consequences: a conditional Brenier theorem in the Monge setting (Appendix C) and the fact that W_γ is a distance (Appendix A.3). The latter can also be obtained from the Benamou–Brenier formulation; see Corollary 4.3.

Definition 3.3 (Representing set $\mathcal{K}_\gamma$). A representing set for γ is a convex compact set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ such that, for every $S \succeq 0$,

$$\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QS).$$

As recalled in Appendix A, monotonicity of γ on $\mathbb{S}_+^d$ allows the representing set to be chosen inside $\mathbb{S}_+^d$. Remark A.4 gives examples of such sets $\mathcal{K}_\gamma$ for the Schatten norms $\gamma = \gamma_p$.

Theorem 3.4 (Max-min and cost-robust representation). For every positive semidefinite matrix $Q \in \mathbb{S}_+^d$, let W^Q denote the quadratic transport functional associated with the cost $\langle Q(x - y), x - y \rangle$, namely

$$W^Q(\mu, \nu)^2 := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)^\top Q(y - x) d\pi(x, y).$$

In particular, $W^{\text{Id}} = W_2$. Then, for any representing set $\mathcal{K}_\gamma$ for γ as in Definition 3.3, for every $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} \inf_{\pi \in \Pi(\mu, \nu)} \int (y - x)^\top Q(y - x) d\pi(x, y) = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu)^2.$$

Denoting $\Sigma_\pi := \int (y - x)(y - x)^\top d\pi(x, y)$, any saddle point (Q^*, π^*) satisfies $Q^* \in \partial\gamma(\Sigma_{\pi^*})$, where the subdifferential is understood on the cone $\mathbb{S}_+^d$.

Proof sketch. Details are given in Appendix A.2. The support representation of γ gives $\gamma(\Sigma_\pi) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(Q\Sigma_\pi)$, with $\text{tr}(Q\Sigma_\pi) = \int (y - x)^\top Q(y - x) d\pi(x, y)$. Hence $W_\gamma(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \max_{Q \in \mathcal{K}_\gamma} F(\pi, Q)$, where $F(\pi, Q) := \int (y - x)^\top Q(y - x) d\pi(x, y)$. The conclusion follows by exchanging the infimum and the maximum. Indeed, $\Pi(\mu, \nu)$ is convex, $\mathcal{K}_\gamma$ is compact and convex, and F is affine in both variables, continuous in Q , and lower semicontinuous in π . Sion's minimax theorem therefore gives $\inf_\pi \max_Q F(\pi, Q) = \max_Q \inf_\pi F(\pi, Q)$. The inner infimum is precisely $W^Q(\mu, \nu)^2$, yielding $W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu)^2$. $\square$

Corollary 3.5 (Kantorovich potentials). For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} \sup_{\varphi(x) + \psi(y) \leq (x - y)^\top Q(x - y)} \left(\int \varphi d\mu + \int \psi d\nu \right),$$

where the supremum ranges over integrable admissible potentials.

Proof. For each fixed $Q \succeq 0$, Kantorovich duality for the continuous, nonnegative quadratic cost identifies the inner supremum with $W^Q(\mu, \nu)^2$. This remains valid for singular Q : equivalently, one may transport the pushforwards by $Q^{1/2}$. Apply Theorem 3.4. The outer maximum is attained by compactness of $\mathcal{K}_\gamma$ and upper semicontinuity of the infimum of linear functions of Q . $\square$

Discrete setting. This subsection spells out the finite-dimensional version used in computations and shows how the abstract coupling problem becomes a concrete convex program. We consider two discrete measures $\mu := \sum_{i=1}^n a_i \delta_{x_i}$ and $\nu := \sum_{j=1}^m b_j \delta_{y_j}$, with weights $a_i, b_j \geq 0$ summing to one. As in classical Kantorovich transport, a coupling is represented by a transport matrix $P \in \mathbb{R}_+^{n \times m}$ satisfying $P\mathbf{1}_m = a$ and $P^\top \mathbf{1}_n = b$. The discrete static Spectral Wasserstein problem can then be written as the convex optimization problem

$$W_\gamma(\mu, \nu)^2 = \min_{P \geq 0, P\mathbf{1}_m = a, P^\top \mathbf{1}_n = b} \gamma\left(\sum_{i=1}^n \sum_{j=1}^m P_{ij} (y_j - x_i)(y_j - x_i)^\top\right).$$

For $\gamma = \gamma_p$ and $p = 1$, this reduces to the usual linear optimal-transport problem with quadratic cost, hence to a classical assignment problem in the equal-weight case. For $p = \infty$, it admits a semidefinite programming formulation through an epigraph constraint on the largest eigenvalue of the displacement covariance. Figure 1 illustrates couplings obtained for the trace and operator objectives. The trace assignments are exact; the operator panels use a rank-one alternating heuristic, not a certified SDP optimizer (Appendix G). Proposition A.10 explains this effect in the regime where the two measures have well-separated means: while the trace case remains isotropic, every Schatten

geometry with $p > 1$ asymptotically selects a quadratic cost aligned with the mean displacement direction, making the matching primarily sensitive to displacements parallel to the vector joining the two means. As explained in Appendix C, for $p = \infty$ an active matrix Q^* is often rank one, so the conditional Brenier argument neither ensures uniqueness nor forces every optimizer to be induced by a map. Figure 1 illustrates the resulting freedom in directions orthogonal to the active one; see also Remark 5.2. Additional image-defined shape pairs are shown in Appendix G.

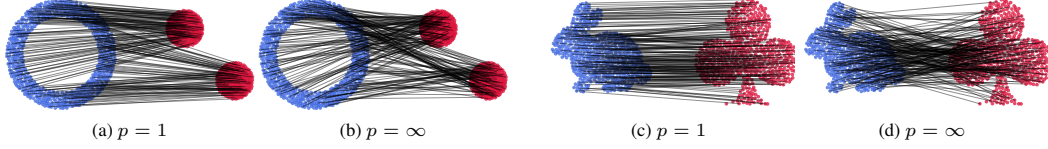


Figure 1: Static coupling illustrations for two source-target pairs: exact trace assignments ($p = 1$) and heuristic operator couplings ($p = \infty$). Only a subsample of matched segments is displayed.

Gaussian setting and Spectral Bures distance. Gaussian marginals compress the transport problem to a finite-dimensional optimization over admissible covariance blocks. The proof of Proposition 3.6 and the closed commuting formula for Schatten norms are collected in Appendix D.

Proposition 3.6 (Gaussian reduction). *Let $\mu = \mathcal{N}(m_0, \Sigma_0)$ and $\nu = \mathcal{N}(m_1, \Sigma_1)$. Then*

$$W_\gamma(\mu, \nu)^2 = \inf_{K: \begin{pmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{pmatrix} \succeq 0} \gamma((m_1 - m_0)(m_1 - m_0)^\top + \Sigma_0 + \Sigma_1 - K - K^\top).$$

4 Dynamic Formulation and Geodesics

We now turn to the Benamou–Brenier formulation [Benamou and Brenier, 2000]. Throughout this section, γ is monotone. Appendix A then allows us to choose $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$; this positive-semidefinite representation is the key ingredient in the static–dynamic equivalence below.

Definition 4.1 (Dynamic Spectral Wasserstein). *For $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$, define*

$$W_\gamma^{\text{BB}}(\mu_0, \mu_1)^2 := \inf_{(\mu_t, v_t)} \int_0^1 \gamma\left(\int v_t(x)v_t(x)^\top d\mu_t(x)\right) dt, \quad (4)$$

where the infimum is over narrowly continuous curves $t \mapsto \mu_t \in \mathcal{P}_2(\mathbb{R}^d)$ and Borel velocity fields satisfying $\partial_t \mu_t + \text{div}(\mu_t v_t) = 0$ and the endpoint conditions $\mu_{t=0} = \mu_0, \mu_{t=1} = \mu_1$.

The equivalent convex momentum formulation is given in Appendix E. The following theorem is the main result of the paper. Its proof, together with proofs of the subsequent corollaries, is given in Appendix E.

Theorem 4.2 (Static–dynamic equivalence). *If γ is monotone, then $W_\gamma^{\text{BB}}(\mu_0, \mu_1) = W_\gamma(\mu_0, \mu_1)$.*

Proof sketch. The detailed argument appears in Appendix E. For the dynamic-to-static inequality, let (μ_t, v_t) solve the continuity equation. By the superposition principle, there is a probability measure η on absolutely continuous paths ω such that $(e_t)_\# \eta = \mu_t$ and $\dot{\omega}_t = v_t(\omega_t)$ for a.e. t , where $e_t(\omega) = \omega_t$. The endpoint coupling $\pi = (e_0, e_1)_\# \eta$ then satisfies $\omega_1 - \omega_0 = \int_0^1 v_t(\omega_t) dt$ for η -a.e. ω . By monotonicity and Appendix A, the dual support set $\mathcal{K}_\gamma$ may be chosen in $\mathbb{S}_+^d$. Hence, for every $Q \in \mathcal{K}_\gamma$, Jensen’s inequality gives $\langle Q, (\omega_1 - \omega_0)(\omega_1 - \omega_0)^\top \rangle \leq \int_0^1 \langle Q, v_t(\omega_t)v_t(\omega_t)^\top \rangle dt$. Integrating with respect to η and taking the supremum over $Q \in \mathcal{K}_\gamma$ yields the static cost of π bounded by the dynamic action. Taking the infimum over admissible (μ_t, v_t) gives one inequality. Conversely, let π be an optimal static coupling, set $z_t(x, y) = (1-t)x + ty$, and define $\mu_t = (z_t)_\# \pi$. An Eulerian velocity field for this curve is

$$v_t(z) = \mathbb{E}_\pi[y - x \mid z_t = z].$$

Conditional Jensen’s inequality yields

$$\int v_t v_t^\top d\mu_t \preceq \int (y - x)(y - x)^\top d\pi \quad \text{for a.e. } t.$$

Monotonicity of γ therefore bounds the dynamic action by the static cost of π , proving the reverse inequality. $\square$

Corollary 4.3 (Metric properties). W_γ is a distance on $\mathcal{P}_2(\mathbb{R}^d)$ inducing the same topology as W_2 .

Corollary 4.4 (Geodesics). If π_* is any optimal coupling for $W_\gamma(\mu_0, \mu_1)$, then $\mu_t = ((1-t)x + ty)_\# \pi_*$ is a constant-speed geodesic and $W_\gamma(\mu_s, \mu_t) = |t-s| W_\gamma(\mu_0, \mu_1)$ for every $0 \leq s \leq t \leq 1$.

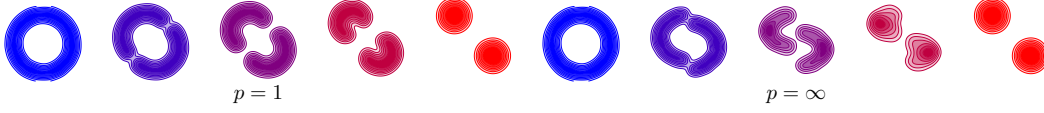


Figure 2: Displacement-interpolation density snapshots induced by the computed static couplings (exact for $p = 1$, heuristic for $p = \infty$), for $p = 1$ (first five panels) and for $p = \infty$ (last five panels). We use $n = 1500$ particles. As in Figure 1, the first and last snapshots correspond to the same source and target distributions ($t = 0$ and $t = 1$).

Figure 2 complements Figure 1: once a coupling is fixed, its displacement interpolation is linear in time at the particle level, but the macroscopic density evolution differs markedly between $p = 1$ and $p = \infty$. Appendix G shows additional examples. For Schatten gauges these effects do not privilege the coordinate axes: for every orthogonal matrix R , simultaneous rotation of the two measures conjugates each displacement covariance by R , leaving W_γ unchanged. The exact transport problem is rotation invariant; the displayed heuristic couplings may additionally reflect the solver initialization.

5 Spectral Wasserstein Gradient Flows

This section is written directly for arbitrary measures. Proposition 2.5 already identifies its empirical version with the normalized matrix ODE, hence with the idealized Muon dynamics at $p = \infty$. The different role of Theorem 5.1 below is to identify this evolution as a curve of maximal slope for a transport metric, including when the distribution is not empirical.

Metric interpretation of the normalized PDE. We now show that (3) is the gradient flow of $\mathcal{F}$ for the metric W_γ , in the metric-gradient-flow sense of Ambrosio et al. [2008]. Recall that the metric derivative of an absolutely continuous curve $t \mapsto \mu_t$ is $|\mu'|_\gamma(t) := \lim_{h \rightarrow 0} W_\gamma(\mu_{t+h}, \mu_t)/|h|$, whenever the limit exists, and the local slope of $\mathcal{F}$ is $|\partial\mathcal{F}|_\gamma(\mu) := \limsup_{\nu \rightarrow \mu} (\mathcal{F}(\mu) - \mathcal{F}(\nu))_+ / W_\gamma(\mu, \nu)$. A W_γ -gradient flow of $\mathcal{F}$ is a curve of maximal slope, i.e., a curve satisfying the energy dissipation identity $\frac{d}{dt} \mathcal{F}(\mu_t) = -\frac{1}{2} |\mu'|_\gamma(t)^2 - \frac{1}{2} |\partial\mathcal{F}|_\gamma(\mu_t)^2$.

Theorem 5.1 (W_γ gradient flow). Fix $T < \infty$ and $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$. Let $v_t(x)$ be continuous in (t, x) , uniformly globally Lipschitz in x , with uniform linear growth on $[0, T]$. Let X_t be its characteristic flow and $\mu_t = (X_t)_\# \mu_0$. Assume that $\mathcal{F}$ is finite on a neighborhood of this curve and admits forces $g_t = g_{\mu_t} \in L^2(\mu_t; \mathbb{R}^d)$ such that, for constants $C, \delta > 0$, every $t \in [0, T]$, and every $\pi \in \Pi(\mu_t, \nu)$ with $\int |y - x|^2 d\pi \leq \delta$,

$$\left| \mathcal{F}(\nu) - \mathcal{F}(\mu_t) - \int g_t(x) \cdot (y - x) d\pi(x, y) \right| \leq C \int |y - x|^2 d\pi(x, y). \quad (5)$$

If $v_t \in \mathcal{J}_\gamma^{\mu_t}(g_t)$, then μ_t is a curve of maximal slope for $\mathcal{F}$ with respect to W_γ . More precisely, almost everywhere,

$$|\mu'|_\gamma(t) = |\partial\mathcal{F}|_\gamma(\mu_t) = \mathcal{N}_{\mu_t}^*(g_t) = \mathcal{N}_{\mu_t}(v_t), \quad \frac{d}{dt} \mathcal{F}(\mu_t) = -\mathcal{N}_{\mu_t}^*(g_t)^2,$$

where $\mathcal{N}_\mu^*$ is the dual norm under the $L^2(\mu)$ pairing.

Proof sketch. The detailed proof is in Appendix E.5. The transport expansion (5) and barycentric projection of an optimal coupling give $|\partial\mathcal{F}|_\gamma(\mu_t) \leq \mathcal{N}_{\mu_t}^*(g_t)$. Perturbing μ_t by $(\text{Id} + hw)_\# \mu_t$, for arbitrary $w \in L^2(\mu_t)$, proves the opposite inequality by duality. The characteristic flow gives the chain rule $\frac{d}{dt} \mathcal{F}(\mu_t) = \int g_t \cdot v_t d\mu_t = -\mathcal{N}_{\mu_t}^*(g_t)^2$. The dynamic formulation bounds the metric speed by $\mathcal{N}_{\mu_t}(v_t) = \mathcal{N}_{\mu_t}^*(g_t)$; the slope inequality and the chain rule give the reverse bound. These equalities imply the energy dissipation identity. $\square$

These are verification assumptions on an existing curve, not an existence or uniqueness theorem. They require neither a density nor invertibility of the force covariance. A concrete smooth class satisfying (5) is $\mathcal{F}(\mu) = \Phi(\int \varphi_1 d\mu, \dots, \int \varphi_m d\mu)$, with C^2 features of bounded Hessian and $\Phi \in C^2$ on a neighborhood of the relevant moments; see Appendix E.5. Regularity of the selected velocity must still be checked. In particular, the theorem does not automatically cover ReLU kinks or changes of rank at the operator endpoint.

Remark 5.2 (Static and infinitesimal views of Muon). *For the Muon gauge $\gamma(S) = \lambda_{\max}(S)$, the static optimality condition in Theorem 3.4 gives $Q^* \in \partial\gamma(\Sigma_{\pi^*})$. Hence Q^* is supported on the top eigenspace of the optimal displacement covariance. When that eigenspace is one-dimensional, with unit eigenvector u^* , one may take $Q^* = u^*(u^*)^\top$, and π^* is optimal for the projected quadratic cost $\langle y - x, u^* \rangle^2$.*

The infinitesimal picture is complementary rather than one-dimensional. If $S_g = \int gg^\top d\mu$, Corollary A.6 gives

$$\mathcal{J}_{\gamma_\infty}^\mu(g) = -\text{tr}(S_g^{1/2})S_g^{\dagger/2}g, \quad \int \mathcal{J}_{\gamma_\infty}^\mu(g)\mathcal{J}_{\gamma_\infty}^\mu(g)^\top d\mu = \text{tr}(S_g^{1/2})^2 P_g,$$

where P_g projects onto $\text{range}(S_g)$. Thus the Measure-LMO whitens the force covariance on its active subspace. This covariance equalization is the mean-field counterpart of Muon’s orthogonalizing matrix update.

Mean-field models for two-layer MLPs. We now specialize the abstract gradient-flow PDE to the standard mean-field parameterization of two-layer MLPs. The mean-field predictor is $H_\mu(z) := \int_{\mathbb{R}^d} \phi(z, x) d\mu(x)$, and the training objective is $\mathcal{F}(\mu) := R(H_\mu)$ for some loss R . For a two-layer network with scalar output and activation σ , one may take parameter $x = (u, v) \in \mathbb{R}^d$ and feature $\phi(z, x) := u \sigma(v \cdot z)$, where u is the outer weight and v is the inner weight. This is the standard mean-field representation of a two-layer multilayer perceptron [Mei et al., 2018, Chizat and Bach, 2018]. The notation is a deliberate departure from usual machine-learning conventions: the network input is denoted by z , while the trainable variable is denoted by x so that it matches the transport notation used throughout the paper. Here d is the sum of the block dimensions of u and v . The experiment normalizes the single $n \times d$ particle matrix. Separate normalization of inner and outer weight blocks is also covered, using the block gauge of Appendix B; these are different choices of geometry. A basic example is the quadratic risk $R(H) = \frac{1}{2} \int |H(z) - y^*(z)|^2 d\rho(z)$, or its empirical version on a dataset. Because H_μ depends linearly on μ , this makes $\mathcal{F}(\mu)$ a quadratic interaction functional. These energies are of MMD-type [Gretton et al., 2012], which is why the MMD experiment below is a natural test case for the Spectral Wasserstein flow. In Figure 3, we train a two-dimensional to one-dimensional ReLU network, with $\sigma(s) = \max(s, 0)$, particles $x = (u, v_1, v_2) \in \mathbb{R}^3$, and covariates $z \in \mathbb{R}^2$. The target y^* is generated by a three-neuron teacher on Gaussian covariates $z \sim \mathcal{N}(0, \text{Id}_2)$. We display both runs on their respective unit-normalized nominal horizons, chosen by hand. This is a qualitative comparison, not a calibration of the intrinsic gradient-flow clocks: the implementation also clips large Euler steps. Appendix G.1 records the exact normalization, safeguards, and an initialization-only dissipation check. We display the particle evolution in the reduced two-dimensional plane $|u|v$. The rays associated with the teacher neurons are shown as dashed half-lines in Figure 3(c)–(d); concentration toward these rays can also be read from the circular histograms. Changing p visibly changes the paths and the loss profile on these normalized plots. We do not infer an intrinsic convergence-rate or wall-clock advantage from this comparison. The ReLU case, and more generally any positively two-homogeneous feature map ϕ , admits a spherical reduction leading to a generalized unbalanced transport geometry; see Appendix J.

Mean-field shallow attention. Using the same spectral-flow construction, we also consider a shallow mean-field multi-head attention model. This is a useful stress test because attention uses softmax nonlinearities, precisely the regime where Muon-type spectral normalization is empirically attractive. A datum is a sequence $z = (z_1, \dots, z_m)$ of tokens in $\mathbb{R}^{d_T}$. A head is a particle $x = (Q, K, V, O)$, with $Q, K, V, O \in \mathbb{R}^{d_H \times d_T}$, so $x \in \mathbb{R}^d$ with $d = 4d_H d_T$. Its elementary feature is the softmax attention map $\phi(z, x)_j := O^\top \sum_{i=1}^m \frac{\exp(\langle Qz_j, Kz_i \rangle / \sqrt{d_H})}{\sum_{\ell=1}^m \exp(\langle Qz_j, Kz_\ell \rangle / \sqrt{d_H})} V z_i$, and the mean-field layer is again $H_\mu(z) = \int \phi(z, x) d\mu(x)$. For an empirical measure μ_X , this is a multi-head attention layer averaged over an arbitrary number n of heads, which is the natural mean-field analogue of increasing the number of heads without fixing a finite architecture. Figure 4 reports an experiment with $m = 16$

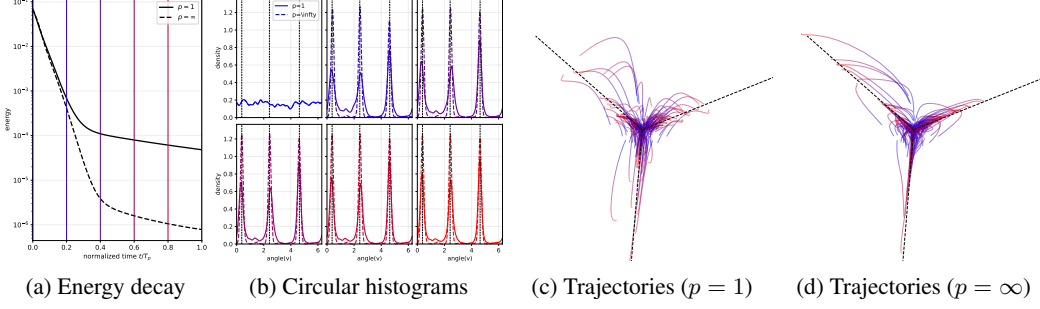


Figure 3: Two-layer ReLU MLP training under spectral flows. (a) Energy decay on separately display coordinates, not necessarily matched intrinsic nominal horizons (solid: $p = 1$, dashed: $p = \infty$), with colored vertical bars marking the snapshot times used in (b). (b) Circular histogram snapshots at matched normalized times. (c)–(d) Particle trajectories in the reduced parameter plane ($|u|v_1, |u|v_2$) for $p = 1$ and $p = \infty$.

tokens, $d_T = 4$, and $d_H = 3$: we fit a random five-head teacher by least squares using a student with $n = 200$ heads, and compare the trace and operator geometries over five paired trials, each with a new teacher and student initialization. The chosen unfolding is a single 200×48 particle matrix. The query/key scale controls the sparsity of the teacher attention matrix. We quantify sparsity through the effective number of attended tokens $N_{\text{eff}} := (\sum_i w_i^2)^{-1}$ averaged over attention rows: the dense high-temperature regime has $N_{\text{eff}} \approx 16$ out of $m = 16$, while the sparse low-temperature regime has $N_{\text{eff}} \approx 2.1$. The $p = 1$ and $p = \infty$ horizons are rescaled so that small-time energy decay rates are comparable; even after this normalization, changing p strongly affects the convergence profile and $p = \infty$ reaches a lower energy during the first phase.

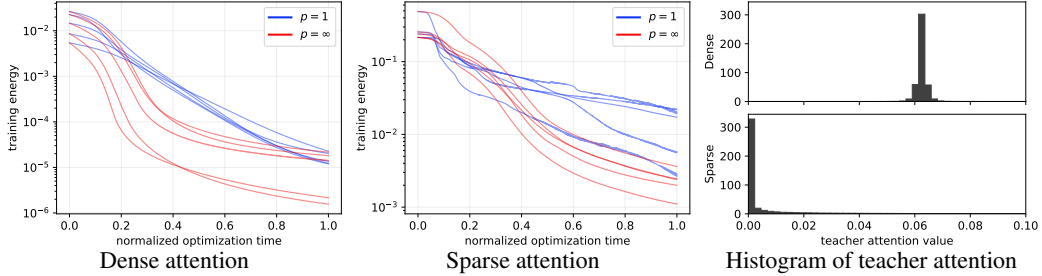


Figure 4: Shallow-attention training under spectral flows ($p = 1$ in blue, $p = \infty$ in red), with five runs per setting. Dense attention corresponds to high temperature and effective support near all tokens ($N_{\text{eff}} \approx 16$), while sparse attention corresponds to low temperature and effective support around two tokens ($N_{\text{eff}} \approx 2.1$). The right panel shows teacher-attention histograms truncated to $[0, 0.1]$ and renormalized on that interval; sparse weights larger than 0.1 are not displayed.

MMD gradient flows. As a simple model for generative fitting and sampling, in the spirit of MMD-based generative models [Li et al., 2015, Bińkowski et al., 2018], we also minimize $\mathcal{F}(\mu) = \text{MMD}(\mu, \nu)^2$ for the energy-distance kernel $k(x, y) = -\|x - y\|_2$. For empirical measures $\mu = n^{-1} \sum_i \delta_{x_i}$ and $\nu = m^{-1} \sum_j \delta_{y_j}$, with $n = m = 200$ particles in $\mathbb{R}^2$, this objective is $\text{MMD}(\mu, \nu)^2 = n^{-2} \sum_{i,i'} k(x_i, x_{i'}) + m^{-2} \sum_{j,j'} k(y_j, y_{j'}) - 2(nm)^{-1} \sum_{i,j} k(x_i, y_j)$, with a small smoothing of $\|x - y\|_2$ in the implementation. The same initial cloud evolves with $p = 1$ and $p = \infty$. Strictly speaking, $-\|x - y\|_2$ is conditionally positive definite on zero-mass signed measures, not a positive-definite kernel on all measures. Adding the usual anchoring terms gives a positive-definite kernel with the same quadratic energy on $\mu - \nu$ [Sejdinovic et al., 2013]. Figure 5 isolates the effect of the spectral tangent norm without the additional architectural structure of a neural network. The trace-norm flow reacts locally to the MMD force field, while the operator-norm flow produces more globally coordinated trajectories; the scalar loss curves remain close on the separately normalized horizons. A plausible explanation for the early outward motion in the operator case is amplification of small transverse force components: the canonical selector multiplies an active

covariance mode of eigenvalue λ_i by $\text{tr}(S_g^{1/2})/\sqrt{\lambda_i}$. Its velocity covariance is isotropic on the active range, as in Remark 5.2. This is a mechanism consistent with the figure, not a separate proof of its transient behavior.

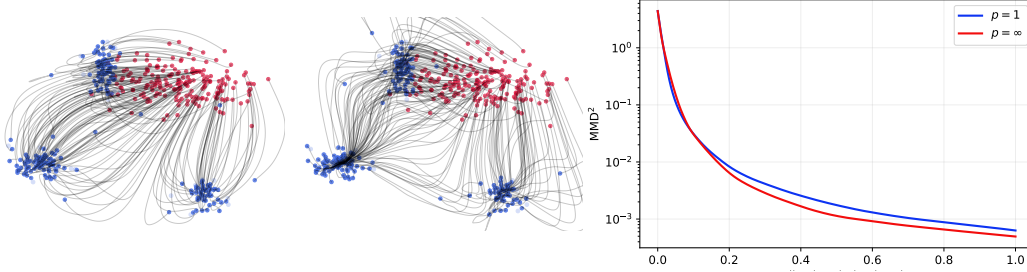


Figure 5: MMD trajectories and objective decay on separately normalized nominal horizons for $p = 1$ and $p = \infty$. The left and middle panels show all particle trajectories, while the right panel shows the decay of $\text{MMD}(\mu_t, \nu)^2$.

Gaussian-preserving linear networks. A particularly tractable example is a linear two-layer network, $H_\mu(z) = \int u(v^\top z) d\mu(u, v)$, trained with quadratic loss. Its force is linear in the parameter $x = (u, v)$. The canonical Measure-LMO preserves this property, because it multiplies the force by a matrix depending only on its covariance. An initially Gaussian distribution therefore remains Gaussian along any sufficiently regular solution, and the PDE closes on its mean and covariance. This illustrates a concrete advantage of the measure formulation: infinitely many neurons can be described by a finite-dimensional moment system. Appendix F derives this system, its modal reductions, and the p -dependent implicit covariance bias illustrated in Figure 6.

Geodesic convexity. Full geodesic convexity, i.e., the condition $\mathcal{F}(\mu_t) \leq (1-t)\mathcal{F}(\mu_0) + t\mathcal{F}(\mu_1)$ along every constant-speed W_γ -geodesic $(\mu_t)_{t \in [0,1]}$, is a central structural notion for metric gradient flows [Ambrosio et al., 2008], as it underlies stability and global convergence results. The following proposition summarizes several basic results along the static displacement geodesics of Corollary 4.4; Appendix H contains the proofs, a refined κ -convexity analysis, and a discussion of the distinction between displacement and full geodesic convexity. These results are, however, of limited direct use in the machine-learning examples considered above, since the quadratic functionals arising in the MLP and MMD settings are typically not geodesically convex. Note also that geodesic convexity of the entropy is stated only along a selected geodesic: uniqueness of geodesics may fail, in particular for $p = \infty$, which makes the Muon setting more delicate.

Proposition 5.3 (Basic displacement-convexity examples). *Assume that γ is monotone. A linear functional $\mathcal{F}(\mu) = \int h d\mu$, with $h : \mathbb{R}^d \rightarrow \mathbb{R}$, is convex along every static displacement geodesic if and only if h is convex. A quadratic interaction functional $\mathcal{F}(\mu) = \int h(x, y) d\mu(x) d\mu(y)$, with $h : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, is convex along every static displacement geodesic if h is convex. Finally, let $d\nu = Z^{-1}e^{-V}dx$, with $V \in C^2(\mathbb{R}^d)$ convex, and define the relative entropy by $\mathcal{F}(\mu) = \int \log(d\mu/d\nu) d\mu$ if $\mu \ll \nu$, and $\mathcal{F}(\mu) = +\infty$ otherwise. If the representing set $\mathcal{K}_\gamma$ contains at least one positive definite matrix, then for every $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$ there exists a constant-speed W_γ -geodesic $(\mu_t)_{t \in [0,1]}$ joining them along which $\mathcal{F}$ is convex.*

A weaker route to convergence is a Polyak–Łojasiewicz (P-L) inequality, which controls the energy gap by the squared metric slope without requiring geodesic convexity. Appendix I proves transfer of such inequalities from W_2 and studies their local, Poincaré counterpart for relative entropy. These results give convergence criteria and explicit examples of spectral gains, not a convergence theorem for the neural-network objectives above.

6 Conclusion

The main message of this paper is that matrix-normalized optimizers for mean-field neural models are naturally encoded by Spectral Wasserstein distances. In this view, matrix normalization is not merely a preconditioning device, but the local steepest-descent geometry generated by a spectral

transport cost. This clarifies the transport side, through coupling-based costs, geodesics, and Gaussian reductions, and the optimization side, through a continuum interpretation of normalized matrix updates. The resulting picture is that the choice of spectral gauge shapes which directions of motion are emphasized or suppressed in the mean-field dynamics, thereby providing a geometric language for comparing matrix-normalized optimizers. A major open question is the rigorous study of the Spectral Wasserstein gradient-flow PDE: global-in-time existence, stability, and the avoidance of local minima for shallow neural-network and MMD losses.

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A Matrix Recap and Explicit LMOs

This appendix collects the matrix facts used in Section 2. The canonical polar representing set is $\mathcal{P}_\gamma := \{Q \in \mathbb{S}^d : \text{tr}(QS) \leq \gamma(S) \text{ for every } S \succeq 0\}$; finite-dimensional convex duality gives $\gamma(S) = \max_{Q \in \mathcal{P}_\gamma} \text{tr}(QS)$ for every $S \succeq 0$.

We first show that the use of positive gauges on the cone $\mathbb{S}_+^d$ is not essentially more restrictive than working with norms on the full space of symmetric matrices: under monotonicity, every such gauge admits a canonical norm extension.

Proposition A.1 (Norm extension of a monotone positive gauge). *Let $\gamma : \mathbb{S}_+^d \rightarrow \mathbb{R}_+$ be a monotone positive gauge, i.e., γ is convex, positively one-homogeneous, $\gamma(S) > 0$ for every nonzero $S \succeq 0$, and $0 \preceq S \preceq S'$ implies $\gamma(S) \leq \gamma(S')$. Define, for $A \in \mathbb{S}^d$,*

$$\|A\|_\gamma := \inf \left\{ \gamma(P) + \gamma(N) : P, N \succeq 0, A = P - N \right\}.$$

Then $\|\cdot\|_\gamma$ is a norm on $\mathbb{S}^d$, and its restriction to $\mathbb{S}_+^d$ coincides with γ .

Proof. We first note that the admissible set is nonempty for every $A \in \mathbb{S}^d$: for instance, the spectral decomposition $A = A_+ - A_-$ gives $A_+, A_- \succeq 0$. Hence $\|A\|_\gamma$ is well-defined and finite.

Positive homogeneity follows directly from the definition. If $a \geq 0$, then

$$\|aA\|_\gamma = \inf_{A=P-N} \{\gamma(aP) + \gamma(aN)\} = a\|A\|_\gamma.$$

If $a < 0$, then $aA = (-a)N - (-a)P$ whenever $A = P - N$, so the same argument gives $\|aA\|_\gamma = |a|\|A\|_\gamma$. In particular, $\|-A\|_\gamma = \|A\|_\gamma$.

We next prove the triangle inequality. Let $A = P_1 - N_1$ and $B = P_2 - N_2$ with $P_i, N_i \succeq 0$. Then

$$A + B = (P_1 + P_2) - (N_1 + N_2),$$

and $P_1 + P_2, N_1 + N_2 \succeq 0$. By subadditivity of γ , which follows from convexity and one-homogeneity,

$$\gamma(P_1 + P_2) + \gamma(N_1 + N_2) \leq \gamma(P_1) + \gamma(P_2) + \gamma(N_1) + \gamma(N_2).$$

Taking the infimum over all decompositions of A and B yields $\|A + B\|_\gamma \leq \|A\|_\gamma + \|B\|_\gamma$.

It remains to prove nondegeneracy. Monotonicity and subadditivity imply $|\gamma(S) - \gamma(T)| \leq \gamma(I)\|S - T\|_{\text{op}}$ on the PSD cone, so γ is continuous. Since it is strictly positive on the compact section $\{S \succeq 0 : \text{tr}(S) = 1\}$, there exists $c > 0$ such that $\gamma(S) \geq c \text{tr}(S)$ for every $S \succeq 0$. Therefore, for every decomposition $A = P - N$,

$$\gamma(P) + \gamma(N) \geq c \text{tr}(P + N).$$

Moreover, from $A = P - N$ we have $P + N \succeq A$ and $P + N \succeq -A$. Let E_+ and E_- denote the spectral projectors of A associated with its positive and negative eigenvalues. Taking traces after compression to these two subspaces gives

$$\text{tr}(E_+(P + N)) \geq \text{tr}(A_+), \quad \text{tr}(E_-(P + N)) \geq \text{tr}(A_-).$$

Since $P + N \succeq 0$, its trace on the kernel of A is nonnegative, and hence $\text{tr}(P + N) \geq \text{tr}|A|$. Therefore

$$\gamma(P) + \gamma(N) \geq c \text{tr}|A|.$$

Taking the infimum gives $\|A\|_\gamma \geq c \text{tr}|A|$. Thus $\|A\|_\gamma = 0$ implies $A = 0$.

Finally, let $S \succeq 0$. The decomposition $S = S - 0$ gives $\|S\|_\gamma \leq \gamma(S)$. Conversely, if $S = P - N$ with $P, N \succeq 0$, then $P = S + N \succeq S$, and by monotonicity, $\gamma(P) \geq \gamma(S)$. Hence

$$\gamma(P) + \gamma(N) \geq \gamma(P) \geq \gamma(S).$$

Taking the infimum over all such decompositions gives $\|S\|_\gamma \geq \gamma(S)$. Therefore $\|S\|_\gamma = \gamma(S)$ for every $S \succeq 0$. $\square$

The next proposition characterizes the monotone norms for which representing sets may be chosen inside the positive-semidefinite cone.

Proposition A.2 (PSD representation and monotonicity). *The following are equivalent:*

- γ is monotone on $\mathbb{S}_+^d$, namely

$$0 \preceq S \preceq T \implies \gamma(S) \leq \gamma(T);$$

- there exists a convex compact representing set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ such that

$$\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QS) \quad \text{for every } S \succeq 0.$$

Moreover, when these properties hold, the canonical choice

$$\mathcal{K}_\gamma = \mathcal{P}_\gamma \cap \mathbb{S}_+^d$$

is admissible.

Proof. If $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, then for $0 \preceq S \preceq T$ one has $\text{tr}(QS) \leq \text{tr}(QT)$ for every $Q \in \mathcal{K}_\gamma$, hence $\gamma(S) \leq \gamma(T)$ by taking suprema.

Conversely, assume γ is monotone and use Proposition A.1 to extend it to a norm on $\mathbb{S}^d$. Define

$$\mathcal{K}_\gamma^+ := \{Q \in \mathbb{S}_+^d : \text{tr}(QS) \leq \gamma(S) \text{ for every } S \succeq 0\}.$$

It is immediate that $\mathcal{K}_\gamma^+$ is convex, closed, bounded, and contained in $\mathcal{P}_\gamma \cap \mathbb{S}_+^d$. We claim that it represents γ . Fix $S \succeq 0$. If $S \succ 0$, choose a subgradient $Q \in \partial\gamma(S)$ in the ambient space of symmetric matrices. Since γ is convex and one-homogeneous, $\text{tr}(QS) = \gamma(S)$ and $\text{tr}(QT) \leq \gamma(T)$ for every $T \succeq 0$. Moreover $Q \succeq 0$: indeed, for every $R \succeq 0$ and every sufficiently small $\varepsilon > 0$ such that $S - \varepsilon R \succeq 0$, the subgradient inequality gives

$$\gamma(S - \varepsilon R) \geq \gamma(S) - \varepsilon \text{tr}(QR),$$

while monotonicity gives $\gamma(S - \varepsilon R) \leq \gamma(S)$. Hence $\text{tr}(QR) \geq 0$ for all $R \succeq 0$, i.e., $Q \succeq 0$. Thus $Q \in \mathcal{K}_\gamma^+$ and $\gamma(S) \leq \sup_{P \in \mathcal{K}_\gamma^+} \text{tr}(PS)$. The reverse inequality follows from the definition of $\mathcal{K}_\gamma^+$. For singular $S \succeq 0$, apply the previous argument to $S + \varepsilon I$. Compactness gives a convergent subsequence of supporting matrices; passing to the limit and using continuity of γ yields equality at S . Therefore $\gamma(S) = \max_{Q \in \mathcal{K}_\gamma^+} \text{tr}(QS)$ for every $S \succeq 0$. $\square$

Proposition A.3 (Norm property of N). *Assume that γ is monotone and admits a positive semidefinite representing set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, so that $\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QS)$ for every $S \succeq 0$. Define $N(V) := \gamma(V^\top V)^{1/2}$. Then N is a seminorm. If, in addition, $\gamma(S) > 0$ for every nonzero $S \succeq 0$, then N is a norm.*

Proof. For every matrix V , one has

$$N(V)^2 = \gamma(V^\top V) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QV^\top V) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(VQV^\top).$$

Since $Q \succeq 0$, we may write $\text{tr}(VQV^\top) = \|VQ^{1/2}\|_F^2$. Hence

$$N(V) = \max_{Q \in \mathcal{K}_\gamma} \|VQ^{1/2}\|_F.$$

For each fixed $Q \in \mathcal{K}_\gamma$, the map $V \mapsto \|VQ^{1/2}\|_F$ is a seminorm. Therefore, for any matrices V, W and scalar a ,

$$N(aV) = \max_{Q \in \mathcal{K}_\gamma} \|aVQ^{1/2}\|_F = |a|N(V),$$

and

$$N(V + W) = \max_{Q \in \mathcal{K}_\gamma} \|(V + W)Q^{1/2}\|_F \leq \max_{Q \in \mathcal{K}_\gamma} (\|VQ^{1/2}\|_F + \|WQ^{1/2}\|_F) \leq N(V) + N(W).$$

Thus N is a seminorm. If moreover $\gamma(S) > 0$ for every nonzero $S \succeq 0$, then $N(V) = 0$ implies $\gamma(V^\top V) = 0$, hence $V^\top V = 0$, and therefore $V = 0$. Thus N is a norm. $\square$

Remark A.4 ($\mathcal{K}_\gamma$ for the Schatten p -norms). *For $\gamma = \gamma_p$, a convenient supporting set is obtained from the dual Schatten unit ball. Namely, if $p \in [1, \infty]$ and $p' \in [1, \infty]$ is the conjugate exponent, $1/p + 1/p' = 1$, one may take*

$$\mathcal{K}_{\gamma_p} = \mathcal{P}_{\gamma_p} \cap \mathbb{S}_+^d := \left\{ G \in \mathbb{S}_+^d : \|G\|_{S_{p'}} \leq 1 \right\}.$$

One can, however, choose a smaller supporting set in the endpoint cases. For $p = 1$, since $\|A\|_{S_1} = \text{tr}(A)$ on $\mathbb{S}_+^d$, it is enough to take $\mathcal{K}_{\gamma_1} = \{I\}$. For $p = \infty$, one may take the set of density matrices,

$$\mathcal{K}_{\gamma_\infty} = \left\{ G \in \mathbb{S}_+^d : \text{tr}(G) = 1 \right\}.$$

The next proposition gives the explicit LMO associated with each Schatten geometry.

Proposition A.5 (Explicit Schatten selectors). *Let $\gamma = \gamma_p = \|\cdot\|_{S_p}$, let $r = 2p$, and let $q = \frac{r}{r-1} = \frac{2p}{2p-1}$ be the dual exponent. If*

$$G = U_r \operatorname{diag}(\sigma_i) W_r^\top$$

is a compact singular value decomposition over the positive singular values, then

$$J_p(G) = -\|G\|_{S_q}^{2-q} U_r \operatorname{diag}(\sigma_i^{q-1}) W_r^\top.$$

For $G = 0$, set $J_p(G) = 0$. In particular,

- $p = 1$: $J_1(G) = -G$, which recovers ordinary gradient flow;
- $p = 2$: $J_2(G) = -\|G\|_{S_{4/3}}^{2/3} U_r \operatorname{diag}(\sigma_i^{1/3}) W_r^\top$, which we call the Frobenius-intermediate normalization;
- $p = \infty$:

$$J_\infty(G) = -\|G\|_{S_1} U_r W_r^\top = -\operatorname{tr}((G^\top G)^{1/2}) G (G^\top G)^{\dagger/2}, \quad (6)$$

which is the Muon normalization.

Proof. The displayed formula is the negative gradient of $\frac{1}{2} \|G\|_{S_q}^2$ when $1 < q \leq 2$. At the endpoint $q = 1$, the compact factor $U_r W_r^\top$ is a canonical partial-polar subgradient; it also fixes the value on the null singular subspaces. $\square$

A.1 Proofs and dual formulas for measure LMOs

Proof of Proposition 2.4.

Proof. Associate to every vector field $h \in L^2(\mu; \mathbb{R}^d)$ the Hilbert–Schmidt operator $T_h : \mathbb{R}^d \rightarrow L^2(\mu)$ defined by $(T_h a)(x) = a \cdot h(x)$. Then

$$T_h^* T_h = \int h h^\top d\mu, \quad \int g \cdot h d\mu = \langle T_g, T_h \rangle_{\text{HS}}.$$

Let $S_g = T_g^* T_g$ and write the polar decomposition

$$T_g = U_g S_g^{1/2}, \quad U_g^* U_g = P_g,$$

where P_g is the orthogonal projector onto $\operatorname{range}(S_g)$. Let $\Pi_g = U_g U_g^*$ be the orthogonal projector onto the range of T_g in $L^2(\mu)$. For any h , replacing T_h by $\Pi_g T_h$ leaves the linear term unchanged and decreases the covariance:

$$\langle T_g, \Pi_g T_h \rangle_{\text{HS}} = \langle T_g, T_h \rangle_{\text{HS}}, \quad (\Pi_g T_h)^* (\Pi_g T_h) = T_h^* \Pi_g T_h \preceq T_h^* T_h.$$

By monotonicity of γ , this projection cannot increase the LMO objective. Hence it is enough to minimize over operators of the form $T_h = U_g A$ with $A = P_g A$. For such fields,

$$\int g \cdot h d\mu = \operatorname{tr}(S_g^{1/2} A), \quad \int h h^\top d\mu = A^\top A.$$

Finally, the operator $T_h = U_g A$ corresponds to the vector field $h(x) = A^\top S_g^{\dagger/2} g(x)$. This proves the finite-dimensional reduction and the lifted selector. $\square$

Corollary A.6 (Schatten measure selectors). *Let $\gamma = \gamma_p = \|\cdot\|_{S_p}$ and set $q = 2p/(2p-1)$. Then*

$$\mathcal{J}_{\gamma_p}^\mu(g)(x) = -\|S_g^{1/2}\|_{S_q}^{2-q} S_g^{(q-2)/2} g(x), \quad S_g = \int g g^\top d\mu,$$

where the power is taken on $\operatorname{range}(S_g)$ and is zero on its kernel; when $S_g = 0$, the selector is zero. In particular, $p = 1$ gives $\mathcal{J}_{\gamma_1}^\mu(g) = -g$, while $p = \infty$ gives

$$\mathcal{J}_{\gamma_\infty}^\mu(g)(x) = -\operatorname{tr}(S_g^{1/2}) S_g^{\dagger/2} g(x).$$

Proof. By Proposition 2.4, one only needs to solve

$$\min_A \left\{ \text{tr}(S_g^{1/2} A) + \frac{1}{2} \|A\|_{S_{2p}}^2 \right\}$$

on the active range. The minimizer is the Schatten duality map

$$A_* = -\|S_g^{1/2}\|_{S_q}^{2-q} S_g^{(q-1)/2}.$$

Lifting by $h(x) = A_*^\top S_g^{\dagger/2} g(x)$ gives the formula. $\square$

The displayed Schatten selector is Borel measurable, including at singular covariances, by its spectral-calculus definition. Measurability does not imply continuity: for example, the canonical matrix selector sends $G_\varepsilon = \text{diag}(1, \varepsilon)$ to $-(1 + \varepsilon)I$ for $\varepsilon > 0$, but sends $G_0 = \text{diag}(1, 0)$ to $-\text{diag}(1, 0)$. Thus rank changes at the Muon endpoint require a separate analysis for ODE or PDE well-posedness. The full argmin is also not necessarily a singleton: for $G = \text{diag}(1, 0)$ and the operator tangent norm, every $V = -\text{diag}(1, a)$ with $|a| \leq 1$ attains the minimum $-1/2$. The displayed canonical selector chooses $a = 0$.

The support representation of γ gives another useful construction. It is less intrinsic than the finite-dimensional LMO above, but when it applies it produces the explicit formula $v = -Q^{-1}g$ and identifies an active quadratic cost. Its limitation is the required attainment at an invertible matrix, which may fail for rank-deficient force covariances, including at $p = \infty$. This is not an inherent rank-one restriction at the operator endpoint: for $S_g \succ 0$ and $\mathcal{K}_\gamma = \{Q \succeq 0 : \text{tr } Q \leq 1\}$, the minimizer is $Q^* = S_g^{1/2} / \text{tr}(S_g^{1/2}) \succ 0$. The rank-one certificates that can appear in the static transport problem play a different role.

Proposition A.7 (Inverse-trace dual certificate). *Assume that $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ is a compact convex representing set for γ and that $\mathcal{K}_\gamma \cap \mathbb{S}_{++}^d \neq \emptyset$. Then*

$$\inf_v \left\{ \int g \cdot v \, d\mu + \frac{1}{2} \gamma \left(\int v v^\top \, d\mu \right) \right\} = -\frac{1}{2} \inf_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \text{tr}(Q^{-1} S_g), \quad S_g = S_\mu(g).$$

If the matrix infimum is attained at

$$Q_\mu^* \in \underset{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d}{\text{argmin}} \text{tr}(Q^{-1} S_g),$$

then $v_*(x) = -(Q_\mu^*)^{-1}g(x)$ is a Measure-LMO.

Proof. For $Q \in \mathcal{K}_\gamma$, define

$$\Phi_Q(v) := \int g \cdot v \, d\mu + \frac{1}{2} \int v^\top Q v \, d\mu.$$

Since $\gamma(R) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QR)$ for $R \succeq 0$, the Measure-LMO objective is $\inf_v \max_{Q \in \mathcal{K}_\gamma} \Phi_Q(v)$. The convex-concave minimax theorem gives

$$\inf_v \max_{Q \in \mathcal{K}_\gamma} \Phi_Q(v) = \max_{Q \in \mathcal{K}_\gamma} \inf_v \Phi_Q(v).$$

Indeed, the matrix set is compact and convex, while any $Q_0 \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d$ provides coercivity in v . For fixed positive definite Q , the inner minimization gives

$$v_Q(x) = -Q^{-1}g(x), \quad \inf_v \Phi_Q(v) = -\frac{1}{2} \text{tr}(Q^{-1} S_\mu(g)).$$

Convexity of $\mathcal{K}_\gamma$ allows singular matrices to be approximated by $(1 - \varepsilon)Q + \varepsilon Q_0 \succ 0$, which proves the displayed inverse-trace identity. Now let Q_μ^* attain the matrix infimum and set $v_* = -(Q_\mu^*)^{-1}g$. The first-order optimality condition for the convex function $Q \mapsto \text{tr}(Q^{-1} S_g)$ over $\mathcal{K}_\gamma$ gives, for every $Q \in \mathcal{K}_\gamma$,

$$\text{tr}(Q (Q_\mu^*)^{-1} S_g (Q_\mu^*)^{-1}) \leq \text{tr}(Q_\mu^* (Q_\mu^*)^{-1} S_g (Q_\mu^*)^{-1}).$$

Since

$$\int v_* v_*^\top \, d\mu = (Q_\mu^*)^{-1} S_g (Q_\mu^*)^{-1},$$

this says precisely that Q_μ^* is active in the support representation of γ at the covariance of v_* . Therefore

$$\int g \cdot v_* d\mu + \frac{1}{2}\gamma\left(\int v_* v_*^\top d\mu\right) = -\frac{1}{2}\text{tr}((Q_\mu^*)^{-1}\mathcal{S}_\mu(g)),$$

which equals the common minimax value. Thus, v_* minimizes the Measure-LMO objective. $\square$

Proof of Proposition 2.5.

Proof. For empirical measures, the continuity equation reduces to the characteristic system for the atoms. The identity $\mathcal{N}_{\mu_X}(v)^2 = n^{-1}\text{N}(V)^2$ and the normalized pairing $n^{-1}\text{tr}(G^\top V)$ reduce the measure-level minimization in Definition 2.3 to

$$\operatorname{argmin}_V \left\{ \text{tr}(G^\top V) + \frac{1}{2}\text{N}(V)^2 \right\} = J_\gamma(G).$$

Stacking the particle velocities yields $\dot{X}_t = J_\gamma(G_t)$. Under the convention of the main text, $F(X) = n\mathcal{F}(\mu_X)$ and $\nabla F(X) = G_t$, giving exactly ODE (2). For the unscaled objective $\tilde{F}(X) = \mathcal{F}(\mu_X)$, one instead has $\nabla \tilde{F}(X) = n^{-1}G_t$ and $\dot{X}_t = n^{-1}J_\gamma(G_t)$ by one-homogeneity. This is the same weak measure equation after rescaling time by n . $\square$

A.2 Comparison between Classical and Spectral Wasserstein Distances

Proof of Proposition 3.2.

Proof. Since γ is a norm on the finite-dimensional space $\mathbb{S}^d$, its restriction to the compact set $\{S \succeq 0 : \text{tr}(S) = 1\}$ is continuous and strictly positive. Hence there exist constants $0 < c_\gamma \leq C_\gamma < \infty$ such that $c_\gamma \text{tr}(S) \leq \gamma(S) \leq C_\gamma \text{tr}(S)$ for every $S \succeq 0$.

Let $\pi \in \Pi(\mu, \nu)$ be any coupling, and define its displacement covariance by

$$\Sigma_\pi := \int_{\mathbb{R}^d \times \mathbb{R}^d} (x - y)(x - y)^\top d\pi(x, y).$$

Then $\Sigma_\pi \succeq 0$, and therefore

$$c_\gamma \text{tr}(\Sigma_\pi) \leq \gamma(\Sigma_\pi) \leq C_\gamma \text{tr}(\Sigma_\pi).$$

Moreover,

$$\text{tr}(\Sigma_\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} \text{tr}((x - y)(x - y)^\top) d\pi(x, y) = \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 d\pi(x, y).$$

Thus, for every coupling $\pi \in \Pi(\mu, \nu)$,

$$c_\gamma \int |x - y|^2 d\pi(x, y) \leq \gamma(\Sigma_\pi) \leq C_\gamma \int |x - y|^2 d\pi(x, y).$$

Taking infima over $\pi \in \Pi(\mu, \nu)$ gives

$$c_\gamma W_2(\mu, \nu)^2 \leq W_\gamma(\mu, \nu)^2 \leq C_\gamma W_2(\mu, \nu)^2,$$

where we used the definitions of W_2 and W_γ . Taking square roots yields

$$\sqrt{c_\gamma} W_2(\mu, \nu) \leq W_\gamma(\mu, \nu) \leq \sqrt{C_\gamma} W_2(\mu, \nu).$$

$\square$

Schatten constants and sharpness. For Schatten norms, $d^{1/p-1} W_2(\mu, \nu)^2 \leq W_{\gamma_p}(\mu, \nu)^2 \leq W_2(\mu, \nu)^2$, equivalently $c_{\gamma_p} = d^{1/p-1}$ and $C_{\gamma_p} = 1$. In particular, $(c_{\gamma_2}, C_{\gamma_2}) = (d^{-1/2}, 1)$ and $(c_{\gamma_\infty}, C_{\gamma_\infty}) = (d^{-1}, 1)$. These constants follow from $\|\lambda\|_{\ell_p} \leq \|\lambda\|_{\ell_1} \leq d^{1-1/p} \|\lambda\|_{\ell_p}$.

The lower Schatten bound is sharp. For $\gamma = \gamma_p$, equality in the lower bound is attained by isotropic Gaussian pairs $\mu = \mathcal{N}(0, \alpha I_d)$ and $\nu = \mathcal{N}(0, \beta I_d)$, $\alpha, \beta > 0$. By the Gaussian computation of Appendix D, the displacement covariance is $(\sqrt{\alpha} - \sqrt{\beta})^2 I_d$, hence $W_{\gamma_p}(\mu, \nu)^2 = d^{1/p-1} W_2(\mu, \nu)^2$.

Proof of Theorem 3.4.

Proof. Let $\pi \in \Pi(\mu, \nu)$ and set

$$\Sigma_\pi := \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)(y - x)^\top d\pi(x, y).$$

Since $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, the matrix Σ_π is well-defined and belongs to $\mathbb{S}_+^d$. By the support representation of γ ,

$$\gamma(\Sigma_\pi) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(Q\Sigma_\pi).$$

Moreover, for every $Q \in \mathcal{K}_\gamma$,

$$\text{tr}(Q\Sigma_\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)^\top Q(y - x) d\pi(x, y).$$

Therefore

$$W_\gamma(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \max_{Q \in \mathcal{K}_\gamma} \int (y - x)^\top Q(y - x) d\pi(x, y).$$

It remains to justify the exchange of $\inf_\pi$ and $\max_Q$. Define

$$F(\pi, Q) := \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x)^\top Q(y - x) d\pi(x, y).$$

The coupling set $\Pi(\mu, \nu)$ is convex and weakly compact, since its marginals are fixed. The set $\mathcal{K}_\gamma$ is convex and compact by assumption. For fixed π , the map $Q \mapsto F(\pi, Q)$ is affine, hence both concave and convex, and is continuous on $\mathcal{K}_\gamma$. For fixed $Q \succeq 0$, the cost $(x, y) \mapsto (y - x)^\top Q(y - x)$ is nonnegative and lower semicontinuous. Hence $\pi \mapsto F(\pi, Q)$ is lower semicontinuous for weak convergence of probability measures, by the Portmanteau theorem. It is also affine in π .

Thus the lower-semicontinuous form of Sion's minimax theorem Sion [1958] applies and gives

$$\inf_{\pi \in \Pi(\mu, \nu)} \max_{Q \in \mathcal{K}_\gamma} F(\pi, Q) = \max_{Q \in \mathcal{K}_\gamma} \inf_{\pi \in \Pi(\mu, \nu)} F(\pi, Q).$$

The only point requiring care is that the quadratic cost is not bounded on $\mathbb{R}^d \times \mathbb{R}^d$. This is precisely why one uses the lower-semicontinuous version of the minimax theorem rather than the bounded continuous one. Consequently,

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} \inf_{\pi \in \Pi(\mu, \nu)} \int (y - x)^\top Q(y - x) d\pi(x, y).$$

Finally, by definition of the quadratic optimal transport cost associated with Q ,

$$W^Q(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \int (y - x)^\top Q(y - x) d\pi(x, y).$$

Therefore

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu)^2,$$

which proves the value formula. Finally, if (Q^*, π^*) is a saddle point, then Q^* maximizes $Q \mapsto \text{tr}(Q\Sigma_{\pi^*})$ over $\mathcal{K}_\gamma$. Since γ is the support function of $\mathcal{K}_\gamma$ on $\mathbb{S}_+^d$, this is equivalent to the cone subgradient relation $Q^* \in \partial\gamma(\Sigma_{\pi^*})$, proving the stated optimality relation. $\square$

A.3 Metric property from the robust-cost formulation

The max–min representation of Theorem 3.4 gives a direct proof of the triangle inequality when γ is monotone. This avoids using the Benamou–Brenier formulation for this specific purpose.

Proposition A.8 (Static proof of the metric property). *Assume that γ is monotone on $\mathbb{S}_+^d$. Then W_γ is a distance on $\mathcal{P}_2(\mathbb{R}^d)$ and induces the same topology as W_2 .*

Proof. By Proposition A.2, monotonicity allows us to choose a representing set $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ such that

$$\gamma(S) = \max_{Q \in \mathcal{K}_\gamma} \text{tr}(QS), \quad S \succeq 0.$$

Hence Theorem 3.4 gives

$$W_\gamma(\mu, \nu) = \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu, \nu).$$

For a fixed $Q \in \mathcal{K}_\gamma$, write

$$d_Q(x, y) := \left\| Q^{1/2}(x - y) \right\|_2.$$

Since $Q \succeq 0$, d_Q is a seminorm distance on $\mathbb{R}^d$, and therefore W^Q is the corresponding quadratic Wasserstein pseudodistance. In particular, for all $\mu_0, \mu_1, \mu_2 \in \mathcal{P}_2(\mathbb{R}^d)$,

$$W^Q(\mu_0, \mu_2) \leq W^Q(\mu_0, \mu_1) + W^Q(\mu_1, \mu_2).$$

Taking the maximum over $Q \in \mathcal{K}_\gamma$ yields

$$\begin{aligned} W_\gamma(\mu_0, \mu_2) &= \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu_0, \mu_2) \\ &\leq \max_{Q \in \mathcal{K}_\gamma} \{W^Q(\mu_0, \mu_1) + W^Q(\mu_1, \mu_2)\} \\ &\leq \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu_0, \mu_1) + \max_{Q \in \mathcal{K}_\gamma} W^Q(\mu_1, \mu_2) \\ &= W_\gamma(\mu_0, \mu_1) + W_\gamma(\mu_1, \mu_2). \end{aligned}$$

Symmetry is immediate from the definition. Moreover $W_\gamma(\mu, \mu) = 0$, and separation follows from Proposition 3.2: if $W_\gamma(\mu, \nu) = 0$, then $W_2(\mu, \nu) = 0$, hence $\mu = \nu$. The same comparison bounds also show that W_γ and W_2 induce the same topology on $\mathcal{P}_2(\mathbb{R}^d)$. $\square$

Remark A.9 (Relation with subspace robust Wasserstein distances). *The proof is the same robust-cost argument used by Paty and Cuturi [2019] for Ky Fan gauges. The only point needed in their argument is that the maximizing cost matrices are positive semidefinite. By Proposition A.2, this positivity holds for any monotone gauge γ , not only for Ky Fan gauges. Thus the Paty–Cuturi triangle-inequality proof extends verbatim to the present monotone-gauge setting.*

A.4 Large-mean asymptotics for Schatten geometries

We now describe the behavior of W_γ when the displacement between the means is much larger than the centered fluctuations of the measures. This regime is useful to interpret the optimizer in the max-min formula: for Schatten geometries with $p > 1$, the optimal certificate becomes increasingly aligned with the direction of the mean displacement, and the residual transport problem is effectively measured only along this direction.

Proposition A.10 (Large-mean asymptotics for Schatten geometries). *Let $\gamma = \gamma_p$ be the Schatten- p norm, with $p \in [1, \infty]$. Fix centered measures $\bar{\mu}, \bar{\nu} \in \mathcal{P}_2(\mathbb{R}^d)$ and a unit vector $u \in \mathbb{R}^d$. For each $R > 0$, let μ_R, ν_R have centered laws $\bar{\mu}, \bar{\nu}$ and means whose difference is Ru . Then, as $R \rightarrow \infty$, the following asymptotics hold. For $p = 1$,*

$$W_\gamma(\mu_R, \nu_R)^2 = R^2 + W_2(\bar{\mu}, \bar{\nu})^2.$$

For $1 < p \leq \infty$,

$$W_\gamma(\mu_R, \nu_R)^2 = R^2 + W^{uu^\top}(\bar{\mu}, \bar{\nu})^2 + o(1).$$

Proof. We first separate the mean displacement from the centered transport. Fix $R > 0$, abbreviate $\mu = \mu_R, \nu = \nu_R$, set $m_\mu = \int x d\mu(x)$, $m_\nu = \int y d\nu(y)$, and let $\Delta = m_\nu - m_\mu = Ru$. Fix $Q \succeq 0$. If $\pi \in \Pi(\mu, \nu)$ and if $\bar{\pi}$ denotes the corresponding coupling of $\bar{\mu}$ and $\bar{\nu}$, then writing $x = m_\mu + \bar{x}$ and $y = m_\nu + \bar{y}$ gives

$$y - x = \Delta + (\bar{y} - \bar{x}).$$

Therefore

$$\begin{aligned} \int (y - x)^\top Q (y - x) d\pi(x, y) &= \Delta^\top Q \Delta + 2\Delta^\top Q \int (\bar{y} - \bar{x}) d\bar{\pi}(\bar{x}, \bar{y}) \\ &\quad + \int (\bar{y} - \bar{x})^\top Q (\bar{y} - \bar{x}) d\bar{\pi}(\bar{x}, \bar{y}). \end{aligned}$$

The middle term vanishes because $\bar{\mu}$ and $\bar{\nu}$ have zero mean. Taking the infimum over couplings yields the exact decomposition

$$W^Q(\mu, \nu)^2 = \Delta^\top Q \Delta + W^Q(\bar{\mu}, \bar{\nu})^2.$$

By Theorem 3.4,

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_{\gamma_p}} \{ \Delta^\top Q \Delta + W^Q(\bar{\mu}, \bar{\nu})^2 \}.$$

For $p = 1$, we use the reduced representing set $\mathcal{K}_{\gamma_1} = \{\text{Id}\}$. Then

$$W_\gamma(\mu, \nu)^2 = \Delta^\top \Delta + W^{\text{Id}}(\bar{\mu}, \bar{\nu})^2 = R^2 + W_2(\bar{\mu}, \bar{\nu})^2,$$

as claimed.

Assume now that $1 < p \leq \infty$, and let p' be the conjugate exponent, with $p' = 1$ when $p = \infty$. We use the positive dual Schatten representing set

$$\mathcal{K}_{\gamma_p} = \{Q \succeq 0 : \|Q\|_{S_{p'}} \leq 1\}.$$

Since $\Delta = Ru$, the max-min formula becomes

$$W_\gamma(\mu, \nu)^2 = \max_{Q \in \mathcal{K}_{\gamma_p}} \{ R^2 u^\top Q u + W^Q(\bar{\mu}, \bar{\nu})^2 \}.$$

By Schatten duality, applied to the rank-one matrix $uu^\top$,

$$\max_{Q \in \mathcal{K}_{\gamma_p}} u^\top Q u = \max_{\substack{Q \succeq 0 \\ \|Q\|_{S_{p'}} \leq 1}} \text{tr}(Q uu^\top) = \|uu^\top\|_{S_p} = 1.$$

For $1 < p < \infty$, the equality case in Hölder's inequality is unique and gives $Q = uu^\top$. For $p = \infty$, the constraint is $\text{tr}(Q) \leq 1$, and $u^\top Q u \leq \text{tr}(Q) \leq 1$; equality again forces $Q = uu^\top$. Thus the exposed face of $\mathcal{K}_{\gamma_p}$ in the direction $uu^\top$ is the singleton $\{uu^\top\}$ for every $1 < p \leq \infty$.

Let Q_R be an optimal certificate for the previous maximization problem. Since $\mathcal{K}_{\gamma_p}$ is compact, any sequence $R_n \rightarrow \infty$ admits a subsequence such that $Q_{R_n} \rightarrow Q_\infty \in \mathcal{K}_{\gamma_p}$. Dividing the optimality relation by R_n^2 and passing to the limit shows that Q_∞ maximizes $Q \mapsto u^\top Q u$ over $\mathcal{K}_{\gamma_p}$. By the uniqueness of the exposed maximizer, $Q_\infty = uu^\top$. Hence every maximizing certificate satisfies

$$Q_R \longrightarrow uu^\top \quad \text{as } R \rightarrow \infty.$$

Set $b(Q) = W^Q(\bar{\mu}, \bar{\nu})^2$. This map is Lipschitz on $\mathbb{S}_+^d$: indeed, comparing the two infima and using $|(\bar{y} - \bar{x})^\top (Q - Q')(\bar{y} - \bar{x})| \leq \|Q - Q'\|_{\text{op}} |\bar{y} - \bar{x}|^2$ gives

$$|b(Q) - b(Q')| \leq 2\|Q - Q'\|_{\text{op}} \left(\int |\bar{x}|^2 d\bar{\mu} + \int |\bar{y}|^2 d\bar{\nu} \right).$$

Consequently,

$$b(Q_R) \longrightarrow b(uu^\top).$$

Convergence of the certificates alone would not yet control the R^2 -weighted term. Optimality of Q_R , compared with the admissible choice $uu^\top$, supplies the missing estimate:

$$0 \leq R^2(1 - u^\top Q_R u) \leq b(Q_R) - b(uu^\top) = o(1).$$

Substituting both estimates into the max-min formula gives

$$W_\gamma(\mu, \nu)^2 = R^2 + b(uu^\top) + o(1).$$

Taking square roots yields

$$W_\gamma(\mu, \nu) = R + \frac{1}{2R} W^{uu^\top}(\bar{\mu}, \bar{\nu})^2 + o(R^{-1}),$$

which completes the proof. $\square$

The proposition shows that all Schatten geometries have the same leading large-mean behavior, namely $W_\gamma(\mu_R, \nu_R) \sim R$. The difference appears at the next order and in the optimizer selected by the max-min formula. For the reduced trace geometry $p = 1$, the certificate is fixed and isotropic, $Q = \text{Id}$, so the centered correction is the usual Wasserstein cost $W_2(\bar{\mu}, \bar{\nu})^2$. By contrast, for every $p > 1$, the optimal certificate aligns with the normalized mean displacement u , and the centered correction becomes $W^{uu^\top}(\bar{\mu}, \bar{\nu})^2$. This is exactly the one-dimensional W_2^2 cost between the projected marginals $(u^\top)_\# \bar{\mu}$ and $(u^\top)_\# \bar{\nu}$. Thus the first centered correction depends only on fluctuations along u ; transverse fluctuations do not enter at this order.

B Connection with Muon, Orthogonalization, and Momentum

This appendix complements Section 2 by linking the matrix ODE for $p = \infty$ to practical Muon variants used at scale, and by showing how Newton–Schulz orthogonalization and momentum fit in the same normalized-flow language.

Taking into account Newton–Schulz orthogonalization. Setting $p = \infty$ in Proposition A.5 yields the LMO (6), so the spectral matrix flow

$$\dot{X}_t = J_\infty(\nabla F(X_t))$$

is exactly the vanishing-step-size, vanishing-momentum, full-batch limit of idealized Muon in the normalization convention used throughout this paper.

An important practical feature of Muon is that one does not compute the exact polar factor $UW^\top$ by an SVD at every step. To make the method scalable and GPU-friendly, practical implementations replace $UW^\top = G(G^\top G)^{\dagger/2}$ by a few Newton–Schulz iterations [Jordan et al., 2024, Liu et al., 2025]. After rescaling G by a scalar $\alpha(G) > 0$ chosen so that the spectrum of $\alpha(G)^{-2}G^\top G$ lies in $[0, 1]$, the classical Newton–Schulz iteration reads

$$Z_{k+1} = \frac{1}{2}Z_k(3I - Z_k^\top Z_k), \quad Z_0 = \alpha(G)^{-1}G.$$

For the discussion below, let us freeze the scale and regard $\alpha > 0$ as fixed a priori. Then after one iteration one obtains

$$\mathcal{U}_{\alpha,1}(G) := -Z_1 = -\frac{1}{2\alpha}G(3I - \alpha^{-2}G^\top G),$$

and after k iterations one obtains more generally a polynomial update of the form

$$\mathcal{U}_{\alpha,k}(G) = -Gp_{\alpha,k}(G^\top G),$$

where $p_{\alpha,k}$ is a matrix polynomial approximating $s \mapsto s^{-1/2}$ on the spectral interval of interest. In this fixed- α setting, the polynomial orthogonalization part does admit a scalar generating function: if

$$p_{\alpha,k}(s) = \sum_{\ell=0}^m a_\ell s^\ell,$$

then

$$\Psi_{\alpha,k}(S) := \sum_{\ell=0}^m \frac{a_\ell}{\ell+1} \text{tr}(S^{\ell+1})$$

satisfies

$$\nabla \Psi_{\alpha,k}(S) = p_{\alpha,k}(S) \quad \text{and} \quad \mathcal{U}_{\alpha,k}(G) = -\nabla_G \left(\frac{1}{2} \Psi_{\alpha,k}(G^\top G) \right).$$

For one Newton–Schulz step this gives the explicit formula

$$\Psi_{\alpha,1}(S) = \frac{3}{2\alpha} \text{tr}(S) - \frac{1}{4\alpha^3} \text{tr}(S^2).$$

This function is not a norm, and it is not globally monotone on $\mathbb{S}_+^d$, because

$$\nabla \Psi_{\alpha,1}(S) = \frac{3}{2\alpha}I - \frac{1}{2\alpha^3}S$$

ceases to be positive semidefinite once S has eigenvalues larger than $3\alpha^2$. It is nevertheless monotone on the truncated spectral region

$$0 \preceq S \preceq 3\alpha^2 I,$$

and in particular on the standard Newton–Schulz regime $0 \preceq S \preceq \alpha^2 I$.

Thus a fixed- α Newton–Schulz update is generated by the spectral potential $\Psi_{\alpha,k}$. It must be interpreted on the *force* side, not substituted for the primal velocity gauge γ . Although this polynomial is generally neither convex in S nor homogeneous, there is a precise convex-dissipation interpretation on the stable region.

Proposition B.1 (Fixed-scale Newton–Schulz dissipation). *For the classical iteration above with fixed α and $k \geq 1$, set $D_{\alpha,k}^*(G) = \frac{1}{2}\Psi_{\alpha,k}(G^\top G)$. On $\{G : \|G\|_{\text{op}} \leq \alpha\}$ this is convex, and $\Psi_{\alpha,k}$ is Loewner-monotone on $0 \preceq S \preceq \alpha^2 I$. There is a global convex extension $\hat{D}_{\alpha,k}^*$ agreeing with $D_{\alpha,k}^*$ on this region. With $D_{\alpha,k} = (\hat{D}_{\alpha,k}^*)^*$, for $\|G\|_{\text{op}} \leq \alpha$ one has*

$$\mathcal{U}_{\alpha,k}(G) = -\nabla \hat{D}_{\alpha,k}^*(G) \in \underset{V}{\operatorname{argmin}} \{ \langle G, V \rangle_F + D_{\alpha,k}(V) \}.$$

Proof. Let $h_0(s) = s/\alpha$ and $h_{j+1}(s) = (3h_j(s) - h_j(s)^3)/2$ for $0 \leq s \leq \alpha$. Induction gives $0 \leq h_j \leq 1$ and $h'_j \geq 0$ there, with $h_j(\alpha) = 1$. The singular-value response is h_k , so $D_{\alpha,k}^*(G) = \sum_i \int_0^{\sigma_i(G)} h_k(s) ds$. Extend h_k by 1 for $s \geq \alpha$. Its primitive is convex and nondecreasing; denote the resulting even convex spectral function by $\hat{D}_{\alpha,k}^*$. Its gradient agrees with $-\mathcal{U}_{\alpha,k}$ in the stable region. Moreover, $p_{\alpha,k}(s) = h_k(\sqrt{s})/\sqrt{s} \geq 0$ there (with the continuous value at $s = 0$), which proves Loewner monotonicity by integration along ordered matrix segments. For $\|G\|_{\text{op}} \leq \alpha$, Fenchel duality gives $\mathcal{U}_{\alpha,k}(G) \in \arg \min_V \{ \langle G, V \rangle_F + D_{\alpha,k}(V) \}$. $\square$

For one step, the singular-value derivative of the dual dissipation is $h_1(s) = 3s/(2\alpha) - s^3/(2\alpha^3)$, with $h'_1(s) = 3(1 - s^2/\alpha^2)/(2\alpha) \geq 0$ on the stable region. This illustrates why convexity of the dissipation, rather than convexity of Ψ as a function of the covariance, is the relevant condition. The same spectral construction works for fields $g \in L^2(\mu; \mathbb{R}^d)$, using $S_g = \int g g^\top d\mu$ and the Hilbert-space operator T_g from Proposition 2.4. It gives a convex dissipation and an energy-decreasing generalized gradient system while $S_{g_{\mu_t}} \preceq \alpha^2 I$.

What does not follow is a gradient flow for a fixed W_γ metric: finite-step fixed-scale maps are not one-homogeneous in the force, whereas the duality map of any squared norm is. A dynamic transport *action* can be defined using the conjugate dissipation, but the squared-distance BB theorem in this paper does not apply to it. Monotonicity alone does not fix this homogeneity obstruction. A priori control keeping the evolving force covariance in the stable region is an additional requirement. Adaptive rescaling and the nonclassical polynomial coefficients used in practical implementations require their own analysis.

Taking into account momentum. A natural momentum extension, in the spirit of Muon with exponential moving average, is obtained by introducing an auxiliary matrix variable A_t and considering

$$\tau \dot{A}_t + A_t = \nabla F(X_t), \quad \dot{X}_t = J_\gamma(A_t), \quad (7)$$

where $\tau \geq 0$ is the memory time scale. The variable A_t plays the role of a continuous-time exponentially averaged gradient. When $\tau = 0$, one recovers exactly the first-order spectral flow studied above.

For finite Schatten geometries ($1 \leq p < \infty$), this system is equivalent to the second-order inertial equation

$$\tau \frac{d}{dt} (\mathbb{I}_p(\dot{X}_t)) + \mathbb{I}_p(\dot{X}_t) + \nabla F(X_t) = 0,$$

where

$$\mathbb{I}_p(V) := \|V\|_{S_r}^{2-r} U \operatorname{diag}(\sigma_i^{r-1}) W^\top, \quad r = 2p,$$

for $V = U \operatorname{diag}(\sigma_i) W^\top$. In particular, for $p = 1$ one has $\mathbb{I}_1(V) = V$ and one recovers the heavy-ball equation

$$\tau \ddot{X}_t + \dot{X}_t + \nabla F(X_t) = 0.$$

We set $\mathbb{I}_p(0) = 0$. At the endpoint $p = \infty$, (7) reads

$$\tau \dot{A}_t + A_t = \nabla F(X_t), \quad \dot{X}_t = -\|A_t\|_{S_1} U_t W_t^\top,$$

which is the exact-polar continuous-time Muon dynamics with momentum.

The corresponding mean-field description naturally lives on phase space. Let

$$\eta_t \in \mathcal{P}_2(\mathbb{R}^d \times \mathbb{R}^d)$$

be the law of pairs (x, a) , where x is the particle position and a is the exponentially averaged force. Its spatial marginal is

$$\mu_t = (\pi_x)_\# \eta_t.$$

For a functional $\mathcal{F}$ on measures, write its Wasserstein force as

$$g_\mu(x) = \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}(x).$$

Given a phase-space law η , define its normalized spatial velocity by the phase-space Measure-LMO

$$u_\eta \in \operatorname{argmin}_{u \in L^2(\eta; \mathbb{R}^d)} \left\{ \int a \cdot u(x, a) d\eta(x, a) + \frac{1}{2} \gamma \left(\int uu^\top d\eta \right) \right\}.$$

The operator proof of Proposition 2.4 applies verbatim with phase space as the underlying probability space and computes u_η from the momentum covariance $S(\eta) = \int aa^\top d\eta$. In particular, this definition remains valid for singular selectors such as $p = \infty$. The phase-space characteristics are

$$\dot{x}_t = u_{\eta_t}(x_t, a_t), \quad \tau \dot{a}_t = g_{\mu_t}(x_t) - a_t. \quad (8)$$

Consequently, the law η_t solves the Liouville-type equation

$$\partial_t \eta_t + \operatorname{div}_x(\eta_t u_t) + \frac{1}{\tau} \operatorname{div}_a(\eta_t(g_{\mu_t}(x) - a)) = 0, \quad (9)$$

where

$$u_t(x, a) = u_{\eta_t}(x, a), \quad \mu_t = (\pi_x)_\# \eta_t.$$

When the inverse-trace problem in Proposition A.7 is attained at an invertible Q_η^* , this constitutive relation reduces to the closed form $u_\eta(x, a) = -(Q_\eta^*)^{-1}a$. When $\tau = 0$, the averaged force instantaneously relaxes to $a = g_{\mu_t}(x)$, and the spatial marginal formally recovers the first-order normalized PDE (3). For $\tau > 0$, the dynamics is no longer a metric gradient flow for W_γ on the position marginal alone; it is a damped kinetic system in which the spectral geometry enters through the constitutive relation between the phase-space force covariance and the spatial velocity.

Stochastic minibatches. An unbiased force estimate need not give an unbiased spectral velocity: $\mathbb{E}[\mathcal{J}_\gamma^\mu(g_\mu + \xi)]$ generally differs from $\mathcal{J}_\gamma^\mu(g_\mu)$ even if $\mathbb{E}\xi = 0$. Bias after nonlinear normalization can prevent convergence; sign-based counterexamples are given by Karimireddy et al. [2019]. These examples do not establish failure for every stochastic Muon variant, but rule out transferring unbiased-gradient arguments without further analysis. Under a fast-resampling closure with fixed-variance random force fields, the candidate transport equation is

$$\partial_t \mu_t + \operatorname{div} v(\mu_t \mathbb{E}_\xi[\mathcal{J}_\gamma^{\mu_t}(g_{\mu_t} + \xi)]) = 0.$$

The covariance inside each LMO is that of the entire random force field; shared minibatch noise is not independent across particles. There is no automatic diffusion term in this scaling. Its gradient-flow structure is open in general.

A different toy model adds independent Brownian noise *after* selecting the deterministic velocity. Its equation is $\partial_t \mu + \operatorname{div}(\mu \mathcal{J}_\gamma^\mu(g_\mu)) = \varepsilon \Delta \mu$. For $\gamma = \operatorname{tr}$ this is the W_2 flow of $\mathcal{F} + \varepsilon \operatorname{Ent}$, where $\operatorname{Ent}(\mu) = \int \rho \log \rho dx$ for $d\mu = \rho dx$. For a general spectral gauge, however, $\mathcal{J}_\gamma^\mu(g_\mu + \varepsilon \nabla \log \rho)$ is not in general $\mathcal{J}_\gamma^\mu(g_\mu) - \varepsilon \nabla \log \rho$. Thus this same entropy-penalization interpretation does not carry over.

Block-wise normalization. When training multilayer perceptrons, Muon is often applied independently to the outer weights u and inner weights v . In the mean-field notation $x = (u, v)$, this corresponds to normalizing the two blocks separately rather than applying one normalization to the concatenated parameter. This is still covered by our framework through a block gauge: for $S = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \succeq 0$, define

$$\gamma_{\text{blk}}(S) := \gamma_1(S_{11}) + \gamma_2(S_{22}),$$

where γ_1 and γ_2 are monotone gauges on the two positive semidefinite block cones. This gauge is convex, positively homogeneous, and positive on every nonzero PSD matrix. It is monotone because $0 \preceq S \preceq T$ implies $S_{ii} \preceq T_{ii}$ for each principal block. Proposition A.1 therefore supplies a norm extension to the whole symmetric space. Furthermore, writing $v = (v_1, v_2)$ gives $\mathcal{N}_\mu(v)^2 =$

$\gamma_1(\int v_1 v_1^\top d\mu) + \gamma_2(\int v_2 v_2^\top d\mu)$; the LMO objective separates into two independent minimizations. Thus the static and dynamic theory and the conditional gradient-flow theorem apply verbatim to independent block normalization, and to any finite number of blocks. This does not separate the underlying joint measure into independent marginals: the static problem still optimizes over a common coupling. The chosen particle/block layout must match the parameterization whose updates are being modeled.

B.1 Time Scales and Discretization

The choice of gauge fixes a time scale. Replacing γ by $a\gamma$ multiplies W_γ by $\sqrt{a}$ and divides the selected velocity by a . Thus an arbitrary acceleration of a continuous-time curve is not itself an optimization gain. Practical iteration counts additionally depend on smoothness in the chosen geometry and on the discretization.

For a smooth matrix objective on a convex region Ω , a precise Lipschitz-gradient constant is

$$L_\gamma := \sup_{X \neq Y \in \Omega} \frac{N^*(\nabla F(X) - \nabla F(Y))}{N(X - Y)}. \quad (10)$$

If it is finite, integration along a segment gives the descent lemma $F(X + V) \leq F(X) + \langle \nabla F(X), V \rangle_F + \frac{1}{2} L_\gamma N(V)^2$. For $V = \tau J_\gamma(\nabla F(X))$, this yields

$$F(X + V) \leq F(X) - \tau \left(1 - \frac{L_\gamma \tau}{2}\right) N^*(\nabla F(X))^2,$$

provided the segment remains in Ω . Hence $0 < \tau < 2/L_\gamma$ is a sufficient descent condition, not a universal numerical accuracy or optimal stability bound. A one-sided quadratic upper model is enough for this conclusion; it need not imply the two-sided gradient bound (10).

For the unscaled trace and operator gauges on $\mathbb{R}^{n \times d}$, let $r_n = \min(n, d)$. Since $\|V\|_{\text{op}} \leq \|V\|_F \leq \sqrt{r_n} \|V\|_{\text{op}}$ and $\|G\|_F \leq \|G\|_* \leq \sqrt{r_n} \|G\|_F$, one obtains on the same region

$$L_1 \leq L_\infty \leq r_n L_1.$$

The operator smoothness constant cannot be smaller under this common normalization. This worst-case comparison does not order convergence in iterations: the operator update also has the larger dissipation $\|G\|_*^2$, and the directions and states visited differ. For fixed particle dimension d , the factor remains bounded by d as $n \rightarrow \infty$; there is no dimension-independent bound when d also grows. Existence of a finite smoothness constant, for either geometry, is a separate property of the objective. Neither ReLU losses nor arbitrary smooth neural-network losses have a global such bound automatically.

A transport analogue, avoiding comparison of covectors at different measures, is the upper estimate

$$\mathcal{F}(\nu) \leq \mathcal{F}(\mu) + \int g_\mu(x) \cdot (y - x) d\pi + \frac{L_\gamma}{2} \gamma \left(\int (y - x)(y - x)^\top d\pi \right)$$

for the couplings and states under consideration. Applied to $\nu = (\text{Id} + \tau v)_\# \mu$ with $v \in \mathcal{J}_\gamma^\mu(g_\mu)$, it gives the same descent bound with $\mathcal{N}_\mu^*(g_\mu)^2$ in place of the matrix dual norm. Establishing this estimate along a neural-network flow, and controlling discretization or particle approximation errors, are beyond the present analysis. In particular, a Lipschitz bound on the energy differential must not be confused with a Lipschitz bound on the Muon velocity map, which can be discontinuous at rank changes.

C Monge Maps and Conditional Brenier Theorem

This appendix collects the map-based statements associated with the static formulation of Section 3.

The next definition records the Monge restriction, which is useful for comparison but is not the correct notion in general.

Definition C.1 (Monge restriction). *Define*

$$W_\gamma^M(\mu, \nu)^2 := \inf_{T_\# \mu = \nu} \gamma \left(\int_{\mathbb{R}^d} (T(x) - x)(T(x) - x)^\top d\mu(x) \right),$$

with value $+\infty$ if no transport map exists.

The next proposition simply records that the map-based problem is a restriction of the coupling-based one.

Proposition C.2 (Monge is a restriction). *For every $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$W_\gamma(\mu, \nu) \leq W_\gamma^M(\mu, \nu).$$

Proof. Every transport map T induces the coupling $(\text{Id}, T)_\# \mu$, and the static cost evaluated on this coupling is exactly the Monge cost associated with T . $\square$

The following remark shows on a two-Dirac example that the relaxation can be genuinely strict.

Remark C.3 (Strictness of the Monge restriction). *Consider*

$$\mu = \frac{1}{2}(\delta_{(-1,0)} + \delta_{(1,0)}), \quad \nu = \frac{1}{2}(\delta_{(0,-1)} + \delta_{(0,1)}).$$

Any transport map from μ to ν is a bijection between the two atoms. The two possible displacement covariance matrices are

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

so for the operator norm and Frobenius norm one gets

$$W_\gamma^M(\mu, \nu)^2 = 2.$$

By contrast, the split coupling assigning mass 1/4 to each source-target pair has displacement covariance equal to the identity matrix, hence

$$W_\gamma(\mu, \nu)^2 = 1 \quad \text{for } \gamma = \|\cdot\|_{S_\infty}, \quad W_\gamma(\mu, \nu)^2 = \sqrt{2} \quad \text{for } \gamma = \|\cdot\|_{S_2}.$$

Therefore the inequality in Proposition C.2 is strict already for two-point measures.

The next remark explains that, for the restriction of an arbitrary matrix norm to $\mathbb{S}_+^d$, the static cost need not be a bona fide metric.

Remark C.4 (Triangle inequality may fail for non-monotone norms). *For Dirac masses, the unique coupling gives*

$$W_\gamma(\delta_x, \delta_y)^2 = \gamma((y-x)(y-x)^\top).$$

Hence if W_γ were always a distance, then the pointwise cost

$$d_\gamma(x, y) := \sqrt{\gamma((y-x)(y-x)^\top)}$$

would in particular have to define a metric on $\mathbb{R}^d$.

This fails for general norms on $\mathbb{S}_+^d$. Fix $d \geq 2$ and a parameter $M > 2$, and define

$$\gamma_{\text{ns}}(S) := \text{tr}(S) + M \sum_{1 \leq i < j \leq d} |S_{ij}|, \quad S \succeq 0.$$

This is the restriction to $\mathbb{S}_+^d$ of the weighted entrywise norm $A \mapsto \sum_i |A_{ii}| + M \sum_{i < j} |A_{ij}|$ on $\mathbb{S}^d$. It is neither spectral nor monotone. Indeed, in dimension two,

$$S = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \preceq T = 2I, \quad \gamma_{\text{ns}}(S) = 2 + M > 4 = \gamma_{\text{ns}}(T).$$

For the displacement

$$\Delta = y - x = (\Delta_1, \dots, \Delta_d)$$

one has

$$d_{\gamma_{\text{ns}}}(x, y)^2 = \|\Delta\|_2^2 + M \sum_{1 \leq i < j \leq d} |\Delta_i \Delta_j|.$$

Taking

$$x_0 = 0, \quad x_1 = e_1, \quad x_2 = e_1 + e_2$$

gives

$$d_{\gamma_{\text{ns}}}(x_0, x_1) = 1, \quad d_{\gamma_{\text{ns}}}(x_1, x_2) = 1, \quad d_{\gamma_{\text{ns}}}(x_0, x_2) = \sqrt{2 + M} > 2,$$

so the triangle inequality fails.

By contrast, if γ is monotone on $\mathbb{S}_+^d$, then Proposition A.2 allows us to choose $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, and therefore

$$d_\gamma(x, y) = \sup_{Q \in \mathcal{K}_\gamma} \left\| Q^{1/2}(x - y) \right\|_2,$$

which is a supremum of seminorms and, by positivity of γ , a norm of $x - y$. For general measures, however, this pointwise argument does not control the cross terms created by gluing couplings. The full metric property of W_γ follows either from the robust static argument in Proposition A.8 or from the dynamic formulation in Corollary 4.3.

The following proposition explains when an optimal coupling is induced by a Monge map of Brenier type.

Proposition C.5 (Conditional Brenier theorem). *Assume μ is absolutely continuous. If a dual maximizing matrix $Q_* \in \mathcal{K}_\gamma$ in Theorem 3.4 is positive definite, then every optimal coupling for $W_\gamma(\mu, \nu)$ is induced by a map. More precisely, there exists a convex function u such that*

$$T_*(x) = Q_*^{-1/2} \nabla u(Q_*^{1/2} x)$$

is optimal.

Without monotonicity, a polar certificate need not be positive semidefinite and may even be indefinite. Under the standing monotonicity assumption, however, Proposition A.2 allows one to choose the representing set $\mathcal{K}_\gamma$ inside $\mathbb{S}_+^d$; the active matrix is then automatically positive semidefinite, and the remaining nondegeneracy assumption is simply that it be invertible. For the trace gauge $p = 1$, one may use the reduced set $\mathcal{K}_\gamma = \{I\}$, so the condition is automatic. For $1 < p < \infty$, the canonical active matrix is $\Sigma_{\pi^*}^{p-1} / \|\Sigma_{\pi^*}\|_{S_p}^{p-1}$ and is therefore positive definite whenever the optimal displacement covariance is. The endpoint $p = \infty$ is sharply different: in that case $\gamma(\Sigma) = \lambda_{\max}(\Sigma)$ and Q_* is supported on the top eigenspace of Σ_{π^*} . When the leading eigenvalue is simple, Q_* is rank one. The associated degenerate quadratic cost does not control directions orthogonal to the active eigenspace, so Brenier's uniqueness argument no longer applies. Monge optimizers may still exist, but the dual certificate alone neither selects one nor implies that every optimizer is map-induced.

Proof. Fix such a positive definite dual maximizer Q_* and let π_* be any primal optimizer. With $F(\pi, Q) = \int (y - x)^\top Q (y - x) d\pi$, minimax equality gives

$$F(\pi_*, Q_*) \leq \max_{Q \in \mathcal{K}_\gamma} F(\pi_*, Q) = W_\gamma(\mu, \nu)^2 = \min_{\pi \in \Pi(\mu, \nu)} F(\pi, Q_*) \leq F(\pi_*, Q_*).$$

Thus π_* is optimal for the quadratic cost induced by Q_* . To identify this optimizer, let

$$\tilde{\pi}_* = (Q_*^{1/2}, Q_*^{1/2})_\# \pi_*,$$

and define

$$\tilde{\mu} = (Q_*^{1/2})_\# \mu, \quad \tilde{\nu} = (Q_*^{1/2})_\# \nu.$$

Then

$$\int (y - x)^\top Q_* (y - x) d\pi_*(x, y) = \int |\tilde{y} - \tilde{x}|^2 d\tilde{\pi}_*(\tilde{x}, \tilde{y}).$$

Because μ is absolutely continuous and $Q_*^{1/2}$ is invertible, $\tilde{\mu}$ is also absolutely continuous. Brenier's theorem therefore yields a convex potential u such that the unique optimal coupling between $\tilde{\mu}$ and $\tilde{\nu}$ is induced by the map

$$\tilde{T}(\tilde{x}) = \nabla u(\tilde{x}).$$

Pulling this map back to the original variables gives

$$T_*(x) = Q_*^{-1/2} \tilde{T}(Q_*^{1/2} x) = Q_*^{-1/2} \nabla u(Q_*^{1/2} x).$$

Since π_* is Q_* -optimal and the latter coupling is unique, π_* must be induced by T_* . The argument applies to every primal optimizer. $\square$

D Gaussian Reductions

This appendix gathers the Gaussian reductions used in the static geometry and in the linear-network flow. Gaussian marginals compress the transport problem to a finite-dimensional optimization over admissible covariance blocks. The next proposition shows that, for Gaussian marginals, the infinite-dimensional transport problem collapses to an optimization over the cross-covariance matrix.

Proposition D.1 (Gaussian reduction). *Let $\mu = \mathcal{N}(m_0, \Sigma_0)$ and $\nu = \mathcal{N}(m_1, \Sigma_1)$. Then*

$$W_\gamma(\mu, \nu)^2 = \inf_K \gamma((m_1 - m_0)(m_1 - m_0)^\top + \Sigma_0 + \Sigma_1 - K - K^\top),$$

where the infimum runs over matrices K such that

$$\begin{pmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{pmatrix} \succeq 0.$$

In particular, for centered Gaussians the covariance cost

$$B_\gamma(\Sigma_0, \Sigma_1)^2 := \inf_{K: \begin{bmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{bmatrix} \succeq 0} \gamma(\Sigma_0 + \Sigma_1 - K - K^\top)$$

is well defined on the cone of covariance matrices. In the monotone regime of Section 4, it is the restriction of the metric W_γ to centered Gaussian laws and therefore defines a metric on covariance matrices.

Proof. Let (X, Y) be any coupling of the two Gaussian marginals and set $K = \mathbb{E}[(X - m_0)(Y - m_1)^\top]$. Its block covariance is

$$\begin{pmatrix} \Sigma_0 & K \\ K^\top & \Sigma_1 \end{pmatrix}.$$

It must be positive semidefinite. Conversely, every feasible block matrix is the covariance of a jointly Gaussian vector with marginals μ and ν . For any coupling with cross-covariance K , the displacement covariance is

$$(m_1 - m_0)(m_1 - m_0)^\top + \Sigma_0 + \Sigma_1 - K - K^\top.$$

The cost therefore depends only on K , and every feasible K is realized by a Gaussian coupling. Optimizing over all couplings is thus equivalent to the stated finite-dimensional problem. $\square$

The next corollary shows that commuting covariances lead to a closed form for the Schatten family, exactly as in the classical Bures case.

Corollary D.2 (Commuting covariances). *If Σ_0 and Σ_1 commute, then for Schatten norms γ_p one has*

$$B_{\gamma_p}(\Sigma_0, \Sigma_1)^2 = \gamma_p((\Sigma_0^{1/2} - \Sigma_1^{1/2})^2).$$

Equivalently, if a_i and b_i are the diagonal entries of Σ_0 and Σ_1 in a common eigenbasis, then, for $p < \infty$,

$$B_{\gamma_p}(\Sigma_0, \Sigma_1)^2 = \left(\sum_{i=1}^d |\sqrt{a_i} - \sqrt{b_i}|^{2p} \right)^{1/p}.$$

For $p = \infty$, the right-hand side is $\max_i |\sqrt{a_i} - \sqrt{b_i}|^2$. When $p = 1$, this is the usual Bures–Wasserstein formula.

Proof. Simultaneously diagonalize $\Sigma_0 = \text{diag}(a_i)$ and $\Sigma_1 = \text{diag}(b_i)$. For any feasible K , each 2×2 principal minor involving the coordinates X_i and Y_i gives $|K_{ii}| \leq \sqrt{a_i b_i}$. Hence the displacement covariance $D = \Sigma_0 + \Sigma_1 - K - K^\top \succeq 0$ satisfies

$$D_{ii} = a_i + b_i - 2K_{ii} \geq (\sqrt{a_i} - \sqrt{b_i})^2.$$

Pinching onto the common eigenbasis is contractive for every unitarily invariant norm, so

$$\|D\|_{S_p} \geq \|\text{diag}(D_{ii})\|_{S_p} \geq \left\| \text{diag}((\sqrt{a_i} - \sqrt{b_i})^2) \right\|_{S_p}.$$

The diagonal choice $K_0 = \text{diag}(\sqrt{a_i b_i})$ is feasible and attains both inequalities, proving the claim. $\square$

The following remark clarifies the scope of Corollary D.2.

Remark D.3. *The commuting formula is stated for Schatten norms because their value depends only on the eigenvalues of the displacement covariance. For more general norms on $\mathbb{S}_+^d$, even when Σ_0 and Σ_1 commute, one should not expect a closed form depending only on the eigenvalues $(\sqrt{a_i} - \sqrt{b_i})^2$, since the norm itself may retain basis-dependent information.*

E Dynamic Formulation Details

This appendix gathers the convex momentum formulation and the proofs deferred from Section 4.

E.1 Convex momentum reformulation

Passing to momenta linearizes the constraint. If $\mu = \rho\lambda$ and $m = w\lambda$ for some reference measure λ , define

$$\mathcal{A}_\gamma(\mu, m) := \sup_{Q \in \mathcal{K}_\gamma} \int_{\mathbb{R}^d} \frac{w(x)^\top Q w(x)}{\rho(x)} d\lambda(x),$$

with the usual perspective convention on $\{\rho = 0\}$.

The following proposition shows that the momentum formulation is convex and exactly matches the velocity action.

Proposition E.1 (Convex momentum action). *The action $\mathcal{A}_\gamma(\mu, m)$ is intrinsic, convex, and weak-* lower semicontinuous. If $m = \mu v$, then*

$$\mathcal{A}_\gamma(\mu, m) = \gamma \left(\int v(x) v(x)^\top d\mu(x) \right).$$

Hence

$$W_\gamma^{\text{BB}}(\mu_0, \mu_1)^2 = \inf_{(\mu_t, m_t)} \int_0^1 \mathcal{A}_\gamma(\mu_t, m_t) dt$$

under the linear constraint

$$\partial_t \mu_t + \text{div}(m_t) = 0.$$

Proof. Let λ be any positive measure dominating both μ and m , and write $\mu = \rho\lambda$ and $m = w\lambda$, with $\rho \geq 0$ and $w : \mathbb{R}^d \rightarrow \mathbb{R}^d$. For $Q \succeq 0$, define the perspective quadratic integrand

$$\phi_Q(\rho, w) := \begin{cases} \frac{w^\top Q w}{\rho}, & \rho > 0, \\ 0, & \rho = 0 \text{ and } Q^{1/2} w = 0, \\ +\infty, & \rho = 0 \text{ and } Q^{1/2} w \neq 0. \end{cases}$$

Then ϕ_Q is convex and lower semicontinuous on $\mathbb{R}_+ \times \mathbb{R}^d$. Indeed, for $\rho > 0$ it is the perspective of the convex quadratic form $w \mapsto w^\top Q w$, and the above convention is its closed extension at $\rho = 0$. Notice that kernel directions have zero recession cost when Q is singular.

The momentum action may be written as

$$\mathcal{A}_\gamma(\mu, m) = \sup_{Q \in \mathcal{K}_\gamma} \int \phi_Q(\rho(x), w(x)) d\lambda(x).$$

This formula is intrinsic, i.e., independent of the dominating measure λ . To see this, choose another dominating measure λ' and write $\mu = \rho'\lambda'$ and $m = w'\lambda'$. Passing to a common dominating measure, for instance $\eta = \lambda + \lambda'$, and using the one-homogeneity of ϕ_Q , namely $\phi_Q(a\rho, aw) = a\phi_Q(\rho, w)$ for $a \geq 0$, gives the same value of $\int \phi_Q d\lambda$ and $\int \phi_Q d\lambda'$. Thus $\mathcal{A}_\gamma(\mu, m)$ depends only on the pair (μ, m) .

We next prove the absolutely continuous representation. If $m = \mu v$, then one may take $\lambda = \mu$, so that $\rho = 1$ and $w = v$. Hence, for every $Q \in \mathcal{K}_\gamma$,

$$\int \phi_Q(1, v(x)) d\mu(x) = \int v(x)^\top Q v(x) d\mu(x) = \text{tr} \left(Q \int v(x) v(x)^\top d\mu(x) \right).$$

Taking the supremum over $Q \in \mathcal{K}_\gamma$ and using the support representation of γ yields

$$\mathcal{A}_\gamma(\mu, \mu v) = \sup_{Q \in \mathcal{K}_\gamma} \operatorname{tr} \left(Q \int v v^\top d\mu \right) = \gamma \left(\int v v^\top d\mu \right).$$

Conversely, strict positivity of γ implies that $\mathcal{K}_\gamma$ contains a positive definite matrix. Indeed, for every unit vector a there is a $Q_a \in \mathcal{K}_\gamma$ with $a^\top Q_a a > 0$; compactness of the sphere gives finitely many such matrices whose average is positive definite and belongs to $\mathcal{K}_\gamma$ by convexity. Fix one such matrix Q_0 . If $\mathcal{A}_\gamma(\mu, m) < +\infty$, then $\int \phi_{Q_0}(\rho, w) d\lambda$ is finite. Since $Q_0 \succ 0$, the singular case $\rho = 0$ and $w \neq 0$ cannot occur on a set of positive λ -measure. Thus m is absolutely continuous with respect to μ , and $m = \mu v$ with $v = w/\rho$ on $\{\rho > 0\}$.

Convexity follows from the same perspective formulation. For fixed $Q \succeq 0$, the map $(\mu, m) \mapsto \int \phi_Q(\rho, w) d\lambda$ is convex, because ϕ_Q is convex and one-homogeneous. Taking the supremum over $Q \in \mathcal{K}_\gamma$ preserves convexity, hence $(\mu, m) \mapsto \mathcal{A}_\gamma(\mu, m)$ is convex.

Finally, we prove weak-* lower semicontinuity. For each fixed $Q \succeq 0$, the functional

$$(\mu, m) \mapsto \int \phi_Q \left(\frac{d\mu}{d\lambda}, \frac{dm}{d\lambda} \right) d\lambda$$

is lower semicontinuous under weak-* convergence of measures, by the standard lower semicontinuity theorem for convex, lower semicontinuous, one-homogeneous integral functionals. Since $\mathcal{A}_\gamma$ is the supremum over $Q \in \mathcal{K}_\gamma$ of these lower semicontinuous functionals, $\mathcal{A}_\gamma$ is also weak-* lower semicontinuous.

The dynamic formula now follows by rewriting the velocity formulation in momentum variables. If (μ_t, v_t) satisfies $\partial_t \mu_t + \operatorname{div}(\mu_t v_t) = 0$ and $m_t = \mu_t v_t$, then the constraint becomes the linear continuity equation $\partial_t \mu_t + \operatorname{div}(m_t) = 0$, and the previous identity gives

$$\mathcal{N}_{\mu_t}(v_t)^2 = \gamma \left(\int v_t v_t^\top d\mu_t \right) = \mathcal{A}_\gamma(\mu_t, m_t).$$

Therefore the dynamic action $\int_0^1 \mathcal{N}_{\mu_t}(v_t)^2 dt$ is exactly $\int_0^1 \mathcal{A}_\gamma(\mu_t, m_t) dt$ in momentum variables. Taking the infimum over all admissible curves gives

$$\mathcal{W}_\gamma^{\text{BB}}(\mu_0, \mu_1)^2 = \inf_{(\mu_t, m_t)} \int_0^1 \mathcal{A}_\gamma(\mu_t, m_t) dt,$$

under the linear constraint $\partial_t \mu_t + \operatorname{div}(m_t) = 0$. $\square$

E.2 Proof of Theorem 4.2 (static/dynamic equivalence)

Proof. Let $\pi \in \Pi(\mu_0, \mu_1)$ and consider the displacement interpolation

$$z_t(x, y) = (1-t)x + ty, \quad \mu_t = (z_t)_\# \pi.$$

Its Eulerian velocity is the conditional mean $v_t(z) = \mathbb{E}_\pi[y - x \mid z_t = z]$. The pair (μ_t, v_t) solves the continuity equation, and conditional Jensen's inequality gives

$$\int v_t v_t^\top d\mu_t \preceq \int (y - x)(y - x)^\top d\pi$$

for a.e. t . By monotonicity of γ , its dynamic action is no larger than the static cost of π . Taking the infimum over π gives

$$\mathcal{W}_\gamma^{\text{BB}}(\mu_0, \mu_1)^2 \leq \mathcal{W}_\gamma(\mu_0, \mu_1)^2.$$

Conversely, take any admissible dynamic plan (μ_t, v_t) . By the superposition principle, there exists a probability measure η on absolutely continuous paths ω such that

$$\mu_t = (e_t)_\# \eta, \quad \dot{\omega}_t = v_t(\omega_t)$$

for η -a.e. path and a.e. t . Let $\pi = (e_0, e_1)_\# \eta$. For every $Q \in \mathcal{K}_\gamma$,

$$\int_0^1 \int v_t^\top Q v_t d\mu_t dt = \int \int_0^1 \dot{\omega}_t^\top Q \dot{\omega}_t dt d\eta(\omega).$$

This is the key point where monotonicity is used: by Proposition A.2 we have chosen $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$, so every test matrix satisfies $Q \succeq 0$. Without this reduction one would have to work with possibly indefinite matrices, and the quadratic Jensen inequality below would no longer apply. Applying Jensen to $t \mapsto Q^{1/2} \dot{\omega}_t$ gives

$$\int_0^1 \dot{\omega}_t^\top Q \dot{\omega}_t dt \geq (\omega_1 - \omega_0)^\top Q (\omega_1 - \omega_0).$$

Hence

$$\int_0^1 \gamma \left(\int v_t v_t^\top d\mu_t \right) dt \geq \int (y - x)^\top Q (y - x) d\pi(x, y)$$

for every $Q \in \mathcal{K}_\gamma$. Taking the supremum over Q yields

$$\int_0^1 \gamma \left(\int v_t v_t^\top d\mu_t \right) dt \geq \gamma \left(\int (y - x)(y - x)^\top d\pi \right).$$

Finally, take the infimum over admissible dynamic plans. $\square$

E.3 Proof of Corollary 4.4 (Spectral Geodesics)

Proof. The upper bound follows by restricting the same displacement interpolation to the time interval $[s, t]$. The coupling $((1-s)x + sy, (1-t)x + ty)_{\# \pi_*}$ has displacement $(t-s)(y-x)$, so the static definition gives

$$W_\gamma(\mu_s, \mu_t)^2 \leq (t-s)^2 \gamma \left(\int (y-x)(y-x)^\top d\pi_*(x, y) \right) = (t-s)^2 W_\gamma(\mu_0, \mu_1)^2.$$

Applying the same argument to the segments $[0, s]$ and $[t, 1]$ yields

$$W_\gamma(\mu_0, \mu_s) \leq s W_\gamma(\mu_0, \mu_1), \quad W_\gamma(\mu_t, \mu_1) \leq (1-t) W_\gamma(\mu_0, \mu_1).$$

By the triangle inequality from Corollary 4.3,

$$W_\gamma(\mu_0, \mu_1) \leq W_\gamma(\mu_0, \mu_s) + W_\gamma(\mu_s, \mu_t) + W_\gamma(\mu_t, \mu_1) \leq W_\gamma(\mu_s, \mu_t) + (1-(t-s)) W_\gamma(\mu_0, \mu_1).$$

Rearranging gives

$$W_\gamma(\mu_s, \mu_t) \geq (t-s) W_\gamma(\mu_0, \mu_1),$$

hence equality holds. $\square$

E.4 Root-norm gauges beyond convexity

The main text assumes that γ is convex, monotone, and positively one-homogeneous. Convexity is essential for the support-function representation $\gamma(S) = \max_Q \text{tr}(QS)$ and hence for the cost-robust formulation. Some parts of the theory, however, only use that the square-root action is a norm.

Definition E.2 (Root-norm gauge). *A positive one-homogeneous function $\gamma : \mathbb{S}_+^d \rightarrow \mathbb{R}_+$ is a root-norm gauge if, for every Hilbert space $\mathcal{H}$, the map*

$$T : \mathbb{R}^d \rightarrow \mathcal{H} \quad \mapsto \quad \mathfrak{N}_\gamma(T) := \gamma(T^*T)^{1/2}$$

is a norm on finite-rank operators.

For a vector field $v \in L^2(\mu; \mathbb{R}^d)$, this gives exactly the tangent norm used in the paper:

$$\mathcal{N}_\mu(v) = \mathfrak{N}_\gamma(T_v), \quad (T_v a)(x) = a \cdot v(x),$$

since $T_v^* T_v = \int v v^\top d\mu$.

Proposition E.3 (Characterization). *A positive one-homogeneous γ is a root-norm gauge if and only if it can be written as*

$$\gamma(S) = \mathfrak{N}(S^{1/2})^2, \quad S \succeq 0,$$

where $\mathfrak{N}$ is a compatible family of norms on finite-rank operators $\mathbb{R}^d \rightarrow \mathcal{H}$, indexed by the target Hilbert space $\mathcal{H}$ and invariant under isometric embeddings of the target. In the spectral case, this is equivalent to

$$\gamma(S) = \varphi \left(\lambda(S)^{1/2} \right)^2$$

for some symmetric gauge norm φ on $\mathbb{R}^d$. Every root-norm gauge is monotone on $\mathbb{S}_+^d$, but it need not be convex.

Proof. If γ is root-norm, take $\mathfrak{N}_{\mathcal{H}} = \mathfrak{N}_{\gamma}$ on each target $\mathcal{H}$; this family is compatible because its value depends only on T^*T . Conversely, such a compatible family gives $\mathfrak{N}_{\gamma}(T) = \mathfrak{N}(T)$ because $\gamma(T^*T) = \mathfrak{N}((T^*T)^{1/2})^2$ and $(T^*T)^{1/2}$ is obtained from T by an isometry on the target in the polar decomposition.

We now prove monotonicity. If $0 \preceq S \preceq T$, Douglas' factorization lemma gives $S^{1/2} = CT^{1/2}$ for some contraction C . Since a contraction is in the closed convex hull of isometries, invariance under isometries and the triangle inequality give

$$\mathfrak{N}(S^{1/2}) = \mathfrak{N}(CT^{1/2}) \leq \mathfrak{N}(T^{1/2}).$$

Thus $\gamma(S) \leq \gamma(T)$. The spectral characterization is the usual description of unitarily invariant norms by symmetric gauge norms. The same convex-hull argument also shows the contraction property $\mathfrak{N}(BT) \leq \mathfrak{N}(T)$ whenever B is a contraction on the target space. $\square$

Corollary E.4 (Schatten threshold). *For $d \geq 2$ and $p \in (0, \infty]$, the function $\gamma_p(S) = \|S\|_{S_p}$ is a root-norm gauge if and only if $p \geq 1/2$. It is convex if and only if $p \geq 1$. In dimension one, all these functions equal $\gamma_p(s) = s$, so both properties hold for every $p > 0$.*

Proof. One has

$$\gamma_p(V^{\top}V)^{1/2} = \|V\|_{S_{2p}}.$$

When $d \geq 2$, this is a norm exactly when $2p \geq 1$. Convexity of the Schatten p -functional on the positive cone holds exactly for $p \geq 1$. When $d = 1$, there is only one singular value, so neither threshold applies. $\square$

For nonconvex root-norm gauges, the static Kantorovich formula still makes sense,

$$W_{\gamma}(\mu, \nu)^2 = \inf_{\pi \in \Pi(\mu, \nu)} \gamma \left(\int (y - x)(y - x)^{\top} d\pi(x, y) \right),$$

but it is no longer a convex optimization problem in the coupling. Moreover, there is no exact cost-robust formula of the form $\gamma(S) = \max_{Q \in \mathcal{K}_{\gamma}} \text{tr}(QS)$, since a maximum of linear forms is convex. Thus the direct robust-cost proof of the triangle inequality from Appendix A.3 is no longer available.

Theorem E.5 (Benamou–Brenier formula for root-norm gauges). *Let γ be a root-norm gauge. Then, for every $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$W_{\gamma}(\mu_0, \mu_1)^2 = \inf_{\partial_t \mu_t + \text{div}(\mu_t v_t) = 0} \int_0^1 \gamma \left(\int v_t v_t^{\top} d\mu_t \right) dt,$$

where the infimum is over narrowly continuous curves with endpoints μ_0, μ_1 and square-integrable velocity fields.

Proof. Let (μ_t, v_t) be admissible. By the superposition principle, there exists a probability measure η on absolutely continuous paths ω such that $\mu_t = (e_t)_{\#} \eta$ and $\dot{\omega}_t = v_t(\omega_t)$ for a.e. t . Set $\pi = (e_0, e_1)_{\#} \eta$ and $\Delta(\omega) = \omega_1 - \omega_0$. Then

$$\Delta = \int_0^1 v_t(\omega_t) dt.$$

At the operator level, $T_{\Delta} = \int_0^1 T_{v_t \circ e_t} dt$ as maps $\mathbb{R}^d \rightarrow L^2(\eta)$. Hence, by the triangle inequality for $\mathfrak{N}_{\gamma}$ and Cauchy–Schwarz,

$$\begin{aligned} \gamma \left(\int \Delta \Delta^{\top} d\eta \right)^{1/2} &\leq \int_0^1 \gamma \left(\int v_t v_t^{\top} d\mu_t \right)^{1/2} dt \\ &\leq \left(\int_0^1 \gamma \left(\int v_t v_t^{\top} d\mu_t \right) dt \right)^{1/2}. \end{aligned}$$

Since Δ depends only on the endpoints, this gives $W_{\gamma}(\mu_0, \mu_1)^2$ bounded above by the dynamic action.

Conversely, fix $\pi \in \Pi(\mu_0, \mu_1)$ and set $z_t = (1-t)x + ty$, $\mu_t = (z_t)_\# \pi$. Let $v_t(z) = \mathbb{E}_\pi[y - x \mid z_t = z]$. Then (μ_t, v_t) solves the continuity equation. The conditional-expectation operator $C_t : L^2(\pi) \rightarrow L^2(\mu_t)$ is a contraction and satisfies $T_{v_t} = C_t T_{y-x}$. Since $\mathfrak{N}_\gamma$ is contractive under target contractions,

$$\mathcal{N}_{\mu_t}(v_t) \leq \gamma \left(\int (y-x)(y-x)^\top d\pi \right)^{1/2}.$$

Integrating over t gives a dynamic action no larger than the static cost of π . Taking the infimum over π proves the reverse inequality. $\square$

Corollary E.6 (Metric property for root-norm gauges). *If γ is a root-norm gauge, then W_γ is a distance on $\mathcal{P}_2(\mathbb{R}^d)$ and induces the same topology as W_2 .*

Proof. Nonnegativity and symmetry are immediate. Since $\mathfrak{N}_\gamma$ is a norm on finite-dimensional operator spaces, it is equivalent to the Hilbert–Schmidt norm, so there are constants $0 < c \leq C < \infty$ such that

$$c \operatorname{tr}(S) \leq \gamma(S) \leq C \operatorname{tr}(S), \quad S \succeq 0.$$

Thus $\sqrt{c} W_2 \leq W_\gamma \leq \sqrt{C} W_2$, which gives separation and topological equivalence.

For the triangle inequality, one may either use Theorem E.5 and concatenate dynamic curves, or argue directly. Given couplings π_{01} and π_{12} , glue them into a measure on triples (x_0, x_1, x_2) . With $\Delta_{ij} = x_j - x_i$, one has $\Delta_{02} = \Delta_{01} + \Delta_{12}$, and the norm property gives

$$\mathfrak{N}_\gamma(T_{\Delta_{02}}) \leq \mathfrak{N}_\gamma(T_{\Delta_{01}}) + \mathfrak{N}_\gamma(T_{\Delta_{12}}).$$

Taking infima over the two input couplings proves $W_\gamma(\mu_0, \mu_2) \leq W_\gamma(\mu_0, \mu_1) + W_\gamma(\mu_1, \mu_2)$. $\square$

E.5 Proof of Theorem 5.1 (Spectral Gradient Flow)

Proof. Write $N = \mathcal{N}_\mu$ and $N^* = \mathcal{N}_\mu^*$ at a fixed state on the curve. First, norm duality gives, for every $v \in \mathcal{J}_\gamma^\mu(g)$,

$$N(v) = N^*(g), \quad \int g \cdot v d\mu = -N^*(g)^2. \quad (11)$$

Indeed, optimizing the radius along a unit-ball minimizing direction reduces the objective to $-rN^*(g) + r^2/2$.

We next prove the slope identity without assuming that μ has a density. Let π be an optimal spectral coupling from μ to ν , and let $w(x) = \int (y-x) d\pi(y \mid x)$ be its conditional mean displacement. Conditional Jensen and monotonicity give $N(w) \leq W_\gamma(\mu, \nu)$. Also, $\int |y-x|^2 d\pi \leq c_\gamma^{-1} W_\gamma(\mu, \nu)^2$ by Proposition 3.2. The transport expansion therefore implies

$$\mathcal{F}(\mu) - \mathcal{F}(\nu) \leq N^*(g) W_\gamma(\mu, \nu) + C c_\gamma^{-1} W_\gamma(\mu, \nu)^2,$$

and hence $|\partial \mathcal{F}|_\gamma(\mu) \leq N^*(g)$. Conversely, for $w \in L^2(\mu; \mathbb{R}^d)$ and $h > 0$, use the deterministic coupling to $\nu_h = (\operatorname{Id} + hw)_\# \mu$:

$$W_\gamma(\mu, \nu_h) \leq hN(w), \quad \mathcal{F}(\nu_h) - \mathcal{F}(\mu) = h \int g \cdot w d\mu + O(h^2 \|w\|_{L^2(\mu)}^2).$$

For $\int g \cdot w d\mu < 0$, division and passage to the limit give $|\partial \mathcal{F}|_\gamma(\mu) \geq -\int g \cdot w d\mu / N(w)$. Taking the supremum proves $|\partial \mathcal{F}|_\gamma(\mu) = N^*(g)$; the case $g = 0$ follows from the upper bound.

The assumed spatial Lipschitz and growth bounds provide a characteristic flow, a uniform second-moment bound, and

$$\frac{X_{t+h} - X_t}{h} \longrightarrow v_t(X_t) \quad \text{in } L^2(\mu_0; \mathbb{R}^d).$$

Apply (5) to the coupling $(X_t, X_{t+h})_\# \mu_0$. Its squared displacement is $O(h^2)$, so (11) yields

$$\frac{d}{dt} \mathcal{F}(\mu_t) = \int g_t \cdot v_t d\mu_t = -\mathcal{N}_{\mu_t}^*(g_t)^2.$$

The energy is locally Lipschitz: $N^*(g_t) = N(v_t)$ is uniformly bounded by the growth and moment bounds, as is $\|g_t\|_{L^2(\mu_t)}$ by norm equivalence.

The BB formula, rescaled from $[0, 1]$ to $[s, t]$, gives $W_\gamma(\mu_s, \mu_t)^2 \leq (t - s) \int_s^t N_{\mu_r}(v_r)^2 dr$. Thus the curve is absolutely continuous and its metric derivative exists almost everywhere, with $|\mu'|_\gamma(t) \leq N_{\mu_t}(v_t)$. At such a time, the definition of descending slope and the energy derivative give $N_{\mu_t}^*(g_t)^2 \leq |\partial \mathcal{F}|_\gamma(\mu_t) |\mu'|_\gamma(t)$. Together with the slope identity this proves equality of speed, slope, and dual norm whenever $g_t \neq 0$. If $g_t = 0$, the speed bound already gives zero. The stated energy dissipation identity follows. $\square$

A concrete smooth class. Let $\varphi_k \in C^2(\mathbb{R}^d)$ have bounded Hessian, and let $\mathcal{F}(\mu) = \Phi(m(\mu))$, where $m(\mu) = (\int \varphi_k d\mu)_{k=1}^m$ and Φ is C^2 near the moment vectors of the curve. The forces are $g_\mu = \sum_k \partial_k \Phi(m(\mu)) \nabla \varphi_k$. On a uniformly second-moment-bounded curve,

$$|m(\nu) - m(\mu)| \leq C_1 \left[\left(\int |y - x|^2 d\pi \right)^{1/2} + \int |y - x|^2 d\pi \right].$$

Taylor expansion of each feature and then of Φ proves (5), uniformly for small displacements. Potential energies are included by taking a single feature. Smooth symmetric interaction energies $\frac{1}{2} \iint K(x, y) d\mu(x) d\mu(y)$ satisfy the same estimate when K has bounded Hessian, by applying its Taylor formula to $\pi \otimes \pi$. These examples concern the differential of the energy; they do not by themselves establish regularity of the selected spectral velocity.

F Gaussian-Preserving Flows and Linear Networks

This section isolates a tractable regime in which the spectral flow reduces from an infinite-dimensional PDE to finite-dimensional covariance dynamics. The general Gaussian-preservation principle comes first, and the rest of the section specializes it to centered Gaussian training of linear two-layer networks.

F.1 Gaussian-Preserving Gradient Flows

This subsection isolates a simple class of spectral gradient flows that can still be analyzed rather explicitly. In general, understanding the long-time behavior of the flow is hard, but when Gaussian measures are preserved the infinite-dimensional evolution reduces to a finite-dimensional ODE on means and covariances.

The next corollary explains why Gaussian preservation transfers from the classical W_2 flow to the spectral flow. The finite-dimensional Measure-LMO reduction is important here because it also covers singular endpoints such as $p = \infty$.

Corollary F.1 (From W_2 Gaussian preservation to spectral Gaussian preservation). *Assume that for every Gaussian state μ the classical Wasserstein gradient*

$$g_\mu(x) = \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}(x)$$

is affine in x . Then the Measure-LMO admits an affine selection at every Gaussian state, and the spectral flow using this selection,

$$\partial_t \mu_t + \operatorname{div}(\mu_t \mathcal{J}_\gamma^{\mu_t}(g_{\mu_t})) = 0$$

admits a closed Gaussian ansatz: every solution of the induced mean and covariance ODE with locally integrable affine coefficients defines a Gaussian solution of the PDE. This does not assert uniqueness among all PDE solutions.

Proof. Fix a Gaussian state μ and write

$$g_\mu(x) = b_\mu + B_\mu x$$

for the affine W_2 gradient provided by the assumption. Set $S_g = \mathcal{S}_\mu(g_\mu)$. By Proposition 2.4, there is a matrix A_g such that

$$\mathcal{J}_\gamma^\mu(g_\mu)(x) = A_g^\top S_g^{\dagger/2} (b_\mu + B_\mu x).$$

This is again affine in x (and is zero when $S_g = 0$). Therefore, the spectral continuity equation is of the form

$$v_t(x) = a_t + A_t x$$

with coefficients depending on the current Gaussian state. A continuity equation with affine drift preserves the Gaussian class, and the associated mean m_t and covariance Σ_t solve the closed ODE system

$$\dot{m}_t = a_t + A_t m_t, \quad \dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top.$$

Conversely, a solution of these moment equations with locally integrable a_t, A_t is transported by their affine characteristic flow, so its Gaussian law solves the PDE. Existence and uniqueness of this state-dependent moment system, particularly at rank changes, are separate questions. $\square$

Two important examples fit into this mechanism. First, for the relative entropy $\text{Ent}_\nu(\mu)$ with Gaussian target ν , the classical W_2 gradient flow is the Fokker–Planck equation, and on the Gaussian class its Wasserstein gradient is affine in x . Second, whenever $\mathcal{F}(\mu)$ depends only on the mean and covariance of μ , its first variation is a quadratic polynomial and its W_2 gradient is affine, so the same transfer principle applies. This is in particular the case for training a linear two-layer network, obtained by taking $\sigma = \text{Id}$ in the MLP model above together with a quadratic loss $R(H)$: the resulting objective depends quadratically on the first two moments of μ .

F.2 Centered Gaussian Linear Two-Layer Model

This subsection develops a concrete Gaussian-preserving example where the predictor is a linear map and the spectral flow closes on covariance matrices. Write m for the input and output dimension; the particle dimension in the general theory is then $d = 2m$. Consider particles

$$x = (u, v) \in \mathbb{R}^m \times \mathbb{R}^m,$$

and the linear two-layer feature map

$$\phi(z, x) = u(v^\top z), \quad z \in \mathbb{R}^m.$$

Here u is the outer weight and v is the inner weight. For a probability measure μ on (u, v) , the predictor is

$$H_\mu(z) = \int u(v^\top z) d\mu(u, v) = M_\mu z, \quad M_\mu := \int uv^\top d\mu(u, v).$$

With a linear teacher $y^*(z) = A_\star z$ and data law ρ , the population least-squares loss is

$$\mathcal{F}(\mu) = \frac{1}{2} \int |H_\mu(z) - A_\star z|^2 d\rho(z).$$

Writing

$$C_\rho := \int zz^\top d\rho(z), \quad B_\rho := \int y^*(z)z^\top d\rho(z) = A_\star C_\rho,$$

one obtains

$$\mathcal{F}(\mu) = \frac{1}{2} \text{tr}(M_\mu C_\rho M_\mu^\top) - \text{tr}(B_\rho M_\mu^\top) + \text{const.}$$

For centered Gaussian states

$$\mu = \mathcal{N}(0, \Sigma), \quad \Sigma = \begin{pmatrix} \Sigma_U & \Sigma_{UV} \\ \Sigma_{VU} & \Sigma_V \end{pmatrix},$$

the predictor is simply $M_\mu = \Sigma_{UV}$. Thus the cross-covariance block carries the learned linear map, while the diagonal covariance blocks influence the normalized metric direction.

The next proposition gives the closed covariance equation for Schatten spectral flows.

Proposition F.2 (Covariance ODE for centered Gaussian states). *Let $p \in [1, \infty]$, let $\gamma = \gamma_p$ be the Schatten- p norm, and let*

$$q = \frac{2p}{2p-1}$$

be the dual exponent of $2p$. Assume that $\mu_t = \mathcal{N}(0, \Sigma_t)$ is a centered Gaussian solution using the canonical Schatten selector and that Σ_t remains positive definite. Then

$$\dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top, \quad A_t = B_t \Sigma_t^{-1/2},$$

where

$$B_t = -\|K_t\|_{S_q}^{2-q} U_t \text{diag}(\sigma_i(t)^{q-1}) W_t^\top, \quad K_t = L_t \Sigma_t^{1/2},$$

$K_t = U_t \text{diag}(\sigma_i(t)) W_t^\top$ is a compact singular value decomposition over its positive singular values, and

$$L_t = \begin{pmatrix} 0 & E_t \\ E_t^\top & 0 \end{pmatrix}, \quad E_t = \Sigma_{UV,t} C_\rho - B_\rho.$$

Equivalently,

$$\dot{\Sigma}_t = B_t \Sigma_t^{1/2} + \Sigma_t^{1/2} B_t^\top.$$

When $K_t = 0$, set $B_t = 0$.

Proof. For centered Gaussian $\mu = \mathcal{N}(0, \Sigma)$, the loss depends on μ only through $M_\mu = \Sigma_{UV}$, so differentiating with respect to this block gives $E = \Sigma_{UV} C_\rho - B_\rho$. Therefore

$$g_\mu(u, v) = \nabla_x \frac{\delta \mathcal{F}}{\delta \mu}(u, v) = \begin{pmatrix} E v \\ E^\top u \end{pmatrix} = Lx.$$

The Wasserstein gradient is affine, hence the Gaussian class is preserved by Corollary F.1. For an affine velocity $v(x) = Ax$, the selected direction solves

$$\min_A \left\{ \text{tr}(L \Sigma A^\top) + \frac{1}{2} \|A \Sigma^{1/2}\|_{S_{2p}}^2 \right\}.$$

Setting $B = A \Sigma^{1/2}$ and $K = L \Sigma^{1/2}$ reduces this to the finite-dimensional Schatten duality-map problem

$$\min_B \left\{ \text{tr}(K B^\top) + \frac{1}{2} \|B\|_{S_{2p}}^2 \right\},$$

whose minimizer is the displayed formula for B_t . Finally, if X_t follows $\dot{X}_t = A_t X_t$ and $X_t \sim \mathcal{N}(0, \Sigma_t)$, then $\dot{\Sigma}_t = A_t \Sigma_t + \Sigma_t A_t^\top$. $\square$

F.3 Modal Reduction Under Simultaneous Diagonalization

This subsection shows that, under a compatible data basis and balanced initialization, the covariance ODE decomposes into interpretable feature modes. Assume that after orthogonal changes of coordinates one has

$$C_\rho = \text{Id}, \quad B_\rho = \text{diag}(\beta_1, \dots, \beta_m),$$

and use a balanced diagonal initialization

$$\Sigma_{U,0} = \text{diag}(\alpha_i), \quad \Sigma_{V,0} = \text{diag}(\alpha_i), \quad \Sigma_{UV,0} = 0, \quad \alpha_i > 0.$$

Then the covariance remains diagonal mode by mode. We write

$$r_i(t) = (\Sigma_{UV,t})_{ii}, \quad s_i(t) = (\Sigma_{U,t})_{ii} = (\Sigma_{V,t})_{ii},$$

so that r_i is the predictor coefficient and s_i measures the corresponding layer magnitude.

Proposition F.3 (Modal ODE). *Let $p \in [1, \infty]$, let $q = 2p/(2p-1)$, and define $\psi_q(a) = a|a|^{q-2}$ for $a \neq 0$ and $\psi_q(0) = 0$. Under the assumptions above, set*

$$a_i = s_i + r_i, \quad b_i = s_i - r_i, \quad \mathcal{N}_t^q = \sum_{j=1}^m |r_j - \beta_j|^q \left(a_j^{q/2} + b_j^{q/2} \right).$$

Then each mode solves

$$\begin{aligned} \dot{r}_i &= -\mathcal{N}_t^{2-q} \psi_q(r_i - \beta_i) \left(a_i^{q/2} + b_i^{q/2} \right), \\ \dot{s}_i &= -\mathcal{N}_t^{2-q} \psi_q(r_i - \beta_i) \left(a_i^{q/2} - b_i^{q/2} \right). \end{aligned}$$

Equivalently,

$$\begin{aligned}\dot{a}_i &= -2\mathcal{N}_t^{2-q} \psi_q(r_i - \beta_i) a_i^{q/2}, \\ \dot{b}_i &= 2\mathcal{N}_t^{2-q} \psi_q(r_i - \beta_i) b_i^{q/2}.\end{aligned}$$

If $\mathcal{N}_t = 0$, all right-hand sides are understood to be zero.

Proof. When Σ_{UV} is diagonal, the error $E_t = \Sigma_{UV,t} - B_\rho$ is diagonal, hence the affine gradient matrix

$$L_t = \begin{pmatrix} 0 & E_t \\ E_t & 0 \end{pmatrix}$$

preserves each coordinate plane. The balanced diagonal covariance structure is therefore invariant. In the i -th plane, the covariance block is

$$\Sigma_t^{(i)} = \begin{pmatrix} s_i & r_i \\ r_i & s_i \end{pmatrix},$$

with eigenvalues $a_i = s_i + r_i$ and $b_i = s_i - r_i$, while the corresponding block of L_t has eigenvalues $\pm(r_i - \beta_i)$. The singular values of $K_t = L_t \Sigma_t^{1/2}$ are therefore $|r_i - \beta_i| a_i^{1/2}$ and $|r_i - \beta_i| b_i^{1/2}$, with no additional factor of two. Applying the Schatten LMO of Proposition F.2 blockwise and collecting the common normalization gives the stated equations. $\square$

F.4 Rank-One and Rank-Two Targets

This subsection records the two simplest modal systems, which are useful for interpreting how the spectral normalization changes learning profiles. If only one target coefficient is nonzero,

$$B_\rho = \text{diag}(\beta, 0, \dots, 0), \quad \beta \neq 0,$$

then all inactive modes remain frozen and the dynamics reduces to the single pair (r, s) governed by Proposition F.3. In the positive-definite interior, the variables $a = s + r$ and $b = s - r$ are positive and encode the two layer variances in the active symmetric and antisymmetric directions. This one-mode system is the basic phase portrait behind the Gaussian experiments: the predictor coefficient r_t moves toward β , while the terminal value of s_t records the implicit covariance bias selected by p . The admissible covariance cone is

$$\{(r, s) \in \mathbb{R}^2 : s \geq |r|\},$$

with positive-definite interior $s > |r|$, equivalently $a > 0$ and $b > 0$. In Figure 6(a), we display interior trajectories from two initial conditions. The next proposition gives the conserved leaves for the two endpoint geometries; these level sets organize the displayed trajectories.

Proposition F.4 (Rank-one endpoint invariants). *Along every interval on which $s > |r|$, the trace geometry $p = 1$ preserves $s^2 - r^2$, while the operator geometry $p = \infty$ preserves $\sqrt{s+r} + \sqrt{s-r}$.*

Proof. Write $a = s + r$ and $b = s - r$. The modal equations of Proposition F.3 give $\dot{a} = -2\Lambda a^{q/2}$ and $\dot{b} = 2\Lambda b^{q/2}$ for a scalar factor Λ depending on the current state. For $p = 1$, one has $q = 2$, hence $\frac{d}{dt}(ab) = \dot{a}b + a\dot{b} = 0$, and $ab = s^2 - r^2$. For $p = \infty$, one has $q = 1$, hence $\frac{d}{dt}(\sqrt{a} + \sqrt{b}) = \dot{a}/(2\sqrt{a}) + \dot{b}/(2\sqrt{b}) = 0$, which gives the stated invariant. $\square$

Figure 6 illustrates these rank-one and rank-two reductions.

For a rank-two target

$$B_\rho = \text{diag}(\beta_1, \beta_2, 0, \dots, 0),$$

the first two modes solve the same equations but are coupled through

$$\mathcal{N}_t^q = \sum_{j=1}^2 |r_j - \beta_j|^q \left(a_j^{q/2} + b_j^{q/2} \right).$$

Thus the trace geometry evolves the two learned directions independently, whereas every $p > 1$ geometry shares a global normalization across them. This is the simplest Gaussian setting where the matrix nature of the spectral flow is visible at the predictor level rather than only through a time rescaling of a single scalar mode. In Figure 6(d), this coupling appears as distinct planar paths toward different targets (β_1, β_2) under the same initialization.

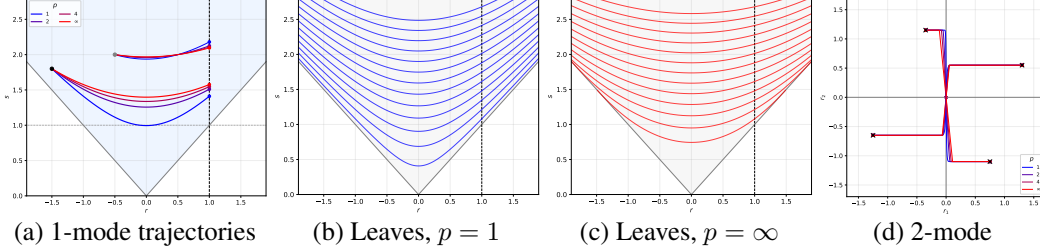


Figure 6: Closed-form Gaussian geometry. Panel (a): one-mode trajectories in (r, s) , for $p \in \{1, 2, 4, \infty\}$. Panels (b)–(c): interior conserved foliations for the two endpoint geometries. Panel (d): two-mode rank-two dynamics in the (r_1, r_2) plane; all trajectories start from the balanced diagonal state $r_1 = r_2 = 0$, $(s_1, s_2) = (0.1, 10)$, and each group targets a different (β_1, β_2) .

G Numerical Protocols and Additional Examples

G.1 Time Normalization and Numerical Safeguards

The experiments are qualitative illustrations of the chosen geometry, not wall-clock or generalization benchmarks. Their notebooks differentiate the empirical loss $\mathcal{F}(\mu_X)$, whereas Proposition 2.5 uses $F(X) = n\mathcal{F}(\mu_X)$. By homogeneity of the LMO, an unclipped notebook step of size Δt therefore corresponds to a mean-field Euler step of size $\Delta t/n$. Table 1 lists the nominal step sizes and counts. Each loss plot independently maps its run’s nominal horizon to $[0, 1]$; these display coordinates are not common intrinsic flow times.

Example	n	steps $(1, \infty)$	Δt_1	Δt_∞	step cap
ReLU MLP	1920	(2000, 2000)	75	50	0.20
Attention	200	(960, 480)	100	180	0.45
MMD	200	(6000, 6000)	5	3.2	none

Table 1: Nominal notebook clocks. Caps apply to the Frobenius norm of the entire particle-matrix increment, not separately to particles.

The MLP and attention implementations multiply an oversized increment by a scalar to enforce the stated cap, and additionally guard coordinates at ± 30 and ± 10 , respectively. The cap makes effective step sizes state-dependent; an active coordinate guard can also change the direction. The MMD experiment uses the smoothed distance $\sqrt{\|x - y\|^2 + 0.01^2}$ and unclipped Euler steps. The archived plots do not record cumulative effective times or all safeguard activations, so no corrected continuum-clock comparison is inferred from them. The slight difference between pre-update loss samples and post-update snapshot labels is also a discretization convention, not a theoretical time rescaling.

An initialization-only check of the MLP notebook (seed 7, double precision, 2000 training covariates) makes this distinction concrete. Writing $G = \nabla_X \mathcal{F}(\mu_X)$, the ideal mean-field dissipations are $D_1 = n\|G\|_F^2$ and $D_\infty = n\|G\|_*^2$. Their initial ratio is $D_\infty/D_1 \simeq 2.55765$, whereas the nominal horizon ratio is 1.5. Both initial updates are already clipped, with factors approximately 0.74469 and 0.40326; their first-order predicted loss-decrease ratio per displayed step is consequently 0.92334, not 2.55765. The script `python/mlp/validate_initialization.py` reproduces these checks directly from the notebook without running training. An initial dissipation ratio alone cannot calibrate the entire trajectories.

For attention, the five paired trials redraw both teacher and student. The effective-token statistics use all attention weights, whereas the displayed weight histograms restrict to $[0, 0.1]$ and renormalize there. The Gaussian modal figures use the ODE of Proposition F.3, integrated in double precision by RK4 with $T = 18$ and $\Delta t = 2 \times 10^{-4}$. Negative numerical covariance eigenvalues are projected to zero; extinct modes are not artificially revived. The validator `python/gaussians/validate_gaussians.py` checks the implemented vector field against the analytical formulas and tests conserved leaves on interior trajectory segments. This is not a proof of

accuracy through a rank change. Their path and interior-invariant interpretation is separate from the empirical neural-network clocks.

G.2 Couplings and Interpolations

This subsection complements Figures 1 and 2 with additional planar shape pairs. Each example uses an image-defined source and a translated image-defined target. Trace couplings are exact assignment solutions. Operator couplings are obtained by alternating a projected assignment with a rank-one cost update, for at most 40 iterations from a horizontal-axis initialization. This heuristic supplies no primal-dual gap certificate and is not asserted to solve the full convex SDP. Accordingly, its interpolations are illustrative displacement paths, not certified W_γ -geodesics. Rotation invariance holds for the underlying Schatten transport problem, not necessarily for this finitely iterated, fixed-initialization solver.

Figure 7 shows additional static couplings for three image-defined source and target pairs, complementing the annulus-to-disks example in the main text.

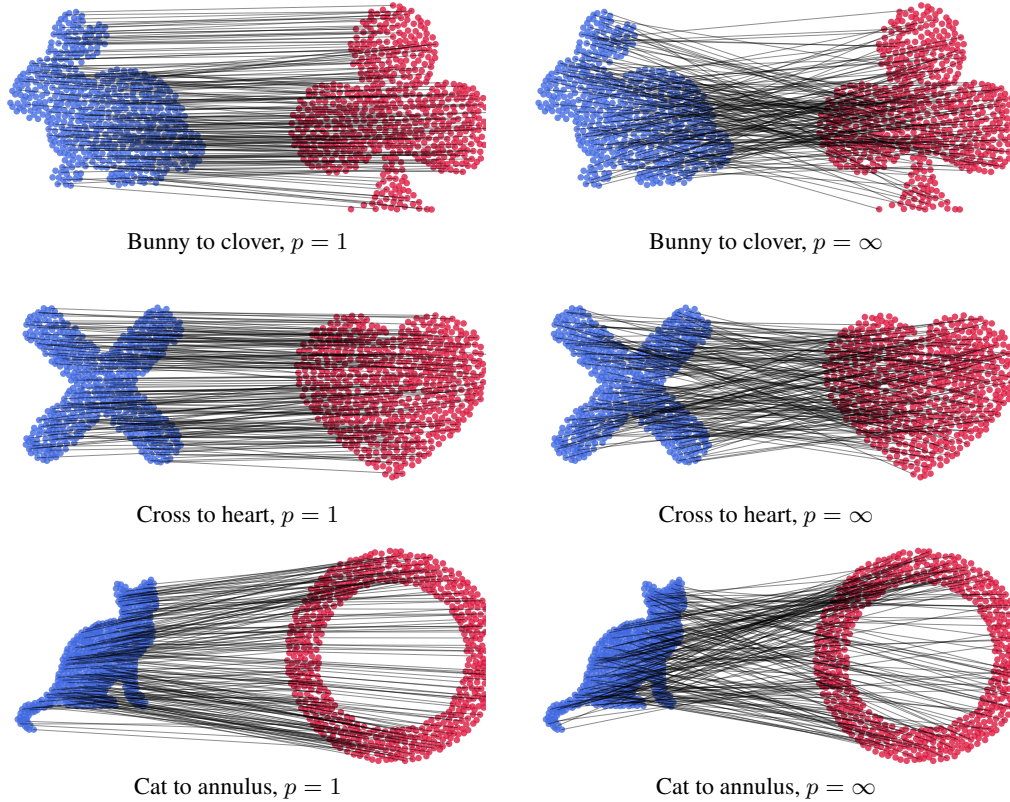


Figure 7: Additional static Spectral Wasserstein couplings. Only a subsample of matching segments is displayed.

Figure 8 displays the corresponding displacement-interpolation density snapshots, using the same computed couplings as in Figure 7.

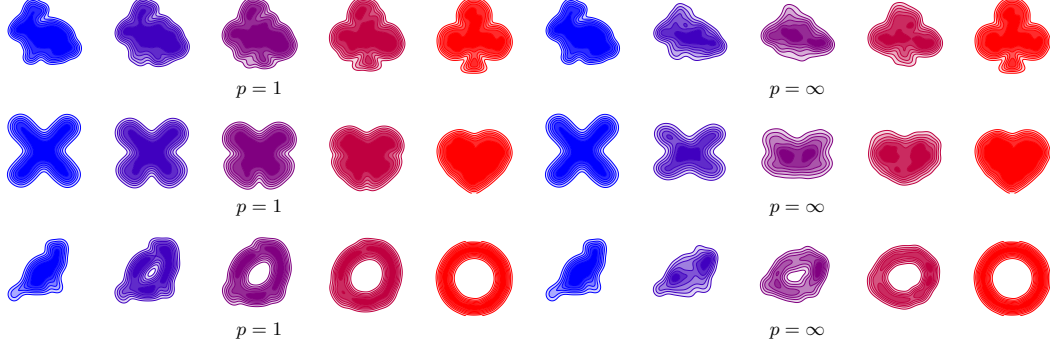


Figure 8: Additional displacement-interpolation density snapshots for the same shape pairs as Figure 7.

H Geodesic Convexity

This section studies one of the basic structural notions behind metric gradient flows, namely geodesic convexity, for the Spectral Wasserstein geometries introduced above. Geodesic convexity is desirable because it supports stability and global convergence statements [Ambrosio et al., 2008], but for many machine-learning objectives it is arguably too strong, so here it is used as a structural benchmark rather than as a blanket modeling assumption.

H.1 Definition and General Setup

This subsection recalls the relevant notion of convexity along Spectral Wasserstein geodesics and explains why it matters for the variational analysis of the flow. When one studies gradient flows in metric spaces, geodesic convexity plays the same role as ordinary convexity in Euclidean optimization: it governs uniqueness, stability, and the variational structure of the dynamics. Corollary 4.4 provides a distinguished class of geodesics from optimal couplings, for which convexity inequalities can be tested directly along Euclidean segments.

The next definition fixes the notion used in the remainder of this section.

Definition H.1 (Geodesic convexity). *A functional $F : \mathcal{P}_2(\mathbb{R}^d) \rightarrow (-\infty, +\infty]$ is called κ -geodesically convex for W_γ if along every constant-speed W_γ -geodesic $(\mu_t)_{t \in [0,1]}$ one has*

$$F(\mu_t) \leq (1-t)F(\mu_0) + tF(\mu_1) - \frac{\kappa}{2}t(1-t)W_\gamma(\mu_0, \mu_1)^2.$$

We call a geodesic a *static displacement geodesic* if it has the form $\mu_t = ((1-t)x + ty)_\# \pi_*$ for a static optimal coupling π_* , as in Corollary 4.4. Full geodesic convexity concerns every constant-speed geodesic and is stronger when the metric admits branching.

Remark H.2 (Branching at the operator endpoint). *This distinction is genuine for the Muon geometry. Let $\gamma(S) = \lambda_{\max}(S)$ on $\mathbb{S}_+^2$, choose a nonzero 1-Lipschitz function $f : [0, 1] \rightarrow \mathbb{R}$ with $f(0) = f(1) = 0$, and set*

$$\mu_t = \frac{1}{2}\delta_{(t, f(t))} + \frac{1}{2}\delta_{(t, -f(t))}.$$

For $s < t$, matching equal signs gives displacement covariance $\text{diag}((t-s)^2, (f(t) - f(s))^2)$ and therefore cost $(t-s)^2$. Conversely, the change of the first-coordinate mean gives the same lower bound for every coupling. Hence (μ_t) is a constant-speed W_γ -geodesic from $\delta_{(0,0)}$ to $\delta_{(1,0)}$, but it is not a static displacement geodesic when $f \neq 0$. For the convex function $h(x) = x_2^2$, one has $\int h d\mu_0 = \int h d\mu_1 = 0$ but $\int h d\mu_t = f(t)^2 > 0$ whenever $f(t) \neq 0$. Thus convexity of h ensures convexity along every static displacement geodesic, but not full geodesic convexity at this endpoint.

H.2 Linear Functionals

This subsection shows that for linear functionals the qualitative and quantitative convexity criteria reduce to explicit pointwise conditions on the integrand. For a Borel function $h : \mathbb{R}^d \rightarrow \mathbb{R}$, define the

linear functional

$$F_h(\mu) := \int_{\mathbb{R}^d} h(x) d\mu(x).$$

Because static displacement geodesics are given by affine interpolation in space, the convexity question for F_h along this class is especially transparent.

The next theorem gives the exact criterion.

Theorem H.3 (Linear functionals along displacement geodesics). *Assume γ is monotone and let $\mathcal{K}_\gamma$ be a convex compact PSD representing set for γ as in Proposition A.2.*

1. *The functional F_h is convex along every static displacement geodesic if and only if h is convex on $\mathbb{R}^d$.*
2. *If in addition $h \in C^2(\mathbb{R}^d)$ and $\kappa \geq 0$, then F_h satisfies the κ -convexity inequality along every static displacement geodesic if and only if*

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } z \in \mathbb{R}^d, Q \in \mathcal{K}_\gamma.$$

Equivalently,

$$\xi^\top \nabla^2 h(z) \xi \geq \kappa \max_{Q \in \mathcal{K}_\gamma} \xi^\top Q \xi \quad \text{for every } z, \xi \in \mathbb{R}^d.$$

For $\kappa < 0$, this displayed scalar reformulation with a maximum is not equivalent to the matrix inequality; multiplication by the negative scalar reverses the extremum, so the corresponding pointwise scalar condition would involve $\min_{Q \in \mathcal{K}_\gamma} \xi^\top Q \xi$. No equivalence for negative κ is asserted here.

Proof. Assume first that h is convex and let

$$\mu_t = ((1-t)x + ty)_\# \pi_*$$

be any static displacement geodesic. Then

$$F_h(\mu_t) = \int h((1-t)x + ty) d\pi_*(x, y).$$

By the convexity of h ,

$$h((1-t)x + ty) \leq (1-t)h(x) + th(y),$$

and integrating yields geodesic convexity of F_h .

Conversely, apply the displacement-convexity assumption to Dirac masses $\mu_0 = \delta_x$ and $\mu_1 = \delta_y$. Their static displacement geodesic is

$$\mu_t = \delta_{(1-t)x + ty},$$

so

$$h((1-t)x + ty) = F_h(\mu_t) \leq (1-t)h(x) + th(y),$$

which proves convexity of h .

Assume now that $h \in C^2$ and that

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } z \in \mathbb{R}^d, Q \in \mathcal{K}_\gamma.$$

Let

$$\mu_t = ((1-t)x + ty)_\# \pi_*$$

be any static displacement geodesic, write $z_t = (1-t)x + ty$ and $\Delta = y - x$. Differentiating under the integral sign gives

$$\frac{d^2}{dt^2} F_h(\mu_t) = \int \Delta^\top \nabla^2 h(z_t) \Delta d\pi_*(x, y).$$

For every $Q \in \mathcal{K}_\gamma$ this implies

$$\frac{d^2}{dt^2} F_h(\mu_t) \geq \kappa \int \Delta^\top Q \Delta d\pi_*(x, y).$$

Since $\kappa \geq 0$, taking the maximum over $Q \in \mathcal{K}_\gamma$ yields

$$\frac{d^2}{dt^2} F_h(\mu_t) \geq \kappa \gamma \left(\int \Delta \Delta^\top d\pi_*(x, y) \right) = \kappa \mathcal{W}_\gamma(\mu_0, \mu_1)^2.$$

Integrating twice in t gives the κ -geodesic convexity estimate.

Conversely, test the midpoint inequality between $\delta_{z-\varepsilon\xi}$ and $\delta_{z+\varepsilon\xi}$. It gives

$$h(z) \leq \frac{h(z - \varepsilon\xi) + h(z + \varepsilon\xi)}{2} - \frac{\kappa}{2} \varepsilon^2 \gamma(\xi\xi^\top).$$

Subtracting $h(z)$, dividing by $\varepsilon^2/2$, and sending $\varepsilon \rightarrow 0$ yields

$$\xi^\top \nabla^2 h(z) \xi \geq \kappa \gamma(\xi\xi^\top) = \kappa \max_{Q \in \mathcal{K}_\gamma} \xi^\top Q \xi.$$

Because $\kappa \geq 0$, this scalar inequality is equivalent to the matrix inequality

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } Q \in \mathcal{K}_\gamma.$$

□

The next remark interprets the condition of Theorem H.3 for the Schatten family.

Remark H.4 (Schatten norms). *For the Schatten geometries used throughout the paper and for $\kappa \geq 0$, the condition*

$$\nabla^2 h(z) \succeq \kappa Q \quad \text{for every } Q \in \mathcal{K}_\gamma$$

is equivalent to

$$\nabla^2 h(z) \succeq \kappa \text{Id}.$$

Indeed, for $p = 1$ we may choose $\mathcal{K}_\gamma = \{\text{Id}\}$. For $1 < p \leq \infty$, a convenient choice is

$$\mathcal{K}_\gamma = \{Q \succeq 0 : \|Q\|_{S_{p'}} \leq 1\}, \quad p' = \frac{p}{p-1},$$

with the usual interpretation $p' = 1$ when $p = \infty$. Under the standing condition $\kappa \geq 0$ in Theorem H.3, every such Q satisfies $Q \preceq \text{Id}$, and every rank-one projector belongs to $\mathcal{K}_\gamma$; hence the condition is equivalent to $\nabla^2 h(z) \succeq \kappa \text{Id}$, exactly as in the classical $\mathcal{W}_2$ geometry. For negative κ , this simplification is not asserted.

H.3 Interaction Functionals

The same segmentwise argument applies to convex interactions.

Proposition H.5 (Convex interactions). *Let $h : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be jointly convex and define*

$$F_h^{\text{int}}(\mu) = \int h(x, x') d\mu(x) d\mu(x').$$

Then F_h^{int} is convex along every static displacement geodesic.

Proof. Let $\mu_t = ((1-t)x + ty)_\# \pi_*$ and take two independent copies $(x, y), (x', y')$ with law π_* . Joint convexity gives

$$h((1-t)x + ty, (1-t)x' + ty') \leq (1-t)h(x, x') + th(y, y').$$

Integrating with respect to $\pi_* \otimes \pi_*$ proves the claim. □

H.4 Relative Entropy

This subsection examines the geodesic convexity of relative entropy, which is a fundamental nonlinear example underlying diffusion-type gradient flows. Section H.5 treats in detail the case of a Gaussian target ν . For the canonical selector, the Gaussian ansatz closes as an ODE for the mean and covariance. Let

$$d\nu(x) = Z^{-1} e^{-V(x)} dx, \quad V \in C^2(\mathbb{R}^d),$$

and define the relative entropy

$$\text{Ent}_\nu(\mu) := \begin{cases} \int \log\left(\frac{d\mu}{d\nu}\right) d\mu, & \mu \ll \nu, \\ +\infty, & \text{otherwise.} \end{cases}$$

The trace case is the classical displacement-convexity theory of entropy. For general spectral geometries the same argument works only when the active quadratic costs remain uniformly elliptic, and otherwise one only gets a weaker statement by approximation.

The next theorem records the all-geodesics result in the uniformly elliptic regime.

Theorem H.6 (Full entropy convexity under uniform ellipticity). *Assume γ is monotone and that $\mathcal{K}_\gamma$ can be chosen so that*

$$\mathcal{K}_\gamma \subset \mathbb{S}_{++}^d.$$

Assume moreover that

$$\nabla^2 V(x) \succeq \kappa Q \quad \text{for every } x \in \mathbb{R}^d, Q \in \mathcal{K}_\gamma.$$

Then Ent_ν is κ -geodesically convex on $(\mathcal{P}_2(\mathbb{R}^d), W_\gamma)$.

Proof. Let (μ_t) be any constant-speed W_γ -geodesic and choose a minimal velocity representation. Its action equals $W_\gamma(\mu_0, \mu_1)^2$. By the superposition principle, it is represented by a path measure η ; let $\pi_* = (e_0, e_1)_\# \eta$. The dynamic-to-static inequalities in the proof of Theorem 4.2 give

$$W_\gamma(\mu_0, \mu_1)^2 \leq \gamma \left(\int (\omega_1 - \omega_0)(\omega_1 - \omega_0)^\top d\eta \right) \leq \int_0^1 \mathcal{N}_{\mu_t}(v_t)^2 dt = W_\gamma(\mu_0, \mu_1)^2.$$

Thus π_* is a static optimal coupling. Choose a dual maximizer Q_* in Theorem 3.4. The pair (π_*, Q_*) is a saddle point, and $Q_* \succ 0$ because $\mathcal{K}_\gamma \subset \mathbb{S}_{++}^d$. Equality throughout the quadratic Jensen estimate for Q_* implies $\dot{\omega}_t = \omega_1 - \omega_0$ for η -a.e. path. Consequently,

$$\mu_t = ((1-t)x + ty)_\# \pi_*, \quad W^{Q_*}(\mu_0, \mu_1)^2 = W_\gamma(\mu_0, \mu_1)^2.$$

In particular, uniform ellipticity rules out the branching phenomenon of the preceding remark.

Since Q_* is positive definite, let $L = Q_*^{1/2}$ and define

$$\tilde{\mu}_i = L_\# \mu_i, \quad \tilde{\nu} = L_\# \nu, \quad \tilde{\pi}_* = (L, L)_\# \pi_*.$$

Then $\tilde{\pi}_*$ is optimal for the Euclidean quadratic cost between $\tilde{\mu}_0$ and $\tilde{\mu}_1$, hence

$$\tilde{\mu}_t = ((1-t)z_0 + tz_1)_\# \tilde{\pi}_*$$

is a W_2 -geodesic and

$$W_2(\tilde{\mu}_0, \tilde{\mu}_1)^2 = W_\gamma(\mu_0, \mu_1)^2.$$

The reference measure $\tilde{\nu}$ has density

$$d\tilde{\nu}(z) = \tilde{Z}^{-1} e^{-\tilde{V}(z)} dz, \quad \tilde{V}(z) = V(L^{-1}z) + \text{constant},$$

so

$$\nabla^2 \tilde{V}(z) = L^{-1} \nabla^2 V(L^{-1}z) L^{-1}.$$

Because $Q_* \in \mathcal{K}_\gamma$, the curvature assumption gives

$$\nabla^2 V(x) \succeq \kappa Q_* = \kappa L^2,$$

hence

$$\nabla^2 \tilde{V}(z) \succeq \kappa \text{Id}.$$

The classical entropy convexity theorem for W_2 therefore yields

$$\text{Ent}_{\tilde{\nu}}(\tilde{\mu}_t) \leq (1-t) \text{Ent}_{\tilde{\nu}}(\tilde{\mu}_0) + t \text{Ent}_{\tilde{\nu}}(\tilde{\mu}_1) - \frac{\kappa}{2} t(1-t) W_2(\tilde{\mu}_0, \tilde{\mu}_1)^2.$$

Finally, relative entropy is invariant under the invertible change of variables L , so

$$\text{Ent}_{\tilde{\nu}}(\tilde{\mu}_t) = \text{Ent}_\nu(\mu_t),$$

and similarly at the endpoints. This gives the result. $\square$

The next theorem explains what remains true in the more relevant but singular regime where the condition $\mathcal{K}_\gamma \subset \mathbb{S}_{++}^d$ fails. This lack of ellipticity is precisely what happens for Schatten- p geometries with $p > 1$, because the corresponding representing sets contain singular matrices, but one still retains a weaker geodesic-convexity statement by approximation.

Theorem H.7 (Weak entropy convexity). *Assume γ is monotone, that $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ is convex compact, and that*

$$\mathcal{K}_\gamma \cap \mathbb{S}_{++}^d \neq \emptyset.$$

Assume moreover that

$$\nabla^2 V(x) \succeq \kappa Q \quad \text{for every } x \in \mathbb{R}^d, Q \in \mathcal{K}_\gamma.$$

Then for every $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$ there exists at least one constant-speed W_γ -geodesic $(\mu_t)_{t \in [0,1]}$ from μ_0 to μ_1 such that

$$\text{Ent}_\nu(\mu_t) \leq (1-t) \text{Ent}_\nu(\mu_0) + t \text{Ent}_\nu(\mu_1) - \frac{\kappa}{2} t(1-t) W_\gamma(\mu_0, \mu_1)^2.$$

Proof. Choose $Q_0 \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d$ and define

$$\mathcal{K}_{\gamma,\varepsilon} := (1-\varepsilon)\mathcal{K}_\gamma + \varepsilon Q_0, \quad \varepsilon \in (0,1).$$

Then every element of $\mathcal{K}_{\gamma,\varepsilon}$ is positive definite. Let γ_ε be the support function of $\mathcal{K}_{\gamma,\varepsilon}$, namely

$$\gamma_\varepsilon(S) = \max_{Q \in \mathcal{K}_{\gamma,\varepsilon}} \text{tr}(QS) = (1-\varepsilon)\gamma(S) + \varepsilon \text{tr}(Q_0 S).$$

Hence

$$(1-\varepsilon)\gamma(S) \leq \gamma_\varepsilon(S) \leq \gamma(S)$$

for all $S \succeq 0$, and therefore

$$(1-\varepsilon)W_\gamma(\mu_0, \mu_1)^2 \leq W_{\gamma_\varepsilon}(\mu_0, \mu_1)^2 \leq W_\gamma(\mu_0, \mu_1)^2.$$

Moreover, convexity of $\mathcal{K}_\gamma$ gives $\mathcal{K}_{\gamma,\varepsilon} \subset \mathcal{K}_\gamma$, so the curvature assumption is unchanged.

By Theorem H.6, each regularized geometry has full entropy convexity. Choose an optimal coupling π_ε for W_{γ_ε} and set

$$\mu_t^\varepsilon = ((1-t)x + ty)_\# \pi_\varepsilon.$$

Then

$$\text{Ent}_\nu(\mu_t^\varepsilon) \leq (1-t) \text{Ent}_\nu(\mu_0) + t \text{Ent}_\nu(\mu_1) - \frac{\kappa}{2} t(1-t) W_{\gamma_\varepsilon}(\mu_0, \mu_1)^2.$$

By compactness of couplings, one may extract a subsequence $\pi_{\varepsilon_n} \rightharpoonup \pi_*$. Setting

$$\mu_t = ((1-t)x + ty)_\# \pi_*,$$

the lower semicontinuity of relative entropy and the convergence

$$W_{\gamma_{\varepsilon_n}}(\mu_0, \mu_1)^2 \rightarrow W_\gamma(\mu_0, \mu_1)^2$$

yield the claimed inequality along the limit geodesic. The limit coupling π_* is W_γ -optimal by continuity of the displacement covariance and of the support functions on trace-bounded sets. $\square$

The next proposition shows that, in the operator case, full geodesic convexity of relative entropy fails in a very concrete way.

Proposition H.8 (Operator-norm counterexample). *Let $\gamma(S) = \lambda_{\max}(S)$ on $\mathbb{S}_+^2$, and let*

$$d\nu(x) = \frac{1}{2\pi} e^{-|x|^2/2} dx.$$

Then Ent_ν is not geodesically convex on $(\mathcal{P}_2(\mathbb{R}^2), W_\gamma)$.

Proof. Fix parameters $L > 0$ and $a > 0$ such that

$$\frac{4a^2}{3} \leq L^2.$$

Set

$$R_0 = [0, 1] \times [-a, a], \quad R_1 = [L, L + 1] \times [-a, a],$$

and let

$$\mu_i = \frac{1}{2a} \mathbf{1}_{R_i} dx, \quad i = 0, 1.$$

Both endpoints are absolutely continuous with bounded support, hence satisfy

$$\text{Ent}_\nu(\mu_0) < +\infty, \quad \text{Ent}_\nu(\mu_1) < +\infty.$$

For any coupling π between μ_0 and μ_1 , define

$$S_\pi = \int (y - x)(y - x)^\top d\pi(x, y).$$

Then

$$\lambda_{\max}(S_\pi) \geq e_1^\top S_\pi e_1 = \int (y_1 - x_1)^2 d\pi(x, y) \geq \left(\int (y_1 - x_1) d\pi(x, y) \right)^2 = L^2.$$

Hence every admissible coupling has operator cost at least L^2 .

The translation map

$$T(x_1, x_2) = (x_1 + L, x_2)$$

pushes μ_0 to μ_1 and has covariance

$$S_T = \text{diag}(L^2, 0),$$

so $W_\gamma(\mu_0, \mu_1)^2 = L^2$.

Now consider the reflected transport

$$R(x_1, x_2) = (x_1 + L, -x_2).$$

It also pushes μ_0 to μ_1 because the second marginal of μ_0 is symmetric on $[-a, a]$. Its displacement is

$$R(x) - x = (L, -2x_2),$$

hence

$$S_R = \text{diag}\left(L^2, \frac{4a^2}{3}\right).$$

By the choice of a ,

$$\lambda_{\max}(S_R) = L^2,$$

so the coupling induced by R is again optimal for W_γ .

Consider the corresponding constant-speed geodesic

$$\mu_t = ((1 - t)\text{Id} + tR)_\# \mu_0.$$

At time $t = \frac{1}{2}$ one gets

$$\mu_{1/2} = \left(x_1 + \frac{L}{2}, 0\right)_\# \mu_0,$$

which is the uniform measure on the horizontal segment

$$\left[\frac{L}{2}, \frac{L}{2} + 1\right] \times \{0\}.$$

Therefore $\mu_{1/2}$ is singular with respect to Lebesgue measure, hence also singular with respect to ν , and thus

$$\text{Ent}_\nu(\mu_{1/2}) = +\infty.$$

So along this optimal W_γ -geodesic the endpoints have finite entropy while the midpoint has infinite entropy, which rules out geodesic convexity. $\square$

The next remark explains why Proposition H.8 does not contradict Theorem H.7.

Remark H.9 (Why there is no contradiction). *Theorem H.7 is an existence statement: for each pair of endpoints, it guarantees at least one W_γ -geodesic along which the entropy inequality holds. Proposition H.8 proves that in the operator geometry there also exist optimal geodesics along which entropy blows up at an interior time. Thus the counterexample rules out convexity along every geodesic, but not the existence of a selected geodesic satisfying the estimate.*

The following remark explains the scope of the entropy results for the Schatten family.

Remark H.10 (Schatten norms). *The full theorem applies to the trace case $p = 1$, where one may choose $K_\gamma = \{\text{Id}\}$ and recover the classical W_2 displacement-convexity theory. For every Schatten geometry with $1 < p \leq \infty$, the natural representing sets contain singular matrices, so the uniformly elliptic argument above does not apply directly. Theorem H.7 nevertheless yields a weak geodesic-convexity statement as soon as K_γ contains one positive definite matrix, which is the case for the standard Schatten choices.*

H.5 Relative-entropy flows over Gaussians

The W_2 -gradient flow of the relative entropy $\text{Ent}_\nu(\mu) = \text{KL}(\mu | \nu)$ is the Langevin/Fokker–Planck equation, whose stochastic single-particle realization is the overdamped Langevin diffusion. This makes it a basic tool for sampling. We now describe what happens for the spectral Schatten geometries. For $p > 1$, the resulting evolution is still explicit on Gaussian measures, but the PDE is nonlocal: its velocity depends on global statistics of the current law, and therefore cannot be represented by an autonomous single-particle SDE with coefficients depending only on the particle position.

Proposition H.11 (Gaussian KL flow for Schatten geometries). *Let $\nu = \mathcal{N}(\bar{m}, \bar{\Sigma})$ with $\bar{\Sigma} \succ 0$, and consider $\mathcal{F}(\mu) = \text{KL}(\mu | \nu)$ with the canonical Schatten selector of Corollary A.6. Given $\mu_0 = \mathcal{N}(m_0, \Sigma_0)$ with $\Sigma_0 \succ 0$, any locally absolutely continuous solution (m_t, Σ_t) of the equations below, with these initial data and $\Sigma_t \succ 0$, defines a Gaussian solution $\mu_t = \mathcal{N}(m_t, \Sigma_t)$ of the normalized continuity equation. Conversely, every sufficiently regular Gaussian solution with this selector satisfies these equations. This is a conditional reduction, not an assertion of existence or uniqueness through changes in the rank of the force covariance. Define the classical $p = 1$ Gaussian velocity by $\dot{m}_t^{(1)} := -\bar{\Sigma}^{-1}(m_t - \bar{m})$ and $\dot{\Sigma}_t^{(1)} := 2I - \bar{\Sigma}^{-1}\Sigma_t - \Sigma_t\bar{\Sigma}^{-1}$. Then the trace geometry $p = 1$ satisfies $\dot{m}_t = \dot{m}_t^{(1)}$ and $\dot{\Sigma}_t = \dot{\Sigma}_t^{(1)}$.*

For a general Schatten geometry $p \in [1, \infty]$, set $q := 2p/(2p - 1)$, with the convention $q = 1$ when $p = \infty$, and define

$$F_t := (\bar{\Sigma}^{-1} - \Sigma_t^{-1})\Sigma_t(\bar{\Sigma}^{-1} - \Sigma_t^{-1}) + \bar{\Sigma}^{-1}(m_t - \bar{m})(m_t - \bar{m})^\top \bar{\Sigma}^{-1}.$$

If $F_t \neq 0$, define

$$\mathcal{N}_t^q := \text{tr}(F_t^{q/2}), \quad P_{p,t} := \mathcal{N}_t^{2-q} F_t^{q/2-1}.$$

For $1 < p \leq \infty$, the negative power is taken on $\text{range}(F_t)$ and set to zero on its kernel; for $p = 1$, take $P_{1,t} = I$. If $F_t = 0$, the velocity and both right-hand sides below are set to zero. Then the Schatten- p Gaussian KL flow is

$$\dot{m}_t = P_{p,t} \dot{m}_t^{(1)}, \quad \dot{\Sigma}_t = P_{p,t}(I - \bar{\Sigma}^{-1}\Sigma_t) + (I - \Sigma_t\bar{\Sigma}^{-1})P_{p,t}.$$

If $\bar{\Sigma}^{-1}$ and Σ_t commute, the covariance equation reduces to

$$\dot{\Sigma}_t = \frac{1}{2} \left(P_{p,t} \dot{\Sigma}_t^{(1)} + \dot{\Sigma}_t^{(1)} P_{p,t} \right).$$

In particular, for $p = 1$, one has $q = 2$, $P_{1,t} = I$, and the classical Langevin covariance equation is recovered. For $p = \infty$, one has

$$P_{\infty,t} = \text{tr}(F_t^{1/2}) F_t^{\dagger/2}.$$

Proof. For $\mu_t = \mathcal{N}(m_t, \Sigma_t)$ and $\nu = \mathcal{N}(\bar{m}, \bar{\Sigma})$, the first variation of $\text{KL}(\mu | \nu)$ has Wasserstein gradient

$$g_t(x) = \nabla_x \log \frac{d\mu_t}{d\nu}(x) = (\bar{\Sigma}^{-1} - \Sigma_t^{-1})(x - m_t) + \bar{\Sigma}^{-1}(m_t - \bar{m}).$$

Thus g_t is affine. Set $H_t := \bar{\Sigma}^{-1} - \Sigma_t^{-1}$ and $b_t := \bar{\Sigma}^{-1}(m_t - \bar{m})$, so that $g_t(x) = H_t(x - m_t) + b_t$. Since $\int (x - m_t) d\mu_t(x) = 0$ and $\int (x - m_t)(x - m_t)^\top d\mu_t(x) = \Sigma_t$, the covariance matrix of g_t is exactly

$$\int g_t(x)g_t(x)^\top d\mu_t(x) = H_t\Sigma_tH_t + b_tb_t^\top = F_t.$$

For the Schatten geometry associated with p , the Measure-LMO gives the velocity

$$v_t(x) = -P_{p,t}g_t(x), \quad P_{p,t} = \mathcal{N}_t^{2-q}F_t^{q/2-1}, \quad \mathcal{N}_t^q = \text{tr}(F_t^{q/2}).$$

This formula is the Schatten duality map applied to the vector field g_t . When $F_t = 0$, one has $g_t = 0$ μ_t -a.e., and the flow is stationary. Because g_t is affine and $P_{p,t}$ is independent of x , the selected velocity v_t is affine in x . The continuity equation for this prescribed affine velocity transports Gaussian laws to Gaussian laws. The moment equations computed below therefore verify the claimed Gaussian solution; they do not establish uniqueness among all solutions of the nonlinear PDE.

We now compute the induced ODE on mean and covariance. Since $v_t(x) = -P_{p,t}H_t(x - m_t) - P_{p,t}b_t$, the mean satisfies

$$\dot{m}_t = \int v_t(x) d\mu_t(x) = -P_{p,t}b_t.$$

But $b_t = \bar{\Sigma}^{-1}(m_t - \bar{m})$, hence $-b_t = \dot{m}_t^{(1)}$. Therefore $\dot{m}_t = P_{p,t}\dot{m}_t^{(1)}$.

For the covariance, using $\dot{\Sigma}_t = \int (v_t(x)(x - m_t)^\top + (x - m_t)v_t(x)^\top) d\mu_t(x)$, we obtain

$$\dot{\Sigma}_t = -P_{p,t}H_t\Sigma_t - \Sigma_tH_tP_{p,t}.$$

On the other hand,

$$\dot{\Sigma}_t^{(1)} = 2I - \bar{\Sigma}^{-1}\Sigma_t - \Sigma_t\bar{\Sigma}^{-1}.$$

Since $H_t = \bar{\Sigma}^{-1} - \Sigma_t^{-1}$ and both $\bar{\Sigma}^{-1}$ and Σ_t^{-1} are symmetric,

$$-H_t\Sigma_t = I - \bar{\Sigma}^{-1}\Sigma_t, \quad -\Sigma_tH_t = I - \Sigma_t\bar{\Sigma}^{-1}.$$

Therefore

$$-P_{p,t}H_t\Sigma_t - \Sigma_tH_tP_{p,t} = P_{p,t}(I - \bar{\Sigma}^{-1}\Sigma_t) + (I - \Sigma_t\bar{\Sigma}^{-1})P_{p,t}.$$

This is the intrinsic covariance equation stated above. Whenever $\bar{\Sigma}^{-1}\Sigma_t = \Sigma_t\bar{\Sigma}^{-1}$, this reduces to the displayed symmetrized formula because $\dot{\Sigma}_t^{(1)} = 2(I - \bar{\Sigma}^{-1}\Sigma_t)$.

Finally, for $p = 1$ one has $q = 2$, hence $\mathcal{N}_t^{2-q} = 1$ and $F_t^{q/2-1} = I$, so $P_{1,t} = I$. The equations therefore reduce to $\dot{m}_t = -\bar{\Sigma}^{-1}(m_t - \bar{m})$ and $\dot{\Sigma}_t = 2I - \bar{\Sigma}^{-1}\Sigma_t - \Sigma_t\bar{\Sigma}^{-1}$, which is the Gaussian restriction of the classical Langevin/Fokker-Planck flow. For $p = \infty$, $q = 1$, giving $P_{\infty,t} = \text{tr}(F_t^{1/2})F_t^{\dagger/2}$. $\square$

I P-L and Poincaré Inequalities

The preceding section shows that geodesic convexity is useful but sometimes too rigid for Spectral Wasserstein geometries. A more direct route to convergence rates is a Polyak-Łojasiewicz (P-L) inequality: a first-order estimate requiring the metric slope to control the energy gap.

I.1 P-L inequalities

Let $\mathcal{F}_\star := \inf_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} \mathcal{F}(\mu) > -\infty$. We use the descending metric slope defined in Section 5. Whenever its identification with the dual force norm holds, as under Theorem 5.1, it can be computed from $g_\mu = \nabla \delta \mathcal{F} / \delta \mu$ by

$$|\partial \mathcal{F}|_\gamma(\mu)^2 = \sup_{\mathcal{N}_\mu(v) \leq 1} \left(\int g_\mu(x) \cdot v(x) d\mu(x) \right)^2.$$

Definition I.1 ($\mathcal{W}_\gamma$ -P-L inequality). *We say that $\mathcal{F}$ satisfies the $\mathcal{W}_\gamma$ -P-L inequality with constant $\kappa > 0$ if*

$$|\partial \mathcal{F}|_\gamma(\mu)^2 \geq 2\kappa(\mathcal{F}(\mu) - \mathcal{F}_\star)$$

for every μ in the domain of $\mathcal{F}$.

Proposition I.2 (W_γ -P-L convergence). *Assume that $\mathcal{F}$ satisfies Definition I.1 and is lower semicontinuous for W_γ convergence. Let $(\mu_t)_{t \geq 0}$ be a global, locally absolutely continuous W_γ gradient flow with $\mathcal{F}(\mu_0) < +\infty$. Assume that $t \mapsto \mathcal{F}(\mu_t)$ is locally absolutely continuous and satisfies the energy dissipation identity*

$$\frac{d}{dt}\mathcal{F}(\mu_t) = -|\dot{\mu}_t|_\gamma^2 = -|\partial\mathcal{F}|_\gamma(\mu_t)^2 \quad \text{for a.e. } t > 0.$$

Set $E(t) = \mathcal{F}(\mu_t) - \mathcal{F}_*$. Then, for $0 \leq s \leq t$,

$$E(t) \leq e^{-2\kappa(t-s)} E(s),$$

and

$$\text{Length}_{W_\gamma}((\mu_r)_{r \in [s,t]}) \leq \sqrt{\frac{2}{\kappa}} (\sqrt{E(s)} - \sqrt{E(t)}).$$

Consequently, μ_t converges to some $\mu_\infty \in \arg\min \mathcal{F}$, and

$$W_\gamma(\mu_t, \mu_\infty) \leq \sqrt{\frac{2}{\kappa}} \sqrt{E(t)} \leq \sqrt{\frac{2E(0)}{\kappa}} e^{-\kappa t}.$$

In particular,

$$\text{dist}_{W_\gamma}(\mu_t, \arg\min \mathcal{F}) \leq \sqrt{\frac{2E(0)}{\kappa}} e^{-\kappa t}.$$

Proof. The energy identity and the P-L inequality give $E'(t) \leq -2\kappa E(t)$, hence the exponential bound by Gronwall's lemma. On the set where $E(t) > 0$, the same two identities give

$$|\dot{\mu}_t|_\gamma = |\partial\mathcal{F}|_\gamma(\mu_t) \leq \frac{|\partial\mathcal{F}|_\gamma(\mu_t)^2}{\sqrt{2\kappa E(t)}} = \frac{-E'(t)}{\sqrt{2\kappa E(t)}}.$$

Integrating from s to t gives the length estimate; intervals on which $E = 0$ are stationary by the energy identity. The tail length is finite, so (μ_t) is Cauchy. Proposition 3.2 and completeness of W_2 imply completeness of W_γ , hence $\mu_t \rightarrow \mu_\infty$. Lower semicontinuity gives $\mathcal{F}(\mu_\infty) = \mathcal{F}_*$, and the last bound follows by sending the final time to $+\infty$ and using that distance is bounded by metric length. $\square$

The following comparison explains why P-L estimates transfer more readily than geodesic convexity: only the tangent norms and their dual slopes enter.

Proposition I.3 (Transfer from W_2 to W_γ). *Assume that*

$$\gamma(S) \leq C_\gamma \text{tr}(S) \quad \text{for every } S \succeq 0.$$

If $\mathcal{F}$ satisfies the W_2 -P-L inequality with constant κ , then it satisfies the W_γ -P-L inequality with constant κ/C_γ .

Proof. The static comparison gives $W_\gamma \leq \sqrt{C_\gamma} W_2$. Both distances induce the same topology, so the definitions of descending slope yield $|\partial\mathcal{F}|_\gamma \geq C_\gamma^{-1/2} |\partial\mathcal{F}|_2$. Combining this with the W_2 -P-L inequality proves the claim, without a differentiability assumption on $\mathcal{F}$. $\square$

The normalization of the gauge matters. Replacing γ by $a\gamma$, $a > 0$, multiplies squared distances and squared tangent norms by a , while dividing squared slopes, P-L constants, and the Poincaré constants below by a . All comparisons in this section use the standard Schatten normalization $\gamma_p(aa^\top) = |a|^2$.

For the Schatten gauges $\gamma_p(S) = \|S\|_{S_p}$ with $p \geq 1$, including the Muon endpoint $p = \infty$, one has $\gamma_p(S) \leq \text{tr}(S)$ on $\mathbb{S}_+^d$, so $C_\gamma = 1$. Thus every W_2 -P-L inequality transfers to these spectral geometries with no loss of constant. A basic instance is the relative entropy $\mathcal{F}(\alpha) = \text{KL}(\alpha|\beta)$: if $d\beta = Z^{-1}e^{-V}dx$ and $\nabla^2 V \succeq \lambda I$ for some $\lambda > 0$, then the classical log-Sobolev inequality gives the λ -P-L inequality in W_2 [Bakry et al., 2014, Villani, 2009]. Hence the same functional is also λ -P-L for the Muon geometry. Thus any global Muon entropy flow satisfying the energy identity above inherits at least the same guaranteed exponential rate in function value as the W_2 flow. This is a comparison of certified rates, not a pointwise ordering of the two trajectories.

I.2 Spectral Poincaré inequalities

Throughout this subsection, assume that $\beta \in \mathcal{P}_2(\mathbb{R}^d)$ has a strictly positive C^1 density. Test functions are taken in the weighted Sobolev space $H^1(\beta)$, the completion of smooth compactly supported functions for the norm $(\int (u^2 + |\nabla u|^2) d\beta)^{1/2}$. These assumptions make the weak gradient well defined and the smooth and Sobolev infima below agree.

The comparison above gives a simple way to transfer a W_2 -P-L estimate, but it does not identify the sharp W_γ -P-L constant in general. Deciding whether a geometry such as Muon yields a strictly better global constant is difficult because its nonlinear slope must be controlled over the entire energy landscape. We therefore study the linearization at a minimizer. This amounts formally to measuring the spectral gap of the Hessian in the spectral tangent norm. For concreteness, consider the relative entropy

$$\mathcal{F}(\alpha) = \text{KL}(\alpha \mid \beta).$$

Its W_2 gradient flow is the Fokker–Planck, or drift–diffusion, process with invariant law β . Its local spectral gap in the W_γ geometry is a γ -Poincaré constant, generalizing the classical Poincaré constant of β .

Let $u \in H^1(\beta)$ be a bounded smooth perturbation with $\int u d\beta = 0$ and set $\alpha_\varepsilon = (1 + \varepsilon u)\beta$ for sufficiently small $|\varepsilon|$. Then

$$\text{KL}(\alpha_\varepsilon \mid \beta) = \frac{\varepsilon^2}{2} \text{Var}_\beta(u) + O(\varepsilon^3), \quad \nabla \log \frac{d\alpha_\varepsilon}{d\beta} = \frac{\varepsilon \nabla u}{1 + \varepsilon u} = \varepsilon \nabla u + O_{L^2(\beta)}(\varepsilon^2).$$

At the level of the dual tangent norm, these expansions give

$$\mathcal{N}_{\alpha_\varepsilon}^* \left(\nabla \log \frac{d\alpha_\varepsilon}{d\beta} \right)^2 = \varepsilon^2 \mathcal{E}_\gamma^\beta(u) + o(\varepsilon^2),$$

where the spectral Dirichlet form is

$$\mathcal{E}_\gamma^\beta(u) := \sup_{\mathcal{N}_\beta(v) \leq 1} \left(\int \nabla u \cdot v d\beta \right)^2.$$

Whenever the entropy slope is identified with this dual norm, the linearization of a κ -P-L inequality is exactly $\kappa \text{Var}_\beta(u) \leq \mathcal{E}_\gamma^\beta(u)$. The spectral Dirichlet form and the inequalities below are defined independently of an existence or slope-identification theorem for the nonlinear entropy flow.

Definition I.4 (γ -Poincaré constant). *The γ -Poincaré constant of β is*

$$\lambda_P^\gamma(\beta) := \inf_{\text{Var}_\beta(u) > 0} \frac{\mathcal{E}_\gamma^\beta(u)}{\text{Var}_\beta(u)}.$$

Equivalently,

$$\lambda_P^\gamma(\beta) \text{Var}_\beta(u) \leq \mathcal{E}_\gamma^\beta(u)$$

for all smooth u .

Equivalently, one may take the infimum over the corresponding Sobolev form domain; the finite-dimensional norm comparisons make the two formulations identical by density.

Proposition I.5 (Robust Poincaré representation). *Let $\mathcal{K}_\gamma \subset \mathbb{S}_+^d$ be a compact convex representing set for γ , and assume that $\mathcal{K}_\gamma \cap \mathbb{S}_{++}^d \neq \emptyset$. Then, for every smooth u ,*

$$\mathcal{E}_\gamma^\beta(u) = \inf_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \int \nabla u(x)^\top Q^{-1} \nabla u(x) d\beta(x).$$

The infimum need not be attained; a singular optimizer is understood through the lower-semicontinuous limit from $\mathcal{K}_\gamma \cap \mathbb{S}_{++}^d$. Consequently,

$$\lambda_P^\gamma(\beta) = \inf_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \lambda_P^Q(\beta), \quad \lambda_P^Q(\beta) := \inf_{\text{Var}_\beta(u) > 0} \frac{\int \nabla u^\top Q^{-1} \nabla u d\beta}{\text{Var}_\beta(u)}.$$

Proof. Let $\ell_u(v) = \int \nabla u \cdot v \, d\beta$ and $\mathcal{N}_\beta(v)^2 = \gamma(\int vv^\top d\beta)$. Fenchel duality for a squared norm gives

$$\inf_v \left\{ \ell_u(v) + \frac{1}{2} \mathcal{N}_\beta(v)^2 \right\} = -\frac{1}{2} \sup_{\mathcal{N}_\beta(v) \leq 1} \ell_u(v)^2 = -\frac{1}{2} \mathcal{E}_\gamma^\beta(u).$$

On the other hand, set $\Phi_Q(v) = \ell_u(v) + \frac{1}{2} \int v^\top Q v \, d\beta$. The support representation of γ and the inverse-trace minimax argument of Proposition A.7 give

$$\begin{aligned} \inf_v \left\{ \ell_u(v) + \frac{1}{2} \mathcal{N}_\beta(v)^2 \right\} &= \inf_v \max_{Q \in \mathcal{K}_\gamma} \Phi_Q(v) = \max_{Q \in \mathcal{K}_\gamma} \inf_v \Phi_Q(v) \\ &= -\frac{1}{2} \inf_{Q \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d} \text{tr} \left(Q^{-1} \int \nabla u \nabla u^\top d\beta \right). \end{aligned}$$

Here any $Q_0 \in \mathcal{K}_\gamma \cap \mathbb{S}_{++}^d$ provides the coercivity needed for the minimax exchange, and convexity of $\mathcal{K}_\gamma$ lets one approximate every singular Q by $(1 - \varepsilon)Q + \varepsilon Q_0 \succ 0$. Comparing the two displays proves the formula for $\mathcal{E}_\gamma^\beta(u)$. Taking the infimum over all u with positive variance gives the formula for $\lambda_\beta^\gamma(\beta)$; the two infima commute because this is an infimum over the product of admissible functions and matrices. $\square$

Equivalently, $\lambda_\beta^Q(\beta)$ is the usual Poincaré constant after the linear change of variables $(Q^{1/2})_\# \beta$. Hence the spectral Poincaré constant is the worst anisotropic Poincaré constant over the dual ball $\mathcal{K}_\gamma$. At singular endpoints, this worst case may be approached by positive definite matrices without being attained in $\mathbb{S}_{++}^d$.

For the Schatten family γ_p , $p \geq 1$, this becomes especially explicit. With $p' = p/(p-1)$, using the conventions $p' = \infty$ for $p = 1$ and $p' = 1$ for $p = \infty$, one may take

$$\mathcal{K}_p = \{Q \succeq 0 : \|Q\|_{S_{p'}} \leq 1\}.$$

Hence

$$\lambda_\beta^p(\beta) = \inf_{\substack{Q \succ 0 \\ \|Q\|_{S_{p'}} \leq 1}} \lambda_P((Q^{1/2})_\# \beta).$$

For $p = \infty$, this is the Muon formula

$$\lambda_\beta^\infty(\beta) = \inf_{\substack{Q \succ 0 \\ \text{tr } Q \leq 1}} \lambda_P((Q^{1/2})_\# \beta).$$

Since every admissible Q satisfies $Q \preceq I$, one has $Q^{-1} \succeq I$ and therefore

$$\lambda_\beta^p(\beta) \geq \lambda_\beta^1(\beta) \quad (p \geq 1).$$

Thus the local Poincaré rate of the spectrally normalized drift–diffusion flow is no smaller than the classical W_2 rate for all Schatten geometries from the trace case to the Muon endpoint.

The same local computation also gives a simple way to compare examples across the full root-norm Schatten range $p \geq 1/2$ discussed in Appendix E.4. For a smooth test function u , set

$$G_u := \int \nabla u \nabla u^\top d\beta.$$

For $\gamma_p(S) = \|S\|_{S_p}$, $p \geq 1/2$, the tangent norm is the Schatten S_{2p} norm of the operator $T_v : a \mapsto a \cdot v$. By Schatten duality, the spectral Dirichlet form is

$$\mathcal{E}_p^\beta(u) = \left\| G_u^{1/2} \right\|_{S_{\theta_p}}^2, \quad \theta_p = (2p)',$$

with the conventions $\theta_{1/2} = +\infty$ and $\theta_\infty = 1$. Relative to the trace case,

$$\mathcal{E}_p^\beta(u) \geq \mathcal{E}_1^\beta(u) \quad (p > 1), \quad \mathcal{E}_p^\beta(u) \leq \mathcal{E}_1^\beta(u) \quad (1/2 \leq p < 1),$$

and, away from $p = 1$, equality holds precisely when G_u has rank at most one. Thus the difference between the geometries is visible exactly through perturbations with several active gradient directions.

Proposition I.6 (Gaussian targets). *Let $\beta = \mathcal{N}(m, \Sigma)$ be a nondegenerate Gaussian measure on $\mathbb{R}^d$, $d \geq 2$. Then, for every $p \geq 1$,*

$$\lambda_P^p(\beta) = \lambda_P^1(\beta) = \frac{1}{\lambda_{\max}(\Sigma)}.$$

For $1/2 \leq p < 1$, the constant is strictly smaller:

$$d^{1-1/p} \lambda_P^1(\beta) \leq \lambda_P^p(\beta) < \lambda_P^1(\beta).$$

Proof. For $p \geq 1$, $\theta_p \leq 2$, so $\|G^{1/2}\|_{S_{\theta_p}}^2 \geq \text{tr } G$. The classical Gaussian Poincaré inequality gives $\lambda_P^p(\beta) \geq 1/\lambda_{\max}(\Sigma)$. Equality follows by testing with a linear function $u(x) = a^\top x$, where a is an eigenvector of Σ associated with $\lambda_{\max}(\Sigma)$: then $G_u = aa^\top$ is rank one, so all Schatten norms coincide on $G_u^{1/2}$.

Let now $1/2 \leq p < 1$, so $\theta_p > 2$. The Schatten comparison gives

$$\|G^{1/2}\|_{S_{\theta_p}}^2 \geq d^{1-1/p} \text{tr } G,$$

which proves the lower bound. For the strict upper bound, translate the measure so that $m = 0$, diagonalize Σ , and write $\sigma_1 = \lambda_{\max}(\Sigma)$ and $\sigma_2 > 0$ for another eigenvalue. In these coordinates, take

$$u_\varepsilon(x) = x_1 + \varepsilon(x_2^2 - \sigma_2).$$

Then

$$\text{Var}_\beta(u_\varepsilon) = \sigma_1 + 2\varepsilon^2 \sigma_2^2, \quad G_{u_\varepsilon} = e_1 e_1^\top + 4\varepsilon^2 \sigma_2 e_2 e_2^\top.$$

Thus $\mathcal{E}_p^\beta(u_\varepsilon) = 1 + o(\varepsilon^2)$ as $\varepsilon \rightarrow 0$ for every $\theta_p > 2$ (and it is exactly 1 for small ε when $\theta_p = \infty$). Therefore

$$\frac{\mathcal{E}_p^\beta(u_\varepsilon)}{\text{Var}_\beta(u_\varepsilon)} = \frac{1}{\sigma_1} - \frac{2\sigma_2^2}{\sigma_1^2} \varepsilon^2 + o(\varepsilon^2),$$

which is strictly smaller than $1/\sigma_1$ for all sufficiently small $\varepsilon > 0$. □

Proposition I.7 (Generalized Gaussian targets). *Let $d \geq 2$, let $r > 1$, $r \neq 2$, and let*

$$d\beta_r(x) = Z_{r,d}^{-1} e^{-|x|^r} dx.$$

Then the Schatten Poincaré constant is strictly improved for $p > 1$ and strictly decreased for $1/2 \leq p < 1$:

$$\lambda_P^p(\beta_r) > \lambda_P^1(\beta_r) \quad \text{for } p > 1, \quad \lambda_P^p(\beta_r) < \lambda_P^1(\beta_r) \quad \text{for } 1/2 \leq p < 1.$$

Thus, for the family $e^{-|x|^r}$ in dimension $d > 1$, the Gaussian exponent $r = 2$ is the only case with no strict improvement of the local Poincaré rate for $p > 1$.

Proof. The reversible generator is

$$L_r = \Delta - r|x|^{r-2} x \cdot \nabla.$$

Writing $V_r(x) = |x|^r$, the ground-state transform of $-L_r$ has effective potential

$$\frac{1}{4} |\nabla V_r|^2 - \frac{1}{2} \Delta V_r = \frac{r^2}{4} |x|^{2r-2} - \frac{r(r+d-2)}{2} |x|^{r-2},$$

which tends to $+\infty$ as $|x| \rightarrow \infty$. Thus the classical Poincaré problem has compact resolvent and its spectral gap is attained.

We first prove that no first eigenfunction has rank-one gradient covariance. Indeed, if G_u has rank one, then the positive density of β_r implies that ∇u is a.e. parallel to some unit vector a . Hence $u(x) = \psi(a \cdot x)$, and, with $t = a \cdot x$, the eigenvalue equation $-L_r u = \lambda u$ becomes

$$-\psi''(t) + r|t|^{r-2} t \psi'(t) = \lambda \psi(t).$$

Since u is nonconstant, one can choose $t \neq 0$ with $\psi'(t) \neq 0$. Setting $x = ta + z$ and varying $z \perp a$ forces $|ta + z|^{r-2}$ to be independent of $|z|$, which is possible only for $r = 2$. This contradiction proves the claim.

Let $\lambda_1 = \lambda_P^1(\beta_r)$. If $p > 1$, then $\theta_p < 2$ and $\mathcal{E}_p^{\beta_r} \geq \mathcal{E}_1^{\beta_r}$, so $\lambda_P^p(\beta_r) \geq \lambda_1$. Suppose equality holds and take a minimizing sequence (u_n) with zero mean and unit variance. Since

$$\lambda_1 \leq \mathcal{E}_1^{\beta_r}(u_n) \leq \mathcal{E}_p^{\beta_r}(u_n) \longrightarrow \lambda_1,$$

its classical Dirichlet energy also converges to λ_1 . Compactness of the resolvent and the spectral decomposition then give, after extraction, convergence in the form norm to a first classical eigenfunction u_* . Moreover, $\mathcal{E}_p^{\beta_r}(u_n) - \mathcal{E}_1^{\beta_r}(u_n) \rightarrow 0$, and the continuity of $u \mapsto G_u$ along this convergence gives

$$\left\| G_{u_*}^{1/2} \right\|_{S_{\theta_p}} = \left\| G_{u_*}^{1/2} \right\|_{S_2}.$$

For $\theta_p < 2$, equality in the Schatten comparison is possible only when G_{u_*} has rank one, impossible by the rigidity fact above. Hence the inequality is strict for $p > 1$.

If $1/2 \leq p < 1$, take any first classical eigenfunction u . The previous paragraph shows that G_u has rank at least two. Since now $\theta_p > 2$,

$$\mathcal{E}_p^{\beta_r}(u) = \left\| G_u^{1/2} \right\|_{S_{\theta_p}}^2 < \left\| G_u^{1/2} \right\|_{S_2}^2 = \mathcal{E}_1^{\beta_r}(u) = \lambda_1 \text{Var}_{\beta_r}(u),$$

which gives $\lambda_P^p(\beta_r) < \lambda_P^1(\beta_r)$. □

The restriction $d \geq 2$ is essential: in dimension one, every gradient covariance is scalar, so all the Schatten Poincaré constants coincide.

J Positively 2-Homogeneous Case and a Generalized Unbalanced Geometry

This section continues the two-layer mean-field discussion of Section 5 in the positively two-homogeneous case. The point is that such models reduce the ambient transport problem to an unbalanced transport problem on the sphere, and this section explains how the spherical dynamics inherits a spectral action from the ambient flow. Let

$$\mathcal{F}(\mu) = R \left(\int_{\mathbb{R}^d} \Phi(x) d\mu(x) \right), \quad \Phi(\lambda x) = \lambda^2 \Phi(x).$$

Writing $x = r\omega$ with $\omega \in \mathbb{S}^{d-1}$, define the weighted spherical projection

$$\int_{\mathbb{S}^{d-1}} \psi(\omega) d\Pi_2(\mu)(\omega) := \int_{\mathbb{R}^d} |x|^2 \psi\left(\frac{x}{|x|}\right) d\mu(x).$$

The integrand on the right is defined to be zero at $x = 0$. The first point is that two-homogeneous models only depend on this weighted spherical projection.

Proposition J.1 (Exact quotient). *There exists a functional $\bar{f}$ on $\mathcal{M}_+(\mathbb{S}^{d-1})$ such that*

$$\mathcal{F}(\mu) = \bar{f}(\Pi_2(\mu)).$$

Proof. The homogeneity identity $\Phi(r\omega) = r^2 \Phi(\omega)$ immediately gives

$$\int_{\mathbb{R}^d} \Phi(x) d\mu(x) = \int_{\mathbb{S}^{d-1}} \Phi(\omega) d\Pi_2(\mu)(\omega),$$

which defines $\bar{f}$. □

The second point is that the ambient normalized velocity splits into radial and tangential parts on the sphere. If a vector field v is 1-homogeneous, namely $v(\lambda x) = \lambda v(x)$ for every $\lambda > 0$, then for every $x = r\omega \neq 0$ it can be written uniquely as

$$v(r\omega) = r(b(\omega)\omega + \tau(\omega)), \quad \tau(\omega) \in T_\omega \mathbb{S}^{d-1}.$$

Indeed, one defines

$$b(\omega) := v(\omega) \cdot \omega, \quad \tau(\omega) := v(\omega) - (v(\omega) \cdot \omega)\omega,$$

so that $\tau(\omega) \cdot \omega = 0$, and then 1-homogeneity gives

$$v(r\omega) = r v(\omega) = r(b(\omega)\omega + \tau(\omega)).$$

The decomposition is unique because it is simply the orthogonal splitting of $v(\omega)$ into its normal and tangential parts on the sphere. In the positively two-homogeneous setting considered here, the first variation is 2-homogeneous and its spatial gradient is therefore 1-homogeneous; correspondingly, the natural steepest-descent velocity fields for the flow belong to this 1-homogeneous class. The next proposition computes the projected spherical PDE.

Proposition J.2 (Projected continuity-reaction equation). *If μ_t solves the Spectral Wasserstein flow and $\nu_t = \Pi_2(\mu_t)$, then*

$$\partial_t \nu_t + \operatorname{div}_{\mathbb{S}^{d-1}}(\nu_t \tau_t) = 2b_t \nu_t.$$

Proof. Test the ambient continuity equation against $\tilde{\psi}(r\omega) = r^2\psi(\omega)$. For $r > 0$,

$$\nabla \tilde{\psi}(r\omega) = r(2\psi(\omega)\omega + \nabla_{\mathbb{S}^{d-1}}\psi(\omega)).$$

Using $v(r\omega) = r(b(\omega)\omega + \tau(\omega))$ and $\omega \cdot \tau(\omega) = 0$, we obtain

$$\nabla \tilde{\psi}(r\omega) \cdot v(r\omega) = r^2(2b(\omega)\psi(\omega) + \nabla_{\mathbb{S}^{d-1}}\psi(\omega) \cdot \tau(\omega)).$$

The weak form of the ambient continuity equation and the definition of Π_2 therefore give

$$\frac{d}{dt} \int_{\mathbb{S}^{d-1}} \psi d\nu_t = \int_{\mathbb{S}^{d-1}} (2b_t\psi + \nabla_{\mathbb{S}^{d-1}}\psi \cdot \tau_t) d\nu_t,$$

which is precisely the weak form of the stated continuity–reaction equation. $\square$

Motivated by Proposition J.2, define for nonnegative measures ν_0, ν_1 on $\mathbb{S}^{d-1}$

$$\operatorname{UW}_\gamma(\nu_0, \nu_1)^2 := \inf_{(\nu_t, b_t, \tau_t)} \int_0^1 \gamma \left(\int_{\mathbb{S}^{d-1}} (b_t(\omega)\omega + \tau_t(\omega))(b_t(\omega)\omega + \tau_t(\omega))^\top d\nu_t(\omega) \right) dt,$$

under the continuity–reaction constraint

$$\partial_t \nu_t + \operatorname{div}_{\mathbb{S}^{d-1}}(\nu_t \tau_t) = 2b_t \nu_t.$$

The next proposition identifies the ambient homogeneous action with this spherical unbalanced action.

Proposition J.3 (Ambient action equals spherical action). *For positively two-homogeneous models and 1-homogeneous velocities, the ambient Spectral Wasserstein action equals the spherical action defining UW_γ .*

Proof. Write $x = r\omega$ and $v(r\omega) = r(b(\omega)\omega + \tau(\omega))$. By definition of $\Pi_2(\mu)$,

$$\int_{\mathbb{R}^d} v(x)v(x)^\top d\mu(x) = \int_{\mathbb{S}^{d-1}} (b(\omega)\omega + \tau(\omega))(b(\omega)\omega + \tau(\omega))^\top d\Pi_2(\mu)(\omega).$$

Inserting this identity into the Benamou–Brenier action yields the claim. $\square$

The following remark explains how the trace-norm case collapses to the classical Wasserstein–Fisher–Rao geometry.

Remark J.4. *When $\gamma(S) = \operatorname{tr}(S)$, the integrand becomes*

$$\int_{\mathbb{S}^{d-1}} (|\tau_t(\omega)|^2 + b_t(\omega)^2) d\nu_t(\omega).$$

Since the reaction rate is $\alpha_t = 2b_t$, the reaction term is $\alpha_t^2/4$. Thus the trace case is the Wasserstein–Fisher–Rao action with this transport/reaction normalization, whose static formulations are developed in Chizat et al. [2018], Liero et al. [2018]. For general γ , the construction gives a generalized unbalanced transport geometry on the sphere. Whether it admits an analogous static formulation is an interesting open direction.

K Notation

This section collects the main notation used throughout the paper. The first table lists standard background notation, while the second table lists the notation introduced for the Spectral Wasserstein geometry.

Table 2: Classical notation.

Notation	Name	First use
$\mathbb{R}^d$	Euclidean space	Sec. 2
$\mathbb{S}^d$	Symmetric matrices	Sec. 2
$\mathbb{S}_+^d$	PSD matrices	Sec. 2
$\mathbb{S}_+^{d,+}$	Positive definite matrices	Prop. A.7
$S \preceq T$	Loewner order	Sec. 2
I, Id	Identity matrix/map	Sec. 2
$\text{tr}(S)$	Trace	Def. 2.1
$\text{diag}(a_i)$	Diagonal matrix	Sec. 2
$\lambda(S)$	Eigenvalues of S	Sec. 2
$\ \cdot\ _F$	Frobenius norm	Def. 2.2
$\langle \cdot, \cdot \rangle_F$	Frobenius product	Def. 2.2
$\ A\ _{S_p}$	Schatten p -norm	Sec. 2
$A^\dagger$	Pseudoinverse	Eq. (6)
$\mathcal{P}_2(\mathbb{R}^d)$	Measures with finite second moment	Def. 2.3
δ_x	Dirac mass	Sec. 2
$T_\# \mu$	Pushforward	Sec. 4
$\Pi(\mu, \nu)$	Couplings	Def. 3.1
$L^2(\mu; \mathbb{R}^d)$	Square-integrable vector fields	Def. 2.3
∇, ∇_x	Euclidean gradient	Sec. 2
div	Divergence	Eq. (3)
$\delta \mathcal{F} / \delta \mu$	First variation	Sec. 2
W_2	Quadratic Wasserstein distance	Sec. 1
$\mathcal{N}(m, \Sigma)$	Gaussian law	Prop. 3.6
Σ	Covariance matrix	Prop. 3.6
$\mathbb{S}^{d-1}$	Unit sphere	App. J
$\nabla_{\mathbb{S}^{d-1}}, \text{div}_{\mathbb{S}^{d-1}}$	Spherical gradient/divergence	Prop. J.2

Table 3: Main notation introduced in the paper.

Notation	Name	Introduction
γ	Monotone matrix gauge on $\mathbb{S}_+^d$	Sec. 2
γ_p	Schatten- p gauge	Sec. 2
p'	Conjugate exponent of p	Rem. A.4
$q = (2p)'$	Tangent-space dual exponent	Prop. A.5
$\mathcal{K}_\gamma$	Representing set	Def. 3.3
$\mathcal{P}_\gamma$	Polar set	App. A
$\mathcal{K}_{\gamma_p}$	Schatten representing set	Rem. A.4
$N(V)$	Matrix tangent norm	Sec. 2
$J_\gamma(G)$	Matrix LMO	Def. 2.2
$J_p(G)$	Schatten matrix LMO	Prop. A.5
X	Particle matrix	Sec. 2
μ_X	Empirical particle measure	Sec. 2
$F(X)$	Particle objective	Sec. 2
$\mathcal{F}(\mu)$	Mean-field objective	Sec. 2
$\mathcal{N}_\mu(v)$	Measure tangent norm	Def. 2.3
$\mathcal{J}_\gamma^\mu(g)$	Measure-LMO	Def. 2.3

Notation	Name	Introduction
$\mathcal{S}_\mu(g)$	Force covariance	Prop. 2.4
$\mathfrak{N}_\gamma(T)$	Root norm of an operator	Def. E.2
Q_μ^*	Inverse-trace minimizer	Prop. A.7
g_μ	Wasserstein force (before normalization)	Sec. 2
W_γ	Static Spectral Wasserstein distance	Def. 3.1
W^Q	Fixed-metric transport cost	Thm. 3.4
c_γ, C_γ	Trace comparison constants	Prop. 3.2
W_γ^{BB}	Dynamic Spectral Wasserstein distance	Def. 4.1
$\mathcal{A}_\gamma(\mu, m)$	Momentum action	Prop. E.1
W_γ^{M}	Monge-restricted cost	Def. C.1
$\phi(z, x)$	Feature map	Sec. 5
u, v	Outer and inner weights (MLP context)	Sec. 5
H_μ	Mean-field predictor	Sec. 5
$R(H)$	Risk functional	Sec. 5
$\text{MMD}(\mu, \nu)^2$	Squared MMD	Sec. 5
(Q, K, V, O)	Attention weights	Sec. 5
N_{eff}	Effective tokens	Sec. 5
C_ρ	Data covariance	App. F
B_ρ	Teacher cross-moment	App. F
M_μ	Learned linear predictor	App. F
$A_\star$	Teacher matrix	App. F
E_t	Linear prediction error	Prop. F.2
$r_i(t)$	Modal predictor coefficient	Prop. F.3
$s_i(t)$	Modal variance	Prop. F.3
β_i	Teacher mode	Prop. F.3
$\mathcal{N}_t$	Modal normalization	Prop. F.3
γ_{blk}	Blockwise gauge	App. B
L_γ	Matrix smoothness constant	Eq. (10)
$r_n = \min(n, d)$	Matrix rank bound	App. B.1
$\mathbf{I}_p(V)$	Momentum constitutive map	Eq. (7)
$F_h(\mu)$	Linear measure functional	Thm. H.3
κ	Convexity or P-L constant (by context)	Def. H.1
$\text{Ent}_\nu(\mu)$	Relative entropy	App. H
$\text{KL}(\mu \mid \nu)$	Relative entropy / KL divergence	Sec. H.5
$ \partial \mathcal{F} _\gamma$	Metric slope of $\mathcal{F}$	Sec. 5
$\mathcal{E}_\gamma^\beta$	Spectral Dirichlet form	App. I
$\lambda_\text{P}^\gamma(\beta)$	Spectral Poincaré constant	App. I
$\theta_p = (2p)'$	Dirichlet-form Schatten exponent	App. I
Φ	Two-homogeneous feature map	App. J
$\Pi_2(\mu)$	Spherical projection	App. J
UW_γ	Unbalanced spherical action	App. J